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Towards precision cosmology with Type Ia supernova peculiar velocities: from SN modeling to field level inference

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Introduction

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1 Introduction to Modern Cosmology

The quest to understand the natural world and how we are placed in it has existed for many centuries. It has influenced many religious, cultural and scientific ideas throughout time and their evolutions lead us to how we see our Universe today. Physical cosmology (cosmology just from now on) is the study of the nature of our Universe through observations in order to understand its origins, dynamics and components. We consider the beginning of modern cosmology was about a century ago with the discovery of the expansion of the Universe initially predicted by Lemaitre in 1927 ([Lemaître, 1927], [Lemaître, 1931]) and observed by Hubble in 1929 ([Hubble, 1929]). The current cosmological model named ΛCDM (Λ Cold Dark Matter) best describes of our Universe's history, dynamics and components. In this section I will detail the necessary tools that allows us to describe the large scale Universe as we know it today.

1.1 General Relativity

According to modern belief and understanding, the Universe contains four fundamental forces dominating at different scales: at large scales, that force is gravity. The basis of our understanding of modern cosmology is centered around Einstein's gravitational theory: General Relativity (GR) ([Einstein, 1916]), the most tested theory of gravity to date. GR's views on the nature of gravity differs a lot form the classical Newtonian view. In GR, massive objects do not directly exert a force on other massive objects, but instead cause the distortions in spacetime. The theory was established after the success of the theory of Special Relativity, also theorized by Einstein in 1905, where the speed of light was shown to be constant in all observation frames.

Understanding the mathematical foundation of GR requires a couple of fundamental notions. GR treats coordinate transformations in a 4-dimensional space (spacetime) (t, x, y, z) using Lorentz transformations. The formalism will be described using tensorial notations and the Einstein summing convention will also be assumed. We invite the reader to familiarise themselves with these notions as well as an understanding of classical field theory. We will also not be going into a lof details regarding some mathematical and geometrical notions (like geodesics, covectors, covariant derivatives, etc..) and physical concepts found in the Einstein equations such as the Riemann tensor $R^{\mu}_{\nu\kappa\lambda}$, Ricci tensor $R_{\mu\nu}$, the Ricci scalar R scalar an Christoffel symbols $\Gamma^{\mu}_{\nu\lambda}$. If you are interested in learning more on those concepts we recommend David Tong's lectures on General Relativity [1].

Equivalence principle

Einstein's equivalence principle builds up upon previous results obtained in Newtonian physics. The principle states that the gravitational mass m_g defined due to gravitational interactions is equal to the intertial mass m_i subdued to acceleration as a result of a force. This previously lead to the idea that free falling objects have the same velocity regardless of their masses. Einstein's principle extends this idea to accelerated frames and not just inertial frames. This principle has been heavily tested on Earth and in the Solar System and is yet to be invalidated (REFS).

Einstein equations

[1] https://www.damtp.cam.ac.uk/user/tong/gr.html

In order to define the equations of motions of any object in GR, the first thing that must be defined is a metric. Physically, a metric is an infinitesimal "separation" that is conserved. In classical physics, the length between two poins ΔL would be identical for two observers even if both are moving. In GR this is no longer true and only a combination of a spatial separation and a temporal one becomes invariant. If we write down the spacetime displacement as dx^μ with μ and index taking the values 0 if it's a time displacement and $1,2,3$ if it's spatial displacement along the x, y, z, we can define the invariant displacement ds^2 as :

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \tag{1.1}$$

where $g_{\mu\nu}$ is the metric that can depend on the coordinates x^μ. Essentially, the metric expresses the distance between two events in spacetime. An example of a metric is the Minkowski metric used to describe events on Special Relativity and is written as:

$$\eta_{\mu\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{1.2}$$

Since in GR spacetime can go through a series of curvatures, we do not expect that an object exhibiting no forces to move in a straight line. What we see instead is that they follow *geodesics*. One can think of geodesics as the straightest possible path in a curved space. The geodesic equation, which corresponds to the equation of motion for an point-like object, is written as:

$$\frac{d^2 x^\mu}{d\lambda^2} = \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} \tag{1.3}$$

where $\Gamma^\mu_{\alpha\beta}$ are the Christoffel symbols and λ is the evolution parameter which increases along the particle's path. One can derive the geodesic equation by applying the equivalence principle on a free falling particle that does not accelerate (such that $\frac{d^2x^\mu}{d\lambda^2} = 0$).

The Einstein field equations ([Einstein, 1916]) close the theory by relating the geometry of spacetime, encoded in the metric, to the distribution of matter and energy within it. In this framework, gravity is not a force but a manifestation of spacetime curvature, and the motion of particles follows the geodesics of that curved geometry. We note $G_{\mu\nu}$ the Einstein tensor describing the geometry and $T_{\mu\nu}$ the energy-momentum tensor describing the energy content, we get the following Einstein equations:

$$\boxed{G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}} \tag{1.4}$$

where $R_{\mu\nu}$ is the Ricci tensor, R the Ricci scalar, G Newton's gravitational constant and c the speed of light. It's important to note that for a symmetric metric, this represents a set of 10 equations where the coefficients of $g_{\mu\nu}$ are unknowns.

We can always rewrite the energy momentum tensor as $T_{\mu\nu} = T'_{\mu\nu} - \Lambda g_{\mu\nu}$ where Λ is a constant that can take any value, even 0. The Einstein equations would be:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T'_{\mu\nu} \tag{1.5}$$

Physically however, this new expression can lead to different interpretation of the nature behind the accelerated expanding Universe (see Section 5.1).

The Einstein equations could also be determined by varying the Einstein-Hilbert action which is:

$$S_{EH} = \frac{c^4}{16\pi G} \int d^4x\sqrt{-g}(R - 2\Lambda) + \int d^4x\sqrt{-g}\mathcal{L}_m \tag{1.6}$$

$$\frac{\delta S_{EH}}{\delta g_{\mu\nu}} = 0 \tag{1.7}$$

where g is the determinant of the metric and $\mathcal{L}_m$ is the Lagrangian density of a field that we include in our theory (such as matter, radiation ...). This is a mathematically rigorous approach to deriving the field equations and it is commonly used when exploring modified theories of gravity (see Section 5.4).

1.2 The expanding, homogenous Universe

Cosmological principle

A key principle in modern cosmology is the cosmological principle, stating that the Universe on large scales is homogenous and isotropic. This is understood in the sense that the Universe does not possess a privileged center and that the structures in it remain identical regardless of the direction of observation. Observations from the SDSS (Sloan Digital Sky Survey [Sarkar et al., 2009]) and DESI (Dark Energy Spectroscopic Instrument [Adame et al., 2025a]) have shown strong evidence for homogeneity in the Universe at large scales. We represent this in detail with large scale dark matter simulations such as the Uchuu simulation [Ishiyama et al., 2021] seen in Figure 1.1. We can see that on large scales the matter distribution is homogeneous. CMB measurements from the Planck collaboration ([Akrami et al., 2020]) have shown strong evidence for isotropy as shown in Figure 1.2.

Expanding Universe and Hubble relation

Following the publication of GR, discoveries from [Lemaître, 1927]'s work on the Einstein field equations showed that the proper velocity of a celestial object was proportional to its distance. Two years later, [Hubble, 1929] manages to show that the radial velocities of 32 nearby galaxies increase with their distances indicating an expanding Universe, which is shown in Figure 1.3. The Hubble-Lemaitre relation is established as :

$$v = H_0 d \tag{1.8}$$

where v is the velocity, d the distance of the object and H_0 the Hubble constant. Initially, H_0 was measured to be around 500kms^{-1}.Mpc^{-1}, today more precise measurements show that it is around 70kms^{-1}.Mpc^{-1}. We may also define a *Hubble diagram* as the diagram that relates the distance of celestial objects from an observer to their velocities.

An expanding Universe entails that objects emitting any kind of waves will be stretched in space making their wavelengths longer. We can call this the **cosmological redshift** z and it is defined as :

$$1 + z = \frac{\nu_{rest}}{\nu_{obs}} = \frac{\lambda_{obs}}{\lambda_{rest}} \tag{1.9}$$

where ν_{rest} and λ_{rest} are the frequency and wavelength emitted at restframe and ν_{obs} and λ_{obs} are the observed ones. Other sources that can the cosmological are the peculiar motion of celestial objects which can shift the emitted colour to either bluer or redder frequencies (see Section 3.2 where we explain the analysis done with peculiar velocities).

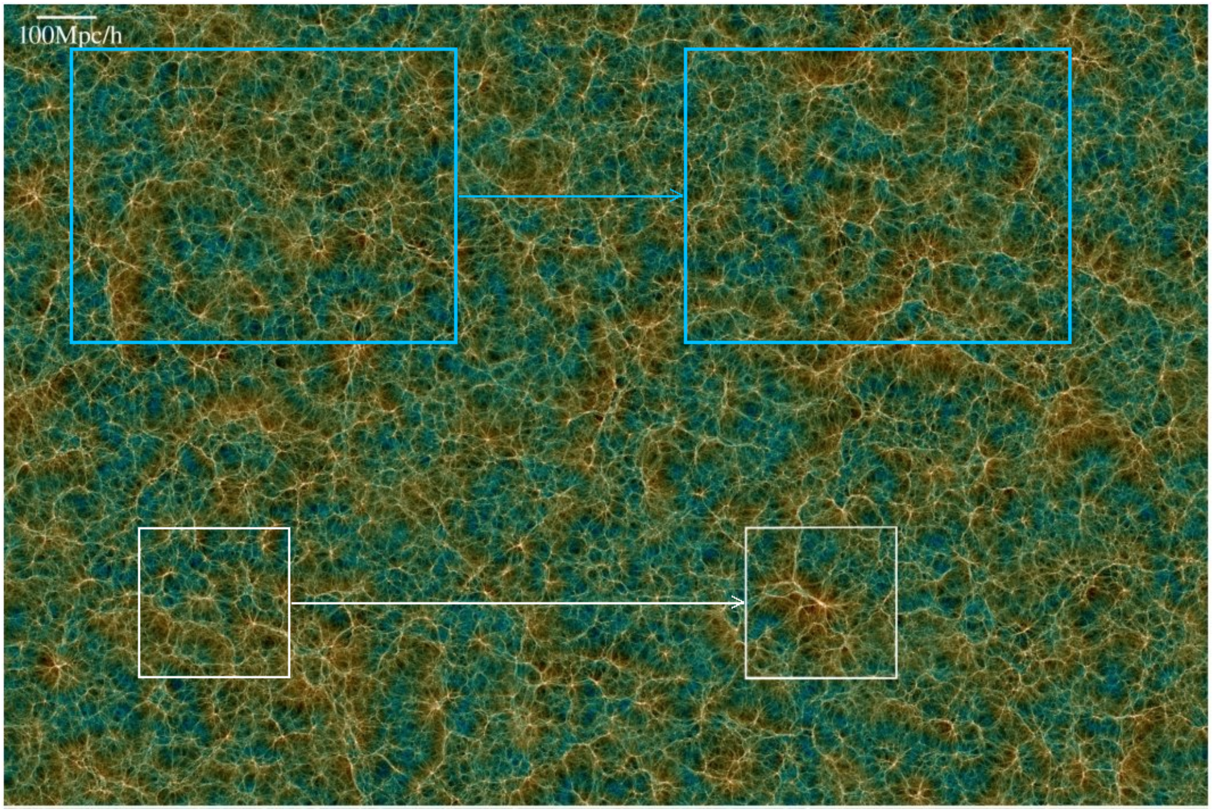


Fig. 1.1 Uchuu dark matter simulation of a box 2000Mpc.h^{-1} $\times$ 1333Mpc.h^{-1} with a thickness of 25Mpc.h^{-1}. The blue boxes highlight the homogeneity at large scales while the white boxes highlight the inhomogeneity at some scales due to structure formation. Adapted from [Ishiyama et al., 2021].

1.3 Friedmann-Lemaitre-Robertson-Walker metric

We present here a metric that satisfies the isotropic homogeneity of the Universe as well as an expanding one called Friedmann-Lemaitre-Robertson-Walker (FLRW). It is more convenient to express the metric in spherical coordinates (t, r, ϕ, θ). Working in natural units $c = 1$, we obtain:

$$ds^2 = dt^2 - a^2(t)\left[\frac{dr^2}{1+kr^2} + r^2(d\theta^2 + \sin^2(\theta)d\phi^2)\right] \tag{1.10}$$

where $a(t)$ is a dimensionless parameter called the "scale" factor, defined such that today at t_0 it is equal to 1, and used to describe the expansion which can be written as :

$$H(t) = \frac{\dot{a}(t)}{a(t)} \tag{1.11}$$

where the dot denotes the derivative with respect to time and $H(t)$ the Hubble parameter. k is a global curvature term such that it can take one of three values $0, -1, 1$ to describe a Universe that is flat, open/hyperbolic or closed/spherical respectively. It is important to note that we can obtain the redshift of an object that emitted a wave at a time t_e from the scale factor as the following relation holds true:

$$\frac{a(t_0)}{a(t_e)} = 1 + z \tag{1.12}$$

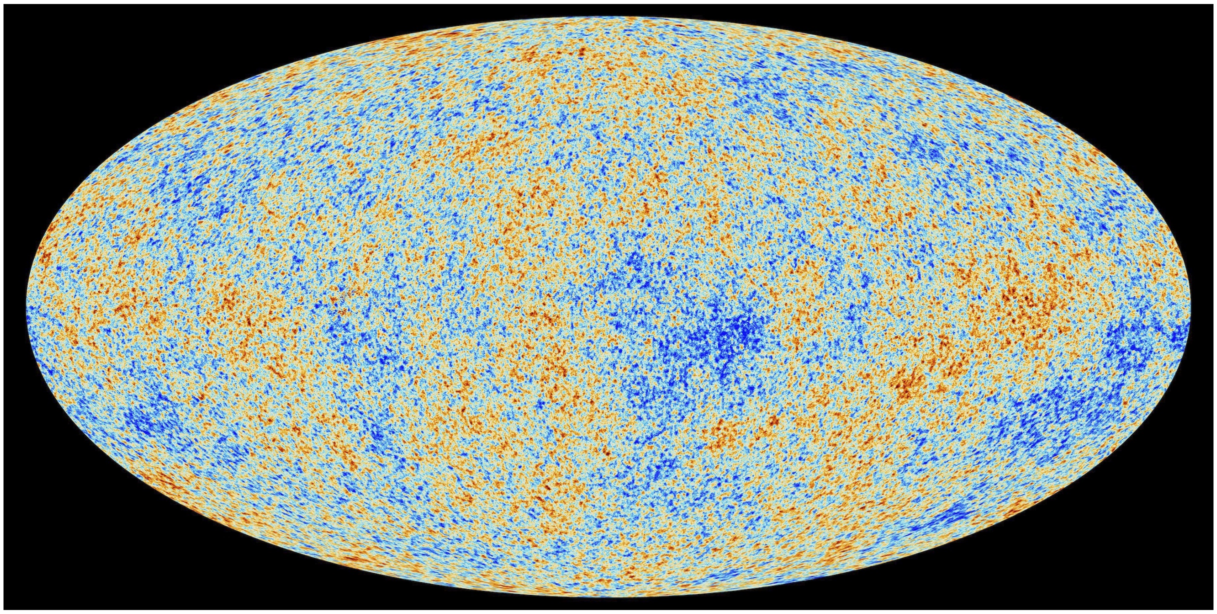


Fig. 1.2 CMB map provided by ESA and the Planck Collaboration.

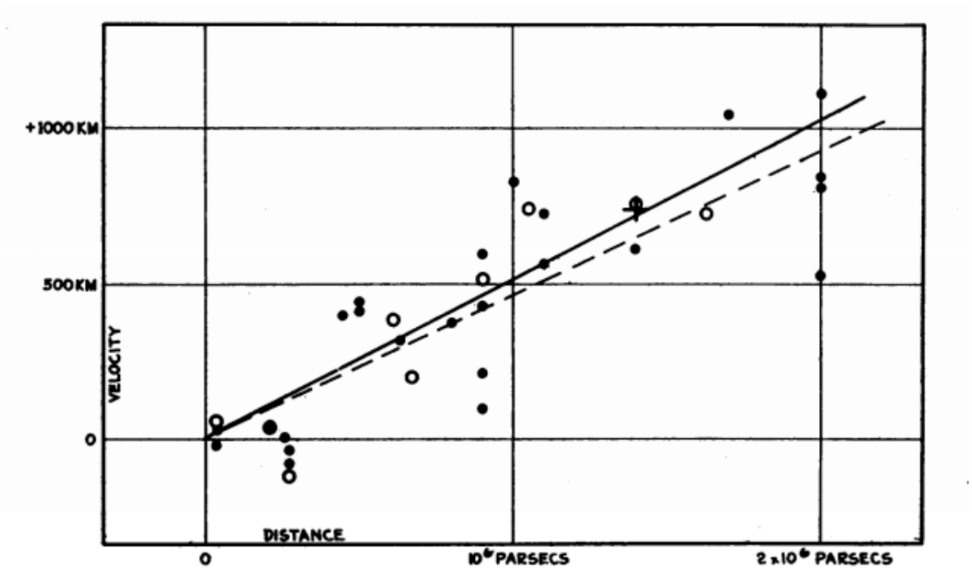


Fig. 1.3 Original Hubble diagram ([Hubble, 1929]) portraying the distance velocity diagram of nearby galaxies.

Oftenly we apply a change of coordinates such that any object would maintain its coordinates regardless of the expansion. These are called comoving coordinates. We define the comoving distance χ such that:

$$r = S_k(\chi) = \begin{cases} S_0(\chi) = \chi \\ S_{-1}(\chi) = \sin(\chi) \\ S_1(\chi) = \sinh(\chi) \end{cases} \tag{1.13}$$

$$d\chi = \frac{dr}{\sqrt{1-kr^2}} \tag{1.14}$$

The metric then becomes :

$$ds^2 = dt^2 - a^2(t)\left[d\chi^2 + S_k^2(\chi)(d\theta^2 + \sin^2(\theta)d\phi^2)\right] \tag{1.15}$$

1.4 Friedmann equations

Since the Universe is assumed to be homogenous and isotropic, we can express the energy-momentum tensor $T^{\mu\nu}$ only as a time-dependent object describing the density ρ and pressure P of every component present in the Universe. We assume that these can be described as a perfect fluid with no viscosity. For each fluid present indexed by i, we can then write down the pressure term as $P_i = w\rho_i c^2$ where w is the equation of state parameter. This gives us the following form for $T^{\mu\nu}$:

$$T_i^{\mu\nu} = \rho_i\left[(w_i + 1)u^\mu u^\nu - w_i g^{\mu\nu}\right] \tag{1.16}$$

with u^μ being the velocity vector of the fluid.

Using this, the Einstein equations and the FLRW metric, Friedmann derived the following equations known as the Friedmann equations in 1924 ([Friedmann, 1924]):

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_i - \frac{kc^2}{a^2} \tag{1.17}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho_i + 3\frac{p_i}{c^2}\right) \tag{1.18}$$

The first equation describes how fast the Universe is expanding knowing the Universe's composition ρ_i and curvature k, while the second describes its acceleration. Combining equations 1.17 and 1.18 we get:

$$\frac{\dot{\rho}_i}{\rho_i} = -3(w_i + 1)\frac{\dot{a}}{a} \tag{1.19}$$

And the solution to this differential equation is:

$$\boxed{\rho_i(t) = \rho_i(t_0)a(t)^{-3(w_i+1)}} \tag{1.20}$$

This entails that the evolution of the scale factor is entirely determined by the nature of the fluids present at a time t. Knowing w of each fluid informs us on the behaviour of $a(t)$. If we consider one fluid that dominates over, the scale factor can be written as:

$$a(t) \sim t^{\frac{2}{3(1+w)}} \tag{1.21}$$

Consequently, the value of w of the dominant fluid determines the behaviour of the expansion:

$$\begin{array}{lll} \ddot{a} > 0 & w < -\frac{1}{3} & \text{accelerated expansion} \\ \ddot{a} = 0 & w = -\frac{1}{3} & \text{constant expansion} \\ \ddot{a} < 0 & w > -\frac{1}{3} & \text{decelerated expansion} \end{array} \tag{1.22}$$

1.5 Energy contents of the Universe

Within the current cosmological framework, the total energy density of the Universe is commonly divided into the following main components:

- **Radiation** ($w_r = \frac{1}{3}$): a relativistic component composed primarily of photons and neutrinos. Its contribution is negligible in the present-day Universe, but it dominated the energy budget during the earliest cosmological epochs.

- **Matter** ($w_m = 0$): a non-relativistic component including both baryonic matter and dark matter. While baryonic matter constitutes the visible content of galaxies, the dominant fraction is dark matter, whose existence was first inferred from galaxy cluster dynamics by Zwicky ([Zwicky, 1933], [Andernach and Zwicky, 2017]). In the standard cosmological model, dark matter is assumed to be cold, meaning that its particles were non-relativistic during the epoch of structure formation.
- **Cosmological constant / Dark energy** ($w_\Lambda = -1$): in the standard ΛCDM model, cosmic acceleration is described by the cosmological constant Λ, interpreted as an intrinsic property of spacetime with equation of state $w = -1$. More generally, dark energy may be viewed as an additional fluid component driving accelerated expansion. Possible deviations from ΛCDM include models with a constant but free equation of state (wCDM), or a time-evolving equation of state such as the CPL parametrization ([Chevallier and Polarski, 2001]).

COMMENT : might be woth mentioning the quantum vaccuum fluctuations and the order of magnitude issue

The equations of state for matter and radiation can be determined using a statistical physics framework. For simplicity, we will not be going into the details of the derivations for each component but simply give the end results.
For non relativistic species, such as matter, we obtain :

$$w_m = 0 \tag{1.23}$$

For relativistic species, such as radiation, we obtain :

$$w_r = \frac{1}{3} \tag{1.24}$$

For the cosmological constant this can be determined without going through the statistical physics approach where by simply applying the Friedmann equations on a constant naturally leads to ([**Review this part**]):

$$w_\Lambda = -1 \tag{1.25}$$

We introduce the critical density ρ_c defined as the energy density of the Universe for $k = 0$ and at $t = 0$:

$$\rho_c = \frac{3H_0^2}{8\pi G} \tag{1.26}$$

This critical density is used to normalize the densities of the other fluids such that the normalized density is written as:

$$\Omega_i = \frac{\rho_i}{\rho_c} \tag{1.27}$$

At $t = 0$, the Friedmann equation yields $\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1$ with Ω_k being the contribution from the curvature by construction:

$$\Omega_k = \frac{-kc^2}{a(t_0)^2 H_0^2} \tag{1.28}$$

Equation 1.17 can be then rewritten as :

$$H(z) = H_0\sqrt{\Omega_{m,0}a(t)^{-3} + \Omega_{r,0}a(t)^{-4} + \Omega_{\Lambda,0} + \Omega_{k,0}a(t)^{-2}} \tag{1.29}$$

Current cosmological observations indicate that the present-day Universe is remarkably close to spatial flatness, corresponding to $\Omega_{k,0} \simeq 0$ [Aghanim et al., 2020]. The dominant contribution to the total energy budget is consistent with a cosmological constant (or dark energy), representing approximately 70% of the total density. Matter contributes roughly 30%, of which about 25% corresponds to dark matter and only $\sim 5\%$ to baryonic matter. The radiation component today is subdominant, contributing less than 1% of the total energy density.

The relative importance of these components evolves with cosmic expansion according to their different scalings with the scale factor. The evolution of each matter component is represented in Figure 1.4. Since radiation scales as a^{-4}, matter as a^{-3}, and the cosmological constant remains constant, the early Universe was initially radiation dominated. As the Universe expanded and cooled, a transition to matter domination occurred, enabling the growth of gravitational instabilities and the formation of large-scale structure. At late times, the cosmological constant became dominant, driving the current phase of accelerated expansion.

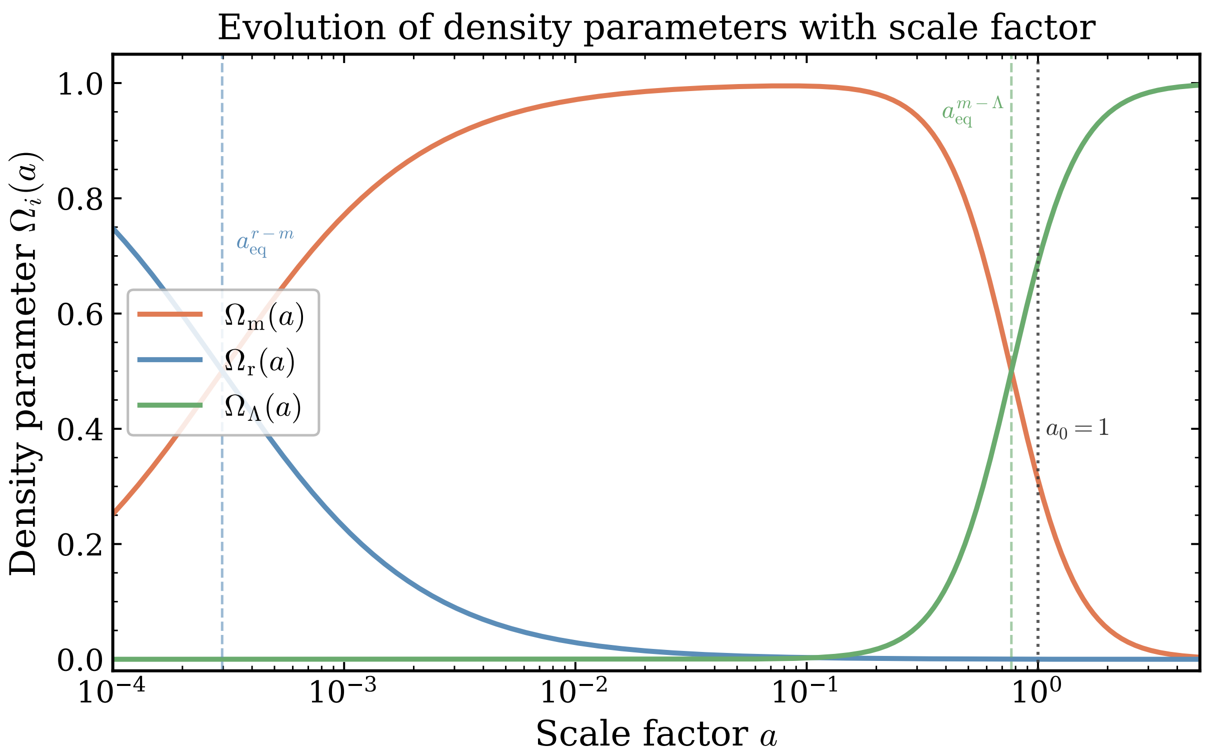


Fig. 1.4 Evolution of the main components of the ΛCDM model with respect to the scale factor. This highlights where each component dominates as well as when they become equal.

The acceleration of the Universe was confirmed by [Riess et al., 1998] and [Perlmutter et al., 1999] when they constructed a Hubble diagram with distant Type Ia Supernovae. Each paper has shown that the supernova Hubble diagram cannot be solely explained with a Universe containing only matter and that an additional energetic component must be accounted for. This is seen in Figure 1.5.

1.6 The ΛCDM Model

The ΛCDM model is the standard model of cosmology. It is built upon the framework of GR and assumes that the Universe began in an extremely hot and dense state, known as the Hot Big Bang, and has

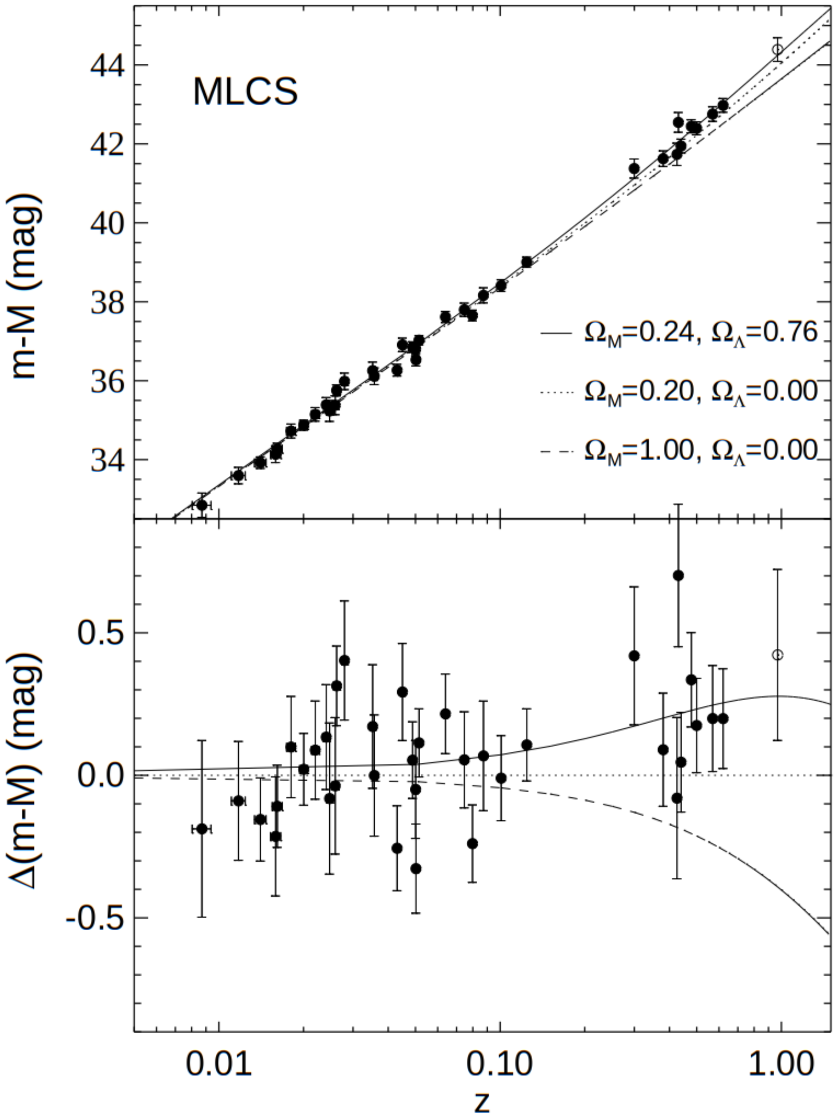


Fig. 1.5 Hubble diagram constructed using 50 type Ia supernovae from [Riess et al., 1998] which confirmed the accelerated expansion. The doted line represents the prediction for a Universe containing only matter that accounts for 20% of the energy content. The dashed line is for a Universe containing matter that accounts for 100% of the energy content of the Universe. The solid line is for a Universe containing matter energy content of 24% and Λ accounting for 76% of the energy content. The data fits the the solid line best.

been expanding ever since. As the Universe expanded and cooled, it underwent three successive epochs of domination by different energy components, each governing the dynamics of expansion through the Friedmann equations: a radiation dominated era at early times, followed by a matter dominated era, and finally the current dark energy dominated era driven by the cosmological constant Λ.

The energy content of the Universe in ΛCDM consists of the components introduced in the previous section: radiation, baryonic matter, cold dark matter and a cosmological constant Λ, with their present-day fractions measured to be approximately $\Omega_r \approx 9 \times 10^{-5}$, $\Omega_b \approx 0.05$, $\Omega_{\rm CDM} \approx 0.26$ and $\Omega_\Lambda \approx 0.69$ respectively ([Aghanim et al., 2020]), in a spatially flat Universe where $\sum_i \Omega_i = 1$. The model makes the additional assumptions that the Universe is homogeneous and isotropic on large scales (the cosmological principle), and that the primordial density perturbations seeding all structures are adiabatic and nearly Gaussian ([Collaboration et al., 2019]).

The ΛCDM model is fully characterised by six independent cosmological parameters:

- A_s — the amplitude of primordial scalar perturbations
- n_s — the spectral index of the primordial power spectrum
- $\Omega_b h^2$ — the physical baryon density
- $\Omega_{\rm CDM} h^2$ — the physical cold dark matter density
- H_0 — the Hubble constant, parametrising the present-day expansion rate
- τ — the optical depth to reionisation

All other cosmological quantities, such as Ω_Λ, σ_8 and S_8, are derived from these six parameters within the model. For example, σ_8 can be related to A_s such that:

$$\sigma_8 \propto \sqrt{A_s} \tag{1.30}$$

Despite its simplicity, ΛCDM has proven remarkably successful in describing a wide range of observations, from the CMB anisotropies ([Aghanim et al., 2020]) to the large-scale distribution of galaxies ([DESI Collaboration et al., 2025]), the BAO standard ruler ([Eisenstein et al., 2005]), and the accelerated expansion of the Universe ([Riess et al., 1998, Perlmutter et al., 1999]). Nevertheless, as discussed in Section 4, emerging tensions between different observational probes suggest that ΛCDM may require extensions or modifications.

The success of the ΛCDM model is rooted in the three observational pillars of the Hot Big Bang model (HBB), which postulates the Universe started 13.8 billion years ago while it was much hotter and denser than it is today. These pillars are:

- Expansion of the Universe as observed by Hubble in [Hubble, 1929].
- Big Bang Nucleosynthesis (BBN) which predicts the primordial abundances of light elements (hydrogen, helium, and lithium) formed at the very early stages of the Universe ([Cyburt et al., 2016], [Wagoner et al., 1967]).
- Cosmic Microwave Background (CMB), the trace of thermal radiation emitted when the Universe cooled sufficiently for protons and electrons to recombine into neutral hydrogen at $z \sim 1100$ ([Penzias and Wilson, 1965a], [Penzias and Wilson, 1965b]) (more details in Section 3.1).

Together, these three pillars provide striong evidence for a hot, dense early Universe that has been expanding and cooling ever since.

Although ΛCDM describes the Universe as homogeneous and isotropic at the largest scales, this description breaks down at smaller scales where structures such as galaxies and galaxy clusters emerge. Explaining the emergence of inhomogeneous structures from an initially homogeneous Universe, while preserving the assumptions of ΛCDM, is an important challenge. We describe this process in the following section.

2 Large Scale Structure Formation

Now that we have understood how modern cosmology came to be, we can move on to understanding the formation of structures. Observations from galaxy surveys such as DESI show that although on large scales the Universe looks homogenous, the smaller scales indicate the existence of an underlying non-homogenous structure. This can be seen through the formation of galaxy clusters and the cosmic web. Understanding how these structures are formed gives us a lot of information regarding the strength of gravity at large scales as well as the amount of matter in the Universe. Since the full derivation of each equation in structure formation is too long to write for this thesis, we recommend reading [Knobel, 2013], [Dodelson, 2003] and [Peebles, 1980] for a more detailed derivation

2.1 Newtonian approach to structure formation

In this introductory section, we will be starting off by describing structure formation by going only to linear scales. This will be the model on which we build up more complex structure formation model.

Theory of structure formation

We start by assuming a Universe is filled with an inhomogeneous, dissipationless, ideal fluid with matter density $\rho(\mathbf{r},t)$, velocity field $\vec{v}(\mathbf{r},t)$, pressure $P(\mathbf{r},t)$ and entropy per unit mass $S(\mathbf{r},t)$ with $\mathbf{r}$ being the fluid's coordinates. We obtain the following hydrodynamical equations:

$$\begin{cases} \frac{\partial\rho}{\partial t} + \vec{\nabla}.(\rho\vec{v}) = 0 & \text{Continuity equation} \\ \frac{\partial\vec{v}}{\partial t} + (\vec{v}.\vec{\nabla})\vec{v} + \frac{\vec{\nabla}P}{\rho} + \vec{\nabla}\Phi = 0 & \text{Euler equation} \\ \nabla^2\Phi = 4\pi G\rho & \text{Poisson equation} \\ \frac{\partial S}{\partial t} + (\vec{v}.\vec{\nabla})S = 0 & \text{Entropy conservation} \end{cases} \tag{1.31}$$

with Φ the gravitational potential. Since matter is approximated as a single-stream fluid, neighboring particles follow nearly parallel trajectories without "shell crossing" (collision of streams). This description breaks down once nonlinear collapse occurs, when multiple streams overlap and complex dynamics emerge. A fully accurate treatment of structure formation would instead require solving the general relativistic Boltzmann equation including all cosmic components and their interactions. This is in itself a difficult problem due to the nonlinear coupling between the fluid equations. To overcome that issue, we assume a FLRW Universe containing cold dark matter (treated as a fluid) and perturb the fluid around the Hubble flow, meaning we separate each variable as a homogenous part and an inhomogeneous perturbation. The cold dark matter assumption justifies the use of Newtonian mechanics since we are dealing with non-relativistic components. Moreover we assume that the perturbations that arise are very small. We can then write down the expansion of the quantities used in the hydrodynamical equation at first order as:

$$\begin{aligned} \rho(\mathbf{r},t) &= \bar{\rho}(t) + \delta\rho(\mathbf{r},t) \\ \mathbf{v}(\mathbf{r},t) &= \bar{\mathbf{v}}(\mathbf{r},t) + \delta\mathbf{v}(\mathbf{r},t) \\ P(\mathbf{r},t) &= \bar{P}(t) + \delta P(\mathbf{r},t) \\ \Phi(\mathbf{r},t) &= \bar{\Phi}(\mathbf{r},t) + \delta\Phi(\mathbf{r},t) \\ S(\mathbf{r},t) &= \bar{S}(t) + \delta S(\mathbf{r},t) \end{aligned}$$

We can then re-introduce our new variables in Equations 1.31 although to slightly simplify the equations we will be working in comoving coordinates $\mathbf{x}$ such that

$$\mathbf{r} = a\mathbf{x}$$
$$\nabla_{\mathbf{r}} = \frac{1}{a}\nabla_{\mathbf{x}}$$
$$\left.\frac{\partial}{\partial t}\right|_{\mathbf{r}} = \left.\frac{\partial}{\partial t}\right|_{\mathbf{x}} - \frac{1}{a}\mathbf{v}.\nabla_{\mathbf{x}}$$

We will also be working in Fourrier space where we get for any perturbation $\delta q(\mathbf{x}, t)$:

$$\delta q(t, \mathbf{k}) = \int \delta q(t, \mathbf{x}) e^{-i\mathbf{k}\mathbf{x}} \mathrm{d}^3 x \tag{1.32}$$

$$\delta q(t, \mathbf{x}) = \frac{1}{(2\pi)^3} \int \delta q(t, \mathbf{k}) e^{-i\mathbf{k}\mathbf{x}} \mathrm{d}^3 k \tag{1.33}$$

With $\mathbf{k}$ are the comoving Fourrier modes. We get the following hydrodynamical equations at first order:

$$\boxed{\begin{cases} \delta\dot{\rho} + 3H\delta\rho + \frac{i\bar{\rho}\mathbf{k}}{a}.\delta\mathbf{v} = 0 & \text{Continuity equation} \\ \delta\dot{\mathbf{v}} + H\delta\mathbf{v} + \frac{i\mathbf{k}}{a\bar{\rho}}(c_s^2\delta\rho + \sigma\delta S) + \frac{i\mathbf{k}}{a}\delta\Phi = 0 & \text{Euler equation} \\ k^2\delta\Phi + 4\pi G a^2 \delta\rho = 0 & \text{Poisson equation} \\ \delta\dot{S} = 0 & \text{Entropy conservation} \end{cases}} \tag{1.34}$$

with $c_s^2 = (\partial\bar{P}/\partial\bar{\rho})_{\bar{S}}$ the speed of sound in the fluid, $\sigma = (\partial\bar{P}/\partial\bar{S})_{\bar{\rho}}$ and $k = |\mathbf{k}|$.

What we are interested in is essentially how do matter perturbations grow as they will give birth to structure by gravitational attraction. The first thing to note is that the entropy fluctuations must exist at early times in order for them to keep existing otherwise they would simply dissipate. Moreover, results from the simplest models of inflation do not support the fluctuations of entropy. So from now on we well set $\delta S = 0$. We can also assume that the velocity perturbations would follow the overdense regions without experiencing any vorticities giving us modes that are parallel to the velocity $\mathbf{k}||\delta\mathbf{v}$. We call the modes that result from these conditions the adiabatic modes.

We will be denoting the overdensity $\delta(\mathbf{k}, t)$ as the matter density contrast which is written as:

$$\delta(\mathbf{k}, t) = \frac{\rho(\mathbf{k}, t)}{\bar{\rho}(t)} - 1 = \frac{\delta\rho(\mathbf{k}, t)}{\bar{\rho}(t)} \tag{1.35}$$

Therefore, we get the following expression for the time derivative of the perturbed density:

$$\dot{\delta\rho} = \frac{d}{dt}(\bar{\rho}\delta) = \dot{\bar{\rho}}\delta + \bar{\rho}\dot{\delta} \tag{1.36}$$

The continuity equation for an ideal fluid can be derived from the Einstein equations and reads for the background matter density:

$$\dot{\bar{\rho}} = -3H\bar{\rho} \tag{1.37}$$

Substituting into the perturbed continuity equation (Equation 1.34):

$$\dot{\delta\rho} + 3H\delta\rho + \frac{i\bar{\rho}\mathbf{k}}{a} \cdot \delta\mathbf{v} = 0 \implies -3H\bar{\rho}\delta + \bar{\rho}\dot{\delta} + 3H\bar{\rho}\delta + \frac{i\bar{\rho}\mathbf{k}}{a} \cdot \delta\mathbf{v} = 0 \tag{1.38}$$

where the first and third terms cancel exactly. Dividing by $\bar{\rho}$, the continuity equation becomes:

$$\dot{\delta} + \frac{ik}{a}\delta v = 0 \tag{1.39}$$

If we then differentiate this equation and combining its results with Euler and Poisson equations we obtain:

$$\ddot{\delta} + 2H\dot{\delta} + \left(\frac{c_s^2 k^2}{a^2} - 4\pi G\bar{\rho}\right)\delta = 0 \tag{1.40}$$

Here we obtain a linear second order differential equation with two solutions to it. This is only valid for collisionless dark matter and not baryonic matter. We denote $\lambda = \frac{2\pi a}{k}$ the physical wavelength of the mode, $\lambda_J = c_s\sqrt{\frac{\pi}{G\rho}}$ the Jeans length and $M_J = \frac{\pi}{6}\lambda_J^3\bar{\rho}$ the Jeans mass. Equation 1.40 tells us that if $\lambda < \lambda_J$, the solution to the equation is an oscillating wave leaving no room for structures to grow. So a condition we must have is $\lambda > \lambda_J$. We place ourselves in the condition where the wavelength is much larger than the Jeans length $\lambda \gg \lambda_J$, Equation 1.40 is now reduced to:

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\bar{\rho}\delta = 0 \tag{1.41}$$

which has the following general solution:

$$\delta(\mathbf{k}, t) = \delta_+(\mathbf{k})D_+(t) + \delta_-(\mathbf{k})D_-(t) \tag{1.42}$$

with the + and - denoting growing and decaying modes. These modes highlight the competing forces due to the internal pressure and the gravitational collapse. We call the growing term $D_+(t)$ the **linear growth function**. In the radiation dominated era, the Mészáros effect suppresses the growth of sub-Hubble matter perturbations, which grow only logarithmically rather than as a power law. In the matter dominated era the expansion rate grows as $H(t) = \frac{2}{3t}$ which gives us:

$$\boxed{D_+ \propto t^{2/3} \propto a(t)} \tag{1.43}$$

$$D_- \propto t^{-1} \tag{1.44}$$

This means that we can get grown modes for matter perturbations in the matter dominated era with D_+ while D_- goes to 0. More general formulations for D_+ can be obtained such as the one found in [Carroll et al., 1992].

The growth of perturbations described by D_+ can be characterised by the *growth rate* f, defined as the logarithmic derivative of the growth factor with respect to the scale factor:

$$f = \frac{d\ln D_+}{d\ln a} \tag{1.45}$$

In the matter dominated era, since $D_+ \propto a$, we get $f = 1$. A useful approximation valid across different epochs and cosmologies is given by [Peebles, 1980]:

$$\boxed{f \approx \Omega_m(z)^\gamma} \tag{1.46}$$

where γ, also called the growth factor, is obtained from the underlying gravitational theory. The growth factor can be related to the dark energy equation of state as seen in [Linder, 2005]:

$$\gamma \approx 0.55 + (1 + w(z)) \tag{1.47}$$

For a flat ΛCDM cosmology and GR it is $\gamma \simeq 0.55$. Deviations from this value would indicate modifications to GR on cosmological scales. In practice, f is not directly observable, and is instead measured in combination with σ_8, the root mean square of matter fluctuations on scales of $8\,h^{-1}$Mpc. The combination $f\sigma_8$ is therefore the key observable quantity that encodes the rate of growth of large-scale structure and provides a direct test of gravity on cosmological scales.

Statistical analysis of overdensities

Practically, we perform a statistical analysis when it comes to the study of the structures of the Universe. δ is looked at as a realization of a stochastic process and by analyzing that we can compare our observations to theories.

COMMENT : primordial fluctuations expected from inflation described in later section implies stochastic nature etc... can relate to CMB observations today (just to make the link)

We introduce $\langle ... \rangle$ as the ensemble average of the stochastic process underlying a random field. The large-scale structure we observe is only one realization of an underlying random cosmic process, so estimating its true statistical properties requires additional assumptions. A common assumption is ergodicity, meaning that averages measured over a sufficiently large volume of the Universe approximate the averages over many hypothetical realizations. In practice, this is related to the fair sample hypothesis, which assumes that widely separated regions of the Universe can be treated as nearly independent samples of the same process. If the survey volume is large enough and correlations become weak on large scales, these approximations are reasonable. However, real galaxy surveys cover finite volumes and therefore their measured averages are affected by statistical fluctuations known as sample variance, or **cosmic variance** when limited by the size of the observable Universe.

We denote the 2-point correlation function as:

$$\xi(\mathbf{x},\mathbf{x}') = \left\langle \delta(\mathbf{x})\delta(\mathbf{x}') \right\rangle \tag{1.48}$$

Our field is homogenous and isotropic therefore the only quantity that affects the correlation function is $r = |\mathbf{x}-\mathbf{x}'|$, $\xi(\mathbf{x},\mathbf{x}') = \xi(r)$. If at $r = +\inf$, the correlations go to 0, then we can decompose it in Fourrier modes: $\delta(\mathbf{k})$:

$$\left\langle \delta(\mathbf{k})\delta^*(\mathbf{k}') \right\rangle = (2\pi)^3 \delta_D(\mathbf{k}-\mathbf{k}')\mathcal{P}(\mathbf{k}) \tag{1.49}$$

$$\mathcal{P}(\mathbf{k}) = \int \xi(\mathbf{x}) e^{-i\mathbf{k}\mathbf{x}} d^3x \tag{1.50}$$

where the $\mathcal{P}(k)$ is the **power spectrum**, which is a real function, and δ_D is the Dirac delta function.
The works from [Harrison, 1970], [Zel'dovich, 1970b] and [Peebles and Yu, 1970] show that there is a preferrable form for the initial primordial power spectrum that takes the form:

$$\mathcal{P}(k) \propto k^{n_s} \tag{1.51}$$

with n_s the **spectral index**. If $n_s = 1$ the power spectrum becomes scale invariant. It is possible to predict the shape of the power spectrum of the initial conditions by studying an inflationary model. Most common models and results from [Collaboration et al., 2019] indicate that the initial perturbations are Gaussian with $n_s \approx 1$.
The power spectrum for the linear overdensity field is:

$$\mathcal{P}_{\text{lin}}(t,k) = A_s k^{n_s} T^2(k) D_+^2(t) \tag{1.52}$$

with A_s the normalization parameter and $T(k)$ is the transfer function which characterizes how each Fourrier mode of the density field evolved from the initial conditions to a later time.

2.2 Relativistic structure formation

Similarly to the previous section, we will be introducing structure formation from a general relativistic framework with a focus on linear perturbation. We will not be deriving every single quantity, however explaination of how we get there and how it relates to the Newtonian approximation will be present.

General description of perturbations

Firstly, we will be working in the FLRW metric for $k = 0$. We will also work in conformal time τ which simplifies this problem. It is defined as $a d\tau = \mathrm{d}t$. We will also work in natural units setting $c = \hbar = 1$. We will denote the FLRW metric as $\bar{g}_{\mu\nu}$ so that for a flat spacetime we have:

$$ds^2 = \bar{g}_{\mu\nu} dx^\mu dx^\nu = a^2(\tau) \left(-d\tau^2 + \bar{\gamma}_{ij} dx^i dx^j\right) \tag{1.53}$$

where $\bar{\gamma}_{ij}$ is the metric for the spatial components and (i, j) can take the values $1, 2, 3$ (in general we will put a bar on top of the parameter related to a flat FLRW metric).

In a relativistic framework, the perturbations are done on the metric in itself such that when we choose a metric $g_{\mu\nu}$ the deviations from the homogenous isotropic background are close enough to $\bar{g}_{\mu\nu}$[2]. We then obtain:

$$\delta g_{\mu\nu} = g_{\mu\nu}(x) - \bar{g}_{\mu\nu}(x) \tag{1.54}$$

Physically, we are splitting $g_{\mu\nu}$ into a background $\bar{g}_{\mu\nu}$ and perturbation $\delta g_{\mu\nu}$. However this perturbation is not a 4-tensor which will require us to set a gauge transformation. Similarly, the perturbation for a general scalar-like object and a vector-like object are defined as:

$$\delta q(x) = q(x) - \bar{q}(x) \tag{1.55}$$

$$\delta q^\mu = q^\mu - \bar{q}^\mu = \delta q_\nu \bar{g}^{\mu\nu} + \bar{q}_\nu \delta g^{\mu\nu} \tag{1.56}$$

We can also perform the same for the energy-momentum tensor.

$$\delta T_{\mu\nu} = T_{\mu\nu} - \bar{T}_{\mu\nu} \tag{1.57}$$

with

$$T_{\mu\nu} = (\rho + P) u_\mu u_\nu + P g_{\mu\nu} + \Pi_{\mu\nu} \tag{1.58}$$

and $\Pi_{\mu\nu}$ is the anisotropic stress which accounts for deviations from an ideal fluid.

The general form for the metric perturbation is written as:

$$\delta g_{\mu\nu} dx^\mu dx^\nu = a^2(\tau) \left[-2A d\tau^2 + B_i d\tau dx^i + 2(H_L \delta_{ij} + H_{ij}) dx^i dx^j\right] \tag{1.59}$$

where $A(\tau, \mathbf{x})$, $B_i(\tau, \mathbf{x})$, $H_L(\tau, \mathbf{x})$ and $H_{ij}(\tau, \mathbf{x})$ are perturbation quantities and δ_{ij} a delta function.

Newtonian gauge

These perturbation quantities will then intervene in the field equations. Determining them necessitates a gauge transformation. We will be showing the results of the Newtonian gauge as it is useful for

[2] Mathematically, this change in metric means we alsothe mainifolds being worked on so simple mathematical operations no longer apply. One way to overcome that issue is to define $\bar{g}_{\mu\nu}$'s functional form as invariant under coordinate transformations.

analyzing the density fluctuations inside of the Hubble horizon ($D_H = 1/H$) as well as it is simple to interpret its results in terms of Newtonian physics. The gauge is defined by:

$$B = H = 0 \tag{1.60}$$

and we will be renaiming the other perturbations as:

$$\Psi = A \tag{1.61}$$

$$\Phi = -H_L \tag{1.62}$$

The field equations for the Newtonian gauge are:

$$\boxed{\begin{cases} -k^2\Phi - 3H(\dot{\Phi} + H\Psi) = 4\pi G a^2 \delta\rho \\ \dot{\Phi} + H\Psi = 4\pi G a^2 \frac{v}{k}(\bar{\rho} + \bar{P}) \\ \ddot{\Phi} + H(\dot{\Psi} + 2\dot{\Phi}) + (2\dot{H} + H^2)\Psi + \frac{k^2}{3}(\Phi - \Psi) = 4\pi G a^2 \delta P \\ k^2(\Phi - \Psi) = 8\pi G a^2 \Pi \end{cases}} \tag{1.63}$$

where the dots are derivatives with respect to the conformal time. If we consider scale much smaller than the Hubble horizon $H/k \ll 1$, the first field equation then becomes:

$$-k^2\Phi = 4\pi G a^2 \delta\rho \tag{1.64}$$

In this limit we recover the Poisson equation in the Newtonian formalism. Therefore, we can say that well within the horizon the Newtonian gauge reduces to Newtonian mechanics. This also means we can have a physical interpretation for Φ acting as the Newtonian gravitational potential (called generalized Newtonian potential). The last equation also informs us that in the matter dominated epoch we have $\Phi \simeq \Psi$ reducing the metric perturbations to only one parameter.

One can only also derive the equations of motion in the Newtonian gauge, however for this thesis we will only be showing the final equations of motions obtained for matter fluctuations in the matter dominated era. We note the matter overdensity $\delta_m = \frac{\delta\rho_m}{\bar{\rho}}$, we obtain:

$$\ddot{\delta}_m + H\dot{\delta}_m - 4\pi G a^2 \bar{\rho}_m \delta_m = 0 \tag{1.65}$$

which is a reduced version of Equation 1.40 where $c_s = 0$.

2.3 Inflation

Inflation is a period of time at the very early stages of the Universe where an accelerated expansion of the Universe occurred ([Guth, 1981]). It was theorized to tackle known issues regarding the fine tuning of the flatness of the Universe and the horizon problem. What we see as well is that through inflation we can have a prediction for the initial density fluctuations. This following condition is imposed on inflation to obtain an accelerated early Universe:

$$\ddot{a} > 0 \tag{1.66}$$

During inflation, we assume that the Universe is dominated by a quantum scalar field ϕ called the inflaton, experiencing a "slow roll" down its potential $V(\phi)$ ([Mukhanov et al., 1992], [Albrecht and Steinhardt, 1982]). The action for a scalar field minimally coupled to gravity is given by:

$$S = \int d^4x\sqrt{-g}\left[\frac{M_{\rm pl}^2}{2}R - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right] \tag{1.67}$$

where $M_{\rm pl}$ is the Planck mass defined as $M_{\rm pl}^2 = \left(\frac{1}{8\pi G}\right)^2$. Varying the action with respect to the metric allows us to obtain the energy momentum tensor $T_{\mu\nu}$ and its components P_ϕ and ρ_ϕ:

$$P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi) \tag{1.68}$$

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi) \tag{1.69}$$

where the kinetic energy of the field is given by $\frac{1}{2}\dot{\phi}^2$.
The dynamics can be recovered with the Klein Gordon equation giving us:

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV(\phi)}{d\phi} = 0 \tag{1.70}$$

We can also apply the Friedmann equations to the inflaton field where we get:

$$H^2 = \frac{1}{M_{\rm pl}^2}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right) \tag{1.71}$$

$$\dot{H} = -\frac{1}{2}\frac{\dot{\phi}^2}{3M_{\rm pl}^2} \tag{1.72}$$

Assuming a slow rolling field entails that the kinetic term is small compared to the potential term V. This assumption yields $a \sim e^{Ht}$, which is a necessary condition to inflation. Additionally we assume coherent oscillations around a minimum that are damped by $H\dot{\phi}$ as seen in the Klein-Gordon equation. The field is then expected to decay into matter and radiation, although the exact physical process behind that is unknown. The inflaton field is then treated as a perturbation around a mean field, that these perturbations give rise to perturbation amplitudes that we can identify in the observations we make today. Some useful parameters to define for a slow-roll inlfation are :

$$\epsilon = -\frac{\dot{H}}{H^2} = \frac{3}{2}\frac{\dot{\phi}^2}{V} \tag{1.73}$$

$$\eta = \epsilon - \frac{\ddot{\phi}}{H\dot{\phi}} \tag{1.74}$$

where the slow rolling conditions are $\epsilon \ll 1$ and $\eta \ll 1$.
We define q as:

$$q(x) = \delta\phi + \frac{\dot{\bar{\phi}}}{H}\Phi \tag{1.75}$$

This quantity q is known as the Mukhanov-Sasaki variable ([Mukhanov, 1988], [Sasaki, 1986]) and is the canonical variable for quantizing inflaton perturbations.

In order to fully treat the inflaton field we have to apply the canonical quantization of the field (quantum field theory) by elevating q to an operator $\hat{q}$ as well as introducing its canonical conjugate momentum $\hat{\pi}$. In Fourrier space, they are expressed as function of the quantum harmonic operator $a_{\mathbf{k}}$ and $a_{\mathbf{k}}^\dagger$. With these operators we can calculate the expected value of the field as well as the power spectrum as a function of k. Once the power spectrum during inflation is determined it is possible to extend this to the primordial power spectrum of matter during the matter dominated era. Outside the Hubble horizon, each mode is frozen with a nearly scale-invariant amplitude. Before writing the primordial power spectrum, we introduce the comoving curvature perturbation $\mathcal{R}$, a gauge-invariant quantity that measures the spatial curvature of constant-density hypersurfaces. In the context of inflation it is related to the inflaton perturbation $\delta\phi$ by:

$$\mathcal{R} = -\frac{H}{\dot{\phi}}\delta\phi \tag{1.76}$$

The key property of $\mathcal{R}$ is that it is conserved on super-Hubble scales for adiabatic perturbations ([Wands et al., 2000]). This means its amplitude is frozen once a mode exits the Hubble radius during inflation and remains unchanged until it re-enters the horizon during the hot Big Bang, directly connecting inflationary predictions to late-time observables. The dimensionless power spectrum of the comoving curvature perturbation $\mathcal{R}$ evaluated at Hubble crossing $k = aH$ is:

$$\mathcal{P}_{\mathcal{R}}(k) = \left.\frac{H^4}{4\pi^2\dot{\phi}^2}\right|_{k=aH} \approx \left.\frac{H^2}{8\pi^2 M_{\mathrm{pl}}^2\epsilon}\right|_{k=aH} \tag{1.77}$$

which is approximately scale-invariant for slow-roll inflation where H and ϵ vary slowly. The spectral index for a slow-roll inflation is given by:

$$n_s = 1 - 6\epsilon + 2\eta \tag{1.78}$$

Once modes re-enter the Hubble horizon during the matter and radiation dominated eras, they begin to evolve dynamically. Their evolution is captured by the transfer function $T(k)$ and the linear growth factor D_+. The linear matter power spectrum at a time τ is therefore:

$$\mathcal{P}_{\mathrm{lin}}(\tau, k) = A_s \left(\frac{k}{k_*}\right)^{n_s} T^2(k)\, D_+^2(\tau) \tag{1.79}$$

where A_s is the primordial amplitude defined at the pivot scale $k_* = 0.05\,\mathrm{Mpc}^{-1}$ ([Planck Collaboration et al., 2020]), $T(k)$ encodes the suppression of small-scale modes that entered the horizon during radiation domination ([Eisenstein and Hu, 1998]), and $D_+(\tau)$ describes the growth of perturbations during matter domination ([Carroll et al., 1992]). The k^{n_s} factor reflects the nearly scale-invariant primordial spectrum, with $n_s \approx 0.965$ ([Planck Collaboration et al., 2020]) consistent with slow-roll inflation.

3 Cosmological Observables

In this section we will briefly present some of the main observables that cosmologists are concerned with.

3.1 The Cosmic Microwave Background

During the radiation dominated era, the Universe was hot and dense, filled with a plasma of baryons and free electrons. Atoms could not form at that time because photons were energetic enough to ionise any neutral hydrogen immediately upon formation. As a result, baryonic matter remained tightly coupled to photons through Thomson scattering off free electrons. As the Universe expanded and cooled, photons eventually became insufficiently energetic to ionise hydrogen, allowing neutral atoms to form for the first time. This epoch is referred to as **recombination** and is associated with the decoupling of baryons and photons. The mean free path of photons then exceeded the Hubble radius, allowing them to travel freely across space without scattering. Today, these photons have been redshifted drastically due to the expansion of the Universe and are observed as a background of microwave radiation. We refer to this relic radiation as the Cosmic Microwave Background (CMB), and it constitutes our primary observational window onto the primordial Universe. The CMB was first discovered by Arno Penzias and Robert Wilson accidentally while calibrating an antenna for telecommunications and astronomy ([Penzias and Wilson, 1965a, Penzias and Wilson, 1965b]).

The CMB has a nearly perfect blackbody spectrum with a mean temperature of $T_{CMB} = 2.7255 \pm 0.0006$ K ([Fixsen, 2009]) and temperature anisotropies at the level of $\Delta T/T \sim 10^{-5}$. The two-point correlation function of these anisotropies, equivalently described by the angular power spectrum C_ℓ (where ℓ denotes the degree if the spherical harmonic function, also known as the multipole number), encodes the imprint of primordial matter density perturbations consistent with a Gaussian random field ([Collaboration et al., 2019]), which later evolved into the large-scale structures we observe today. Secondary anisotropies, arising from effects such as gravitational lensing by galaxy clusters ([Hu, 2000]) and inverse Compton scattering of CMB photons off hot intracluster gas (the Sunyaev-Zel'dovich effect [Sunyaev and Zeldovich, 1972]), provide additional information on the late-time Universe. The polarisation of the CMB, decomposed into curl-free E-modes and divergence-free B-modes, further constrains inflationary models and provides a potential window onto primordial gravitational waves ([Kamionkowski et al., 1997]).

Through the detailed structure of the acoustic peaks in C_ℓ, CMB observations allow precise constraints on multiple cosmological parameters, including Ω_m, Ω_b, H_0, n_s, and A_s ([Aghanim et al., 2020]). Figure 1.6 shows angular power spectrum comparisons between different CMB experiments (such as Planck or WMAP) and the ΛCDM prediction in a dashed grey line.

3.2 Galaxy Surveys

Galaxy surveys map the three-dimensional distribution of galaxies across large volumes of the Universe. Galaxies are biased tracers of the underlying matter distribution, and their spatial statistics allow us to infer a vast amount of information about the structure, dynamics, and history of the Universe. Surveys such as SDSS ([York et al., 2000]) and DESI ([DESI Collaboration et al., 2016]) measure millions of galaxies across a wide redshift range, from which significant cosmological constraints have been extracted.

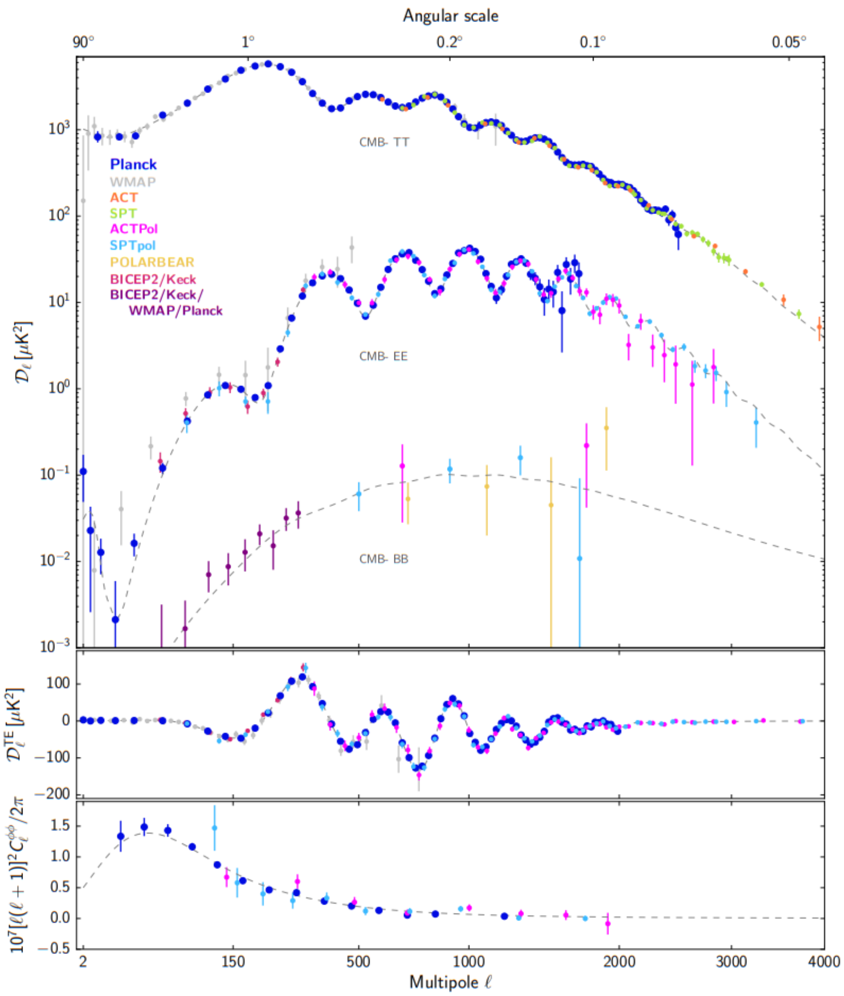


Fig. 1.6 Compilation of CMB angular power spectrum measurements from [Collaboration et al., 2020]. The upper panel shows the power spectra of the temperature and E-mode and B-mode polarization signals, the next panel the cross-correlation spectrum between T and E, while the lower panel shows the lensing deflection power spectrum

3.2.1 The Matter Power Spectrum

As introduced in section 2.1, a key statistical tool for analysing galaxy surveys is the matter power spectrum $P(k)$, defined as the variance of density fluctuations as a function of scale k:

$$\langle \delta(\mathbf{k}) \delta^*(\mathbf{k}') \rangle = (2\pi)^3 \mathcal{P}(k)\, \delta_D^{(3)}(\mathbf{k} - \mathbf{k}') \tag{1.80}$$

The shape and amplitude of $\mathcal{P}(k)$ depend directly on cosmological parameters such as primordial scalar perturbations A_s. The primordial power spectrum can be related to late-time observations through the transfer function ([Eisenstein and Hu, 1998]):

$$\mathcal{P}(k) = A_s \left(\frac{k}{k_*}\right)^{n_s - 1} T^2(k)\, D_1^2(z) \tag{1.81}$$

where $k_* = 0.05\,\mathrm{Mpc}^{-1}$ is the pivot scale at which the primordial amplitude A_s is defined ([Planck Collaboration et al., 2020 $T(k)$ is the transfer function encoding the scale-dependent evolution of perturbations through radiation domination, and $D_1(z)$ is the linear growth factor describing the growth of structure during matter domination.

3.2.2 Baryon Acoustic Oscillations

Baryon Acoustic Oscillations (BAO) are a feature in the matter clustering that corresponds to the imprint of sound waves propagating through the baryon-photon plasma in the early Universe. The radiation pressure of the hot plasma drove acoustic oscillations at a sound speed of $c_s \simeq c/\sqrt{3}$. These oscillations were frozen in at the **drag epoch** $z_d \approx 1060$, shortly after photon decoupling at $z_* \approx 1100$, when baryons decoupled from the photon fluid and the sound waves could no longer propagate. The comoving distance these sound waves had travelled by that time defines the **sound horizon**:

$$r_s(z_d) = \int_{z_d}^{\infty} \frac{c_s(z)}{H(z)}\,dz \approx 147\,\mathrm{Mpc} \tag{1.82}$$

This characteristic scale is imprinted in the distribution of matter as a preferred separation between galaxies, appearing as a distinct peak in the two-point correlation function $\xi(r)$ at $r \approx 147$ Mpc, or equivalently as oscillations in $P(k)$. The BAO scale thus defines a **standard ruler** that can be measured at different redshifts to constrain the expansion history of the Universe through $H(z)$ and the angular diameter distance $D_A(z)$ ([Eisenstein et al., 2005], [Blake and Glazebrook, 2003]). Figure 1.7 shows the BAO signal in the correlation functions for different catalogues.

3.2.3 Redshift Space Distortions

Galaxies are observed in redshift space rather than real space. Their peculiar velocities along the line of sight distort their apparent positions, producing anisotropies in the observed galaxy clustering. On large scales, coherent infall of galaxies into overdensities causes an apparent compression of structures along the line of sight, known as the **Kaiser effect** ([Kaiser, 1987]). On small scales, the virial motions of galaxies within collapsed structures produce an elongation of structures along the line of sight, known as the **Fingers of God** effect ([Jackson, 1972]).

The amplitude of these distortions depends on the rate of structure growth, encapsulated in the parameter $f\sigma_8$, where $f = d\ln D_1/d\ln a \approx \Omega_m(z)^{0.55}$ is the growth rate predicted by General Relativity. Modified gravity theories predict different values of f, making measurements of $f\sigma_8$ a direct test of GR on cosmological scales ([Guzzo et al., 2008], [Aghanim et al., 2020]). Recent use of the RSD allowed to constrain the modified gravity using the DESI data ([Adame et al., 2025a]). This is represented in Figure 1.8 where the parameters constrained are generalized parameters for a modified theory of Gravity. The results show that GR is consistent with our observations.

3.3 Type Ia Supernovae

As discussed in Section 2, it is possible to obtain the distance of an object through its luminosity. However, measuring the intrinsic luminosity of celestial objects is a difficult task in itself. Type Ia Supernovae are thermonuclear explosions involving white dwarf stars that are **standardizable candles**, meaning they're characterized by a reproducible intrinsic luminosity. By measuring their apparent brightness as a function of redshift, one can reconstruct the luminosity distance-redshift relation $d_L(z)$ and constrain the expansion history of the Universe. This led to the discovery of the accelerated expansion of the Universe ([Riess et al., 1998], [Perlmutter et al., 1999]). A detailed discussion of Type Ia Supernovae and their use as cosmological probes is presented in Chapter 2.

3.4 Gravitational Lensing

According to general relativity, mass curves spacetime and light follows the geodesics of that curved geometry. As a result, the path of photons is deflected when passing near massive objects, a phenomenon known as gravitational lensing. In a cosmological context, this deflection can be derived from the geodesic equation applied to the perturbed FLRW metric, and it provides a direct probe of the total matter distribution,both baryonic and dark, along the line of sight.

Two main regimes of gravitational lensing are relevant for cosmology. **Strong lensing** occurs in the vicinity of objects with deep gravitational potentials, such as massive galaxy clusters, producing dramatic and easily identifiable distortions such as multiple images, giant arcs, and Einstein rings. It provides precise measurements of the mass distribution of individual clusters. **Weak lensing** operates in the regime where the distortion of any individual background source is too small to be detected — the induced ellipticity is much smaller than the intrinsic ellipticity of a galaxy [Bartelmann and Schneider, 2001]. However, the lensing signal can be recovered statistically by measuring the coherent distortion of the shapes of large samples of background galaxies, an effect known as **cosmic shear** γ.

Since lensing responds to all matter along the line of sight, cosmic shear surveys provide a direct measurement of the projected matter distribution. The amplitude of the cosmic shear signal is sensitive to both the amplitude of matter fluctuations σ_8 and the matter density Ω_m, but these two parameters are largely degenerate in lensing observations. This degeneracy is characterised by the parameter $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$, which current surveys constrain to percent-level precision ([Heymans et al., 2021] [DES Collaboration et al., 2022] [Collaboration et al., 2026], [Dalal et al., 2023]).

3.5 The Lyman-α Forest

When observing the spectra of distant quasars, a dense series of absorption lines is observed, a feature known as the **Lyman-α forest**. These absorption lines are produced by neutral hydrogen clouds in the intergalactic medium (IGM) through which the quasar photons pass on their way to the observer, typically at redshifts $2 \lesssim z \lesssim 5$. Each absorbing cloud at redshift z produces an absorption line at an observed wavelength:

$$\lambda_{obs} = 1216\,(1+z)\,\text{Å} \tag{1.83}$$

where 1216 Å is the restframe Lyman-α wavelength. It's possible to show that the neutral hydrogen fraction in the IGM traces the underlying total matter density field, making the Lyman-α forest an effective map of the matter distribution along the line of sight ([Gunn and Peterson, 1965]).

The cosmological information in the Lyman-α forest is extracted through the power spectrum of the transmitted flux fluctuations, which can be directly related to the three-dimensional matter power spectrum ([Croft et al., 1998]). A particularly powerful feature of this probe is that it is sensitive to scales $k \sim 0.1$–$10\,h\,\mathrm{Mpc}^{-1}$ at high redshift, well beyond the reach of galaxy surveys, making it highly complementary to both CMB and galaxy clustering measurements. This sensitivity to small scales and high redshifts allows the Lyman-α forest to constrain the sum of neutrino masses $\sum m_\nu$ through the small-scale suppression of the power spectrum ([Palanque-Delabrouille et al., 2015]), the spectral index n_s and the BAO standard ruler at $z \sim 2.3$ ([Adame et al., 2025b]).

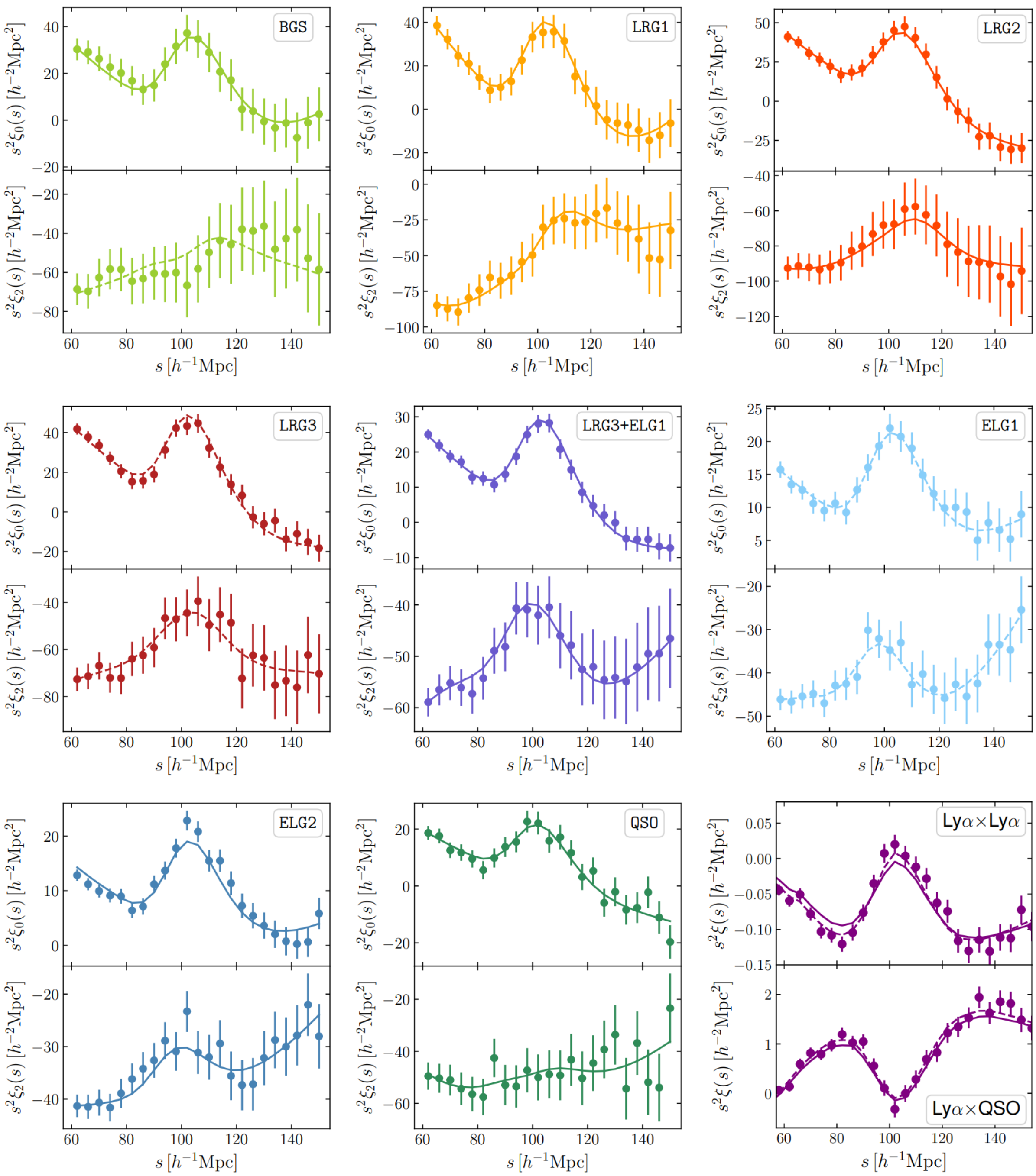


Fig. 1.7 Correlation functions from the DESI DR2 results [DESI Collaboration et al., 2025] for different galaxy catalogues (BGS, LRG, etc...). The BAO signal can be clearly seen as a bump around 100 Mpc.h^{-1} accross all catalogues

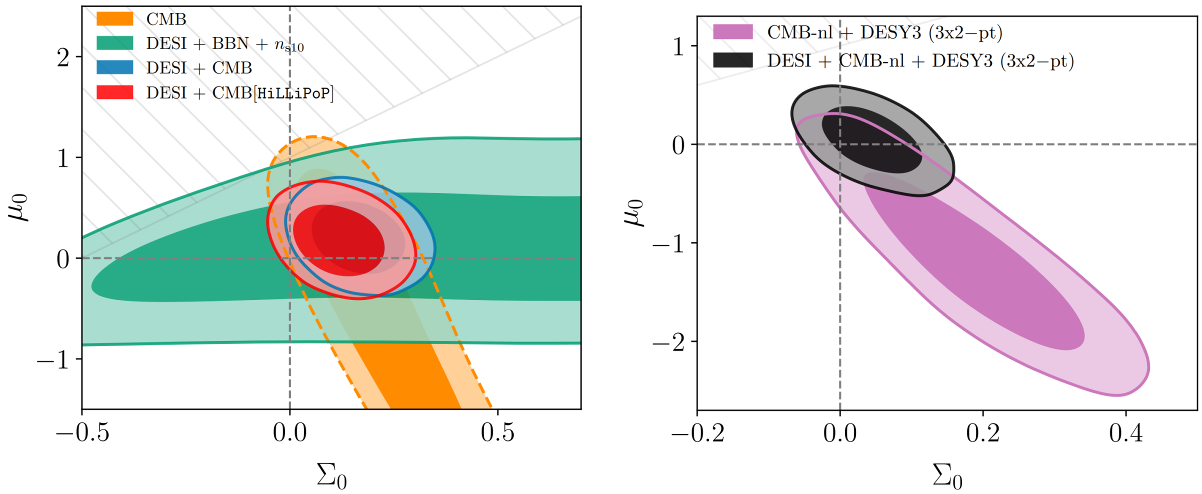


Fig. 1.8 Constraints on generalized parameters of modified gravity. The left figure shows CMB, DESI and DESI + CMB results. The right figure shows the addition of weak lensing results. Taken from [Adame et al., 2025a]

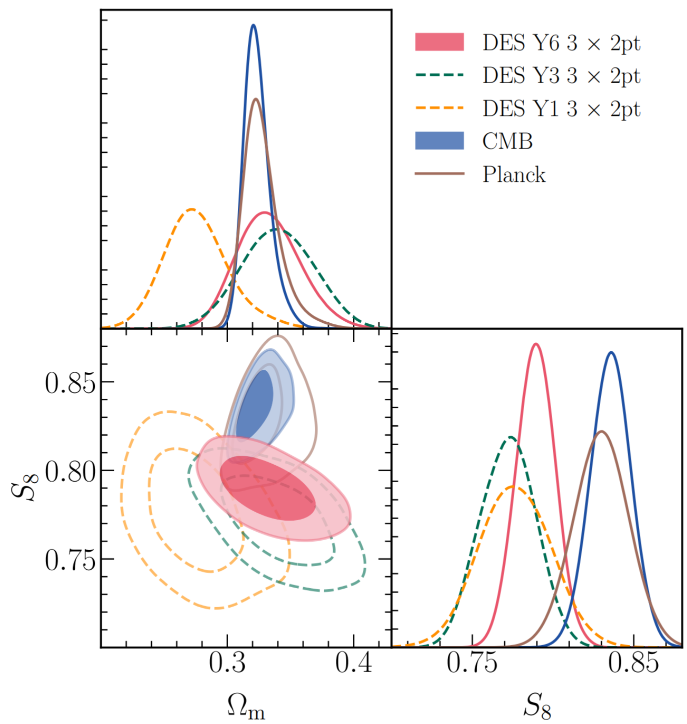


Fig. 1.9 A comparison of the marginalized parameter constraints in the ΛCDM model from the Dark Energy Survey (DES) Year 6([Collaboration et al., 2026]) in red in comparison with previous data releases from DES (green year 3 and orange year 1) and Planck (brown) as well as a combination of CMB measurements (blue).

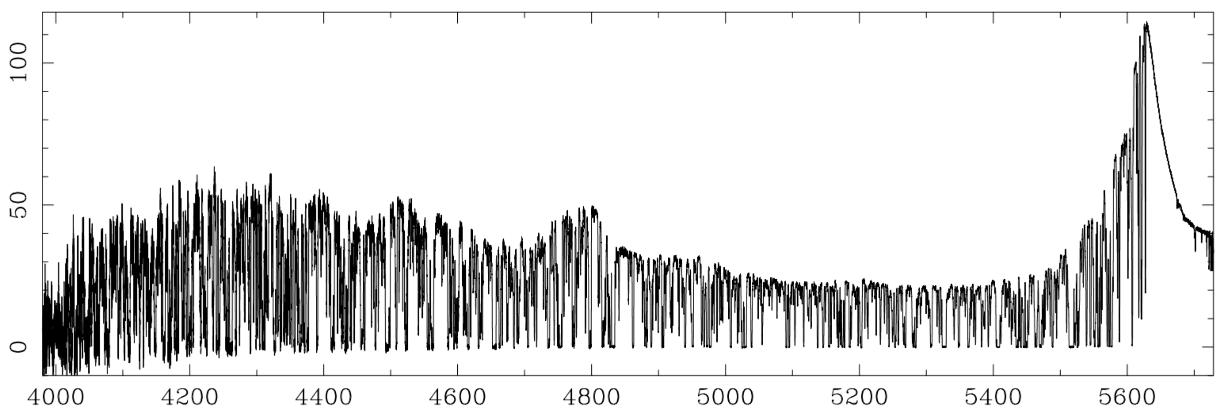


Fig. 1.10 High resolution spectrum of the Lyman-α forest part of a quasar measured at $z = 3.63$ taken with the HIRES spectrograph on the 10 m Keck telescope in Hawaï.

4 Tensions in The Cosmological Model

The previous sections served to introduce and demonstrate the power of the ΛCDM model in describing the Universe. Nonetheless, tensions have emerged between cosmological parameter measurements obtained from different probes, suggesting that the discrepancies may be of physical or systematic origin. Here we present the three main tensions.

4.1 H_0 Tension

The Hubble constant H_0 can be inferred in two ways: through direct measurements via the cosmic distance ladder, or indirectly through model-dependent fitting of early-Universe observables. The direct measurement is obtained by calibrating standard candles such as Cepheid variables and Type Ia Supernovae. The latest results from the SH0ES collaboration give $H_0 = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ ([Riess et al., 2022]). The indirect method relies on fitting the acoustic peak structure of the CMB angular power spectrum within the ΛCDM framework. The Planck 2018 collaboration reports $H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ ([Aghanim et al., 2020]), yielding a $\sim 5\sigma$ discrepancy with the SH0ES result. We see this in Figure 1.11. This tension persists across multiple independent measurements of H_0 ([Di Valentino et al., 2021]), raising the question of whether it reflects unknown systematic errors, or new physics beyond ΛCDM present at different epochs of the Universe. Recent results from the Chicago-Carnegie Hubble Program (CCHP) ([Freedman et al., 2025]) have measured using independent data from the James Webb Space Telescope with three independent methods (Tip of the Red Giant Branch(TRGB), J-Region Asymptotic Giant Branch (JAGB) and Cepheids) the H_0 value. Their results show that with TRGB and JAGB the H_0 values are compatible with Planck measurements indicating that the tension arises from systematic effects. Moreover, a recent study by [Stiskalek et al., 2025] explicitly accounted for peculiar velocities when measuring H_0 using the same data as the one used in SH0ES and obtained a smaller value than the one reported by SH0ES of $H_0 = 71.1 \pm 1.4\mathrm{km\,s^{-1}\,Mpc^{-1}}$ inciting the importance of modeling peculiar velocities. The paper also highlights that whether the sample selection is done on supernova magnitude or host galaxy redshift does not significantly impact the value of H_0.

4.2 S_8 Tension

The parameter $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ combines the amplitude of matter fluctuations σ_8 with the matter density Ω_m, and characterises the growth of large-scale structure. It can be measured both through the spatial correlations of galaxies and weak gravitational lensing, and through the CMB anisotropy power spectrum, where it is related to the primordial amplitude A_s through the linear growth of structure. The Planck 2018 collaboration measures $S_8 = 0.832 \pm 0.013$ ([Aghanim et al., 2020]), while the Dark Energy Survey Year 6 measures $S_8 = 0.789^{+0.012}_{-0.012}$ from a combination of cosmic shear, galaxy clustering, and galaxy-galaxy lensing ([Collaboration et al., 2026]), indicating a 2.3σ discrepancy.

4.3 (w_0, w_a) Tension

Within the ΛCDM paradigm, the accelerated expansion of the Universe is driven by a cosmological constant with equation of state $w = -1$, corresponding to a constant dark energy density $\rho_\Lambda = \mathrm{const}$. If instead the dark energy density evolves with time, the simplest extension is the Chevallier-Polarski-Linder

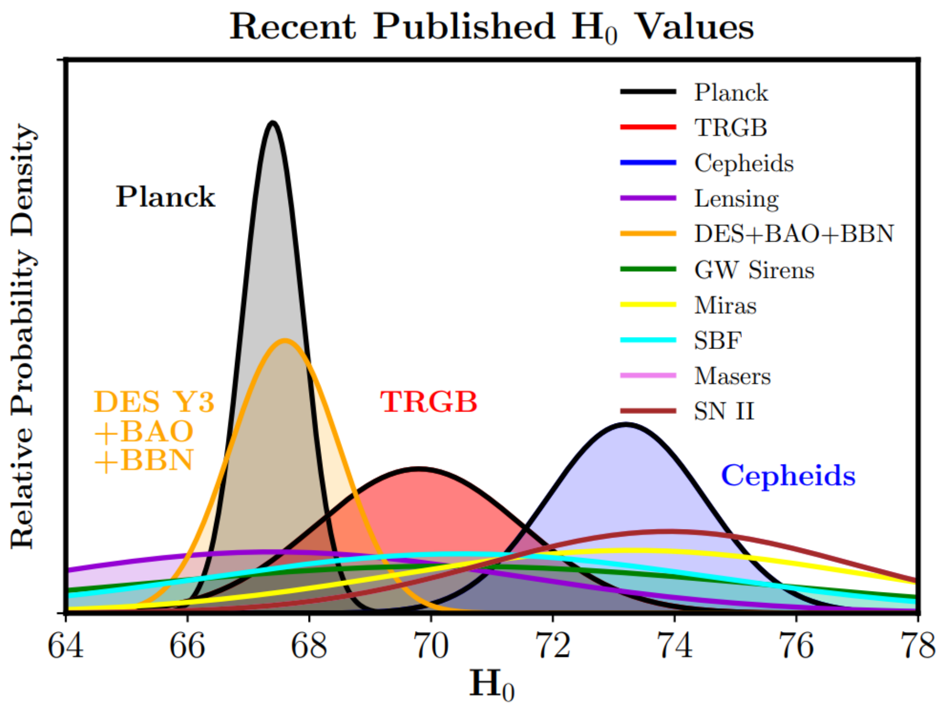


Fig. 1.11 Probability density functions of H_0 from different measurements probed at different redshifts, taken from [Freedman, 2021]. The CMB measurements prefer a lower H_0 value while Cepheids prefer higher values.

(CPL) parametrisation ([Chevallier and Polarski, 2001], [Linder, 2003]):

$$w(a) = w_0 + w_a(1 - a) \tag{1.84}$$

where w_0 is the present-day equation of state and w_a parametrises its time evolution. The ΛCDM limit is recovered for $w_0 = -1$ and $w_a = 0$.

Recent results from DESI DR2, when combined with Planck CMB data and Type Ia Supernova datasets, show deviations from ΛCDM in the (w_0, w_a) plane ([DESI Collaboration et al., 2025]). The significance of the deviation depends on the supernova sample used: 2.8σ when combined with Pantheon+, 3.8σ with UNION3, and 4.2σ with DES Year 5 ([DESI Collaboration et al., 2025]). A subsequent recalibration of the DES Year 5 supernova sample ([Popovic et al., 2025]) reduced this DES discrepancy to 3.2σ.

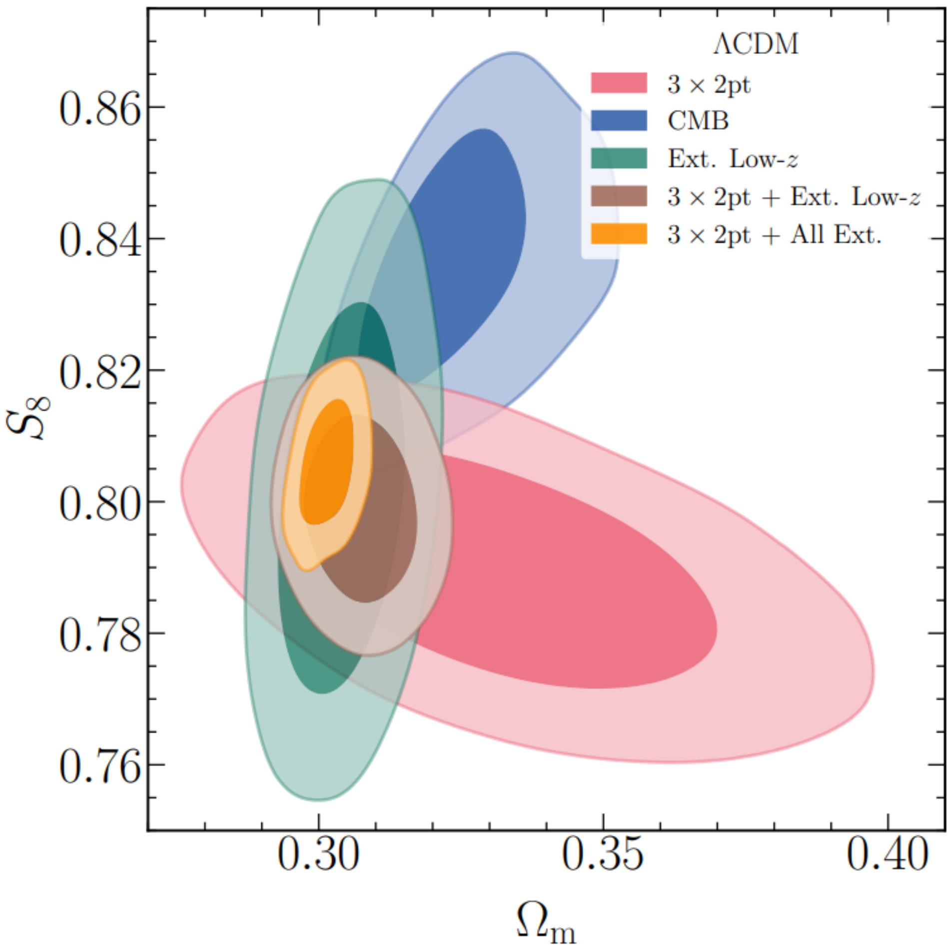


Fig. 1.12 S_8 and Ω_m contours for different galaxy surveys and CMB taken from [Collaboration et al., 2026] enclosing the 68% and 95% posterior probability. The combination of the DES results with all of the other probes is in orange while the CMB measurements are in blue.

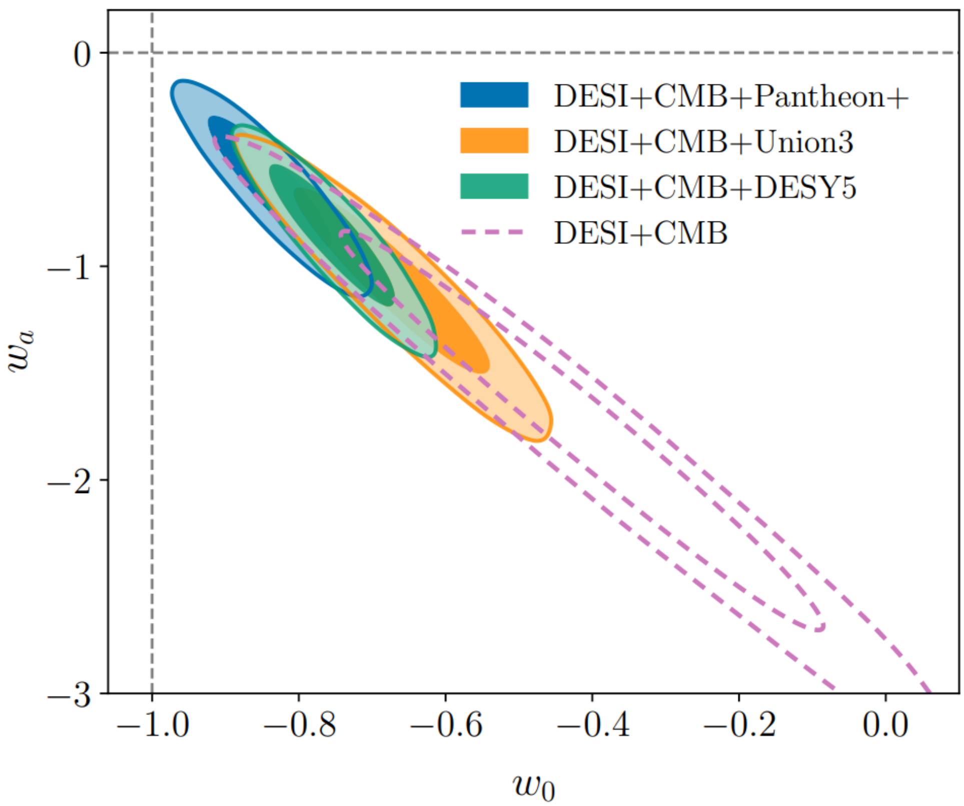


Fig. 1.13 w_0 and w_a contours from the latest [DESI Collaboration et al., 2025] measurements enclosing the the 68% and 95% posterior probability when combined with Pantheon+, UNION3 and DES year 5 data and CMB.

5 Physical explanations for the accelerated Universe

Since the discovery of the acceleration of the Universe by [Riess et al., 1998] and [Perlmutter et al., 1999], the quest to understand the underlying nature of this acceleration has seen many theories arise on what it could be. We've seen one way this can be done by via the addition of a cosmological constant that introduces a new type of energy in the Einstein equations. Due to the fact that we do not know enough about the nature of this energy it is called "dark energy". In the next subsections we will be discussing some of the theoretical models that were proposed to explain this acceleration.

5.1 Cosmological constant

The simplest way to account for the acceleration as we have mentioned before is by adding a cosmological constant Λ to the Einstein equations 1.5, which is what is done in the ΛCDM model. We can look at Λ in two different lights, either a modification to the geometry of spacetime or as a part of the energy content of the Universe. Mathematically they are equivalent since the energy-momentum tensor $T_{\mu\nu}$ can include the $\Lambda g_{\mu\nu}$ and their equations of state are indistinguishable in ΛCDM.

5.2 Dynamical dark energy

One could naturally assume that if the acceleration of the Universe is driven by a fluid that its energy density would change throughout time. The common way to model this is with the Chevallier-Polarski-Linder (CPL) model ([Chevallier and Polarski, 2001]) characterized by two parameters w_0 and w_a such that:

$$w(z) = w_0 + w_a \frac{z}{1+z} = w_0 + w_a(1-a) \tag{1.85}$$

Therefore the Hubble parameter becomes:

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_k(1+z)^2 + e^{\left(-3w_a \frac{z}{1+z}\right)} \Omega_\Lambda (1+z)^{3(1+w_0+w_a)}} \tag{1.86}$$

ΛCDM then becomes a particular case where $w_0 = -1$ and $w_a = 0$.

5.3 Quintessence field

The quintessence model uses an approach similar to inflation where one assumes the accelerated expansion is driven by a scalar field ϕ, here called the quintessence field. The action associated to that field is written as:

$$S(\phi) = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right) \tag{1.87}$$

The equations of motion of ϕ are determined by the Klein-Gordon equation:

$$\ddot{\phi} + 3H\dot{\phi} + \frac{d}{d\phi} V = 0 \tag{1.88}$$

For a FLRW metric, the pressure and energy density become:

$$P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi) \tag{1.89}$$

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi) \tag{1.90}$$

Giving us the following equation of state:

$$w = \frac{P_\phi}{\rho_\phi} = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)} \tag{1.91}$$

Under a slow-roll approximation, the field evolves slowly such that $\dot{\phi} \simeq 0$ giving us $w \approx -1$ where we recover the results of a cosmological constant. If we consider a fast rolling field where the potential is negligible in front of the evolution, the equation of state becomes $w \approx 1$ however this is not physical since we know that for an accelerating Universe $w < -\frac{1}{3}$. Results from [DESI Collaboration et al., 2025] show a preference for a dynamical dark energy in the CPL parameterization.

5.4 Modified Theories of Gravity

An alternative to dark energy is to modify the theory of GR on cosmological scales. We call the models that do so Modified Gravity models (MG). Multiple classes of MG theories exist each modifying a certain aspect of GR. We will be presenting some of those models in each class.

Additional scalar fields

The first class consists of adding an additional scalar field to the Einstein-Hilbert action. These fields need to have equations of motions of second-order time derivatives at most to avoid any unphysical solutions.

One of the models that do that are the $f(R)$ models ([Sotiriou and Faraoni, 2010]) which replaces the Ricci scalar R by a general function of R. The action for an $f(R)$ gravity becomes:

$$S = \frac{c^4}{16\pi G} \int \sqrt{-g} f(R) d^4x + \int \sqrt{-g} \mathcal{L}_m d^4x \tag{1.92}$$

And the field equations are:

$$f'(R)R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + [g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu] f'(R) = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1.93}$$

For the simplest case $f(R) = R - 2\Lambda$ we recover GR. For a FLRW metric, the Friedmann equations become (with $c = 1$):

$$3f'H^2 = 8\pi G\rho_m + \frac{1}{2}(f'R - f) - 3H\dot{f}' \tag{1.94}$$

$$-2f'\dot{H} = 8\pi G(\rho_m + P_m) + \ddot{f}' - H\dot{f}' \tag{1.95}$$

If one rewrite Equation 1.94 to match what we get in GR $3H^2 = 8\pi G\rho$, we can identify a and effective "curvature fluid" that under certain conditions would cause an accelerated Universe. This gives us an effective equation of state on w_{eff}:

$$w_{eff} = -1 + \frac{1}{3f'H^2}\left[8\pi G(\rho_m + P_m) + \ddot{f}' - H\dot{f}'\right] \tag{1.96}$$

Since acceleration is possible for $w < -\frac{1}{3}$ this means that a condition on any $f(R)$ theory would be:

$$8\pi G(\rho_m + P_m) + \ddot{f}' - H\dot{f}' < 2f'H^2 \tag{1.97}$$

$f(R)$ must also be able to reproduce GR during the early Universe and at high matter density regions to remain consistent with observational results. This screening effect is called the chameleon mechanism and the most well known $f(R)$ theories do possess it.

Massive gravity

In quantum field theory, forces are mediated by particles — for instance photons for electromagnetism and gluons for the strong force. By analogy, gravity is expected to be mediated by a spin-2 particle called the graviton.The graviton is predicted to be massless particle with an infinite interaction range, that can recover GR in a quantum field theory formalism. If we now assume that the graviton does in fact possess a mass, that would decrease the range of its gravitational interaction as well as its strength between distant objects ([Fierz and Pauli, 1939], [de Rham et al., 2011]). The gravitational potential is then modified as follows:

$$\Phi = -\frac{GM}{r}e^{-mr} \tag{1.98}$$

where r is the distance to an object, M the mass of the object and m the mass of the graviton. However, the introduction of a massive gravitons produces multiple instabilities such as the inability to recover GR when m goes to 0. It also suffers to produce flat FLRW solutions.

Breaking assumptions

The last class of models we will present here are the ones that break physical assumptions. Here we present an example of a model proposed by Gia Dvali, Gregory Gabadadze and Massimo Porrati ([Dvali et al., 2000]) where gravity can propagate in a 5-dimensional space time with an additional spatial dimension. We can think of it as our 4D world embedded in an infinite volume 5D space. The idea being that the gravity on large scales appears to be in 5D while on small scales in 4D, this is due to the Vainshtein effect. The action for such a theory looks like :

$$S = 2M_5^3 \int d^5X\sqrt{-\tilde{g}}\tilde{R} + 2M_{pl}^2 \int d^4x\sqrt{-g}R + \int \sqrt{-g}\mathcal{L}_m + S_{GH-} + S_{GH+} \tag{1.99}$$

where M_5 is the Planck mass M_{pl} in 5D, $\tilde{g}$ and $\tilde{R}$ are the norm of the metric and the Ricci scalar in 5D. The S_{GH-} and S_{GH+} are Gibbons-Hawkings terms that serve as boundary terms commonly found in, but not limited to, quantum gravity theories.

The theory does predict a late time acceleration ([Deffayet, 2001]), however it suffers theoretical instabilities and unphysical solutions ([Koyama, 2008]). Observations have also ruled the theory out DGP theories ([Fang et al., 2008]).

CHAPTER 2

Cosmology with type Ia Supernova

Contents

1 Type Ia Supernova

For many centuries, astronomers all over the globe would observe the emergence of a bright light in the sky thinking it was the birth of a new star in the Universe. In 1572, an astronomer by the name of Tycho Brahe developped an interest in such event and named the phenomenon *nova stella*, latin for new star. His work inspired later astronomers suhc as Kepler and eventually lead to the Copernicean revolution.

It wasn't until 1934 when Fritz Zwicky and Walter Baade when the term supernova (from the latin "super" and "novae" meaning new bright stars) was coined to describe the brightest novae observed. The supernova classifies phenomena related to the end of a star's life cycle when they go through explosions. This process is temproary therefore taking part in the class of **transients**. Their fluxes are initially weak and then increases over time to a point where it becomes detectable from the Earth, then it diminishes slowly till it disapears.

In the next sections we will be explaining the different classifications of supernovae as well as our understanding behind their explosion mechanisms. A particular focus will be on a specific class of supernovae called "type Ia" and why they are heavily studied by cosmologists.

1.1 Classification

Two main categories of supernovae exists: type I and tpye II each with their own subcategories of supernovae. These main categories were discovered by Rudolph Minkowskia and Walter Baade in 1941 when studying the spectral features of 14 supernovae. Type I supernovae's absorption spectra is characterized by having no Hydrogen, contrary to type II. Type I supernova are then divided in two three classes ([Wheeler and Levreault, 1985, Elias et al., 1985], [Wheeler and Harkness, 1990]):

- **Ia**: contains Silicon line at 615.0 nm
- **Ib**: does not contain Silicon and contains Helium line at 587.6 nm
- **Ic**: does not contain Silicon or Helium lines

Type II supernova are divided in following way ([Hillier, Desmond John and Dessart, Luc, 2019]):

- **IIn**: contains narrow absoption lines
- **IIb**: contains Helium like Ib
- **IIP**: Hydrogen dominates the spectra and its integrated flux becomes constant
- **IIL**: Hydrogen dominates the spectra and its integrated flux decreases in time

The classification of supernovae can be summarized in Figure 2.2.

1.2 Explosion mechanisms

Different explosion mechanisms and progenitors exist for the different classes of supernovae. We will be separating them into two cases: core-collapse and thermonuclear.

Core collapse

This explosion concerns supernovae of types Ib, Ic and all types II. The occurence of this explosion happens at the end of the life cycle of a star after it has completed the fusion of its light elements. The star

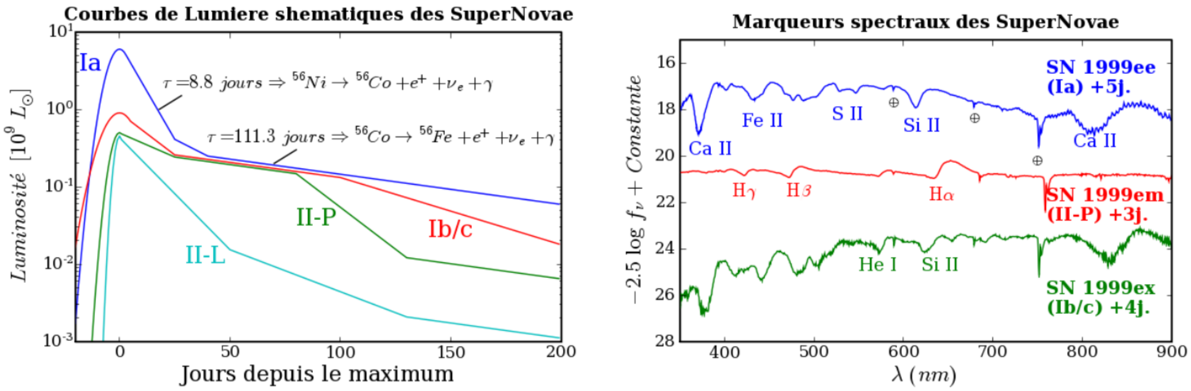


Fig. 2.1 Photometric and spectroscopic curves for different classes taken from [Baumont, 2007]. The left are the schematics for the integrated fluxes (lightcurves) for SN Ia (blue), SN Ib or Ic (red), SN IIP (green) and SN IIL(cyan). On the right we have the spectra of three supernovae observed in 1999, SN Ia (blue), SN IIP (red) and SN Ic (green).

then does generate enough energy to compensate for the gravitational pull and then collapes on itself. The material of the star collapses until it reaches scales at which it is stopped by the strong nuclear interaction, which defines the limiting distance between atoms. This causes a collision between the different layers of the star creating a shock wave that propulses the star material and ignites the supernova explosion. If the star has a mass inferior to $\sim 3.3 M_\odot$ (Oppenheimer-Volkoff limit [Oppenheimer and Volkoff, 1939]), where $\odot$ denotes the Solar mass, the star will become neutron star. Larger mass stars turn to black holes.

Thermonuclear

Thermonuclear explosions concern tpye Ia supernovae. This occurs in white dwarf stars comprised primarily of Oxygen and Carbon when their masses approach the Chandrasekhar limit of $\sim 1.44 M_\odot$ ([Chandrasekhar, 1931]). White dwarf stars form from the remains of low mass stars (typically under $8M_\odot$) and are dense degenerated systems. This term comes from the fact that the density of a white dwarf is limited by electron degeneracy pressure that comes from the Pauli exclusion principle which states that two electron within the same energy levels cannot share the same quantum state. Accreting more mass from external sources induces Carbon and Oxygen fusion which increases the temperature of the white dwarf stars. If the accretion keeps on happening, the fusion of Carbon and Oxygen becomes a chain reaction at a constant pressure causing a thermonuclear explosion ([Röpke and Hillebrandt, 2004]). The explosion ends up releasing around 10^{51}J of energy, mainly kinetic energy. Only a fraction of that is turned into luminous energy of 10^{44}J ([Khokhlov et al., 1993]) meaning tpye Ia supernovae have a luminosity at the same order of magnitude of its host galaxy.

Observations such as SN2011fe have demonstrated that type Ia supernovae are originated from compact and dense stars, which are white dwarfs. It also important to note that tpye Ia supernova spectra are characterized by Silicium and Calcium absoption lines which are the product of the Carbon and Oxygen fusion. Due to the fact that type Ia supernovae occur in quasi-identical environments, we tend to observe similar luminosities between them. Finally, they are observed in environments with older stars which is justified due to the fact that white dwarfs happen at the final stages of a star.

Fig. 2.2 Classification of supernovae. The left column are type I (blue) and the right column are tpye II (red).

White dwarf stars aren't very luminous to be observed at long distances. Usually they are detected a short while after their explosions. Therefore the progenitors of type Ia supernovae have never been observed. Multiple scenarios exist to explain the explosions. We will only be presenting the two main scenarios:

- **Single degenerate** ([Nomoto, 1982]): This occurs when white dwarfs accrete mass from nearby more massive non-degenerate stars will low surface gravity, like red giants. Arguments against that scenario is that the accreted Helium and Hydrogen must all be transformed into Carbon and Oxygen with a fusion reaction, which would explain why they do not appear as absorption line in the spectrum of the supernova. The other is that the speed of the accretion can neither be too slow nor too fast or the explosion doesn't happen. Lastly, the companion stars weren't observed near the explosion area ([Li et al., 2011]).

- **Double degenerate** ([Iben and Tutukov, 1984]): This occurs when a white dwarf star accretes its mass from a nearby less massive white dwarf star in close orbit. This scenario does explain the absence of Helium and Hydrogen absoption lines and the observation of superluminous supernovae (like [Andrew Howell et al., 2006]). It also allows for the case of a double detonation which when it happens for two white dwarfs who's masses are inferor to the Chandrasekhar mass can explain the existence of subluminous supernovae ([Tanikawa et al., 2019]).

1.3 Explosion rate

The explosion rate of type Ia supernovae is the measurement of the number of explosion at a certain redshift in year. It can be divided into two categories: low redshift up to $z \sim 0.3$ and hight redshift. At low redshift the explosion is nearly constant and barely evolves. Low redshift samples such as ZTF allow us to measure that rate. A study done by [Perley et al., 2020] on the ZTF sample, assuming an isotropic distribution, have measured a rate of

$$R_{ZTF} = (2.35 \pm 0.24) \times 10^4 \text{Gpc}^{-3}.\text{yr}^{-1} \tag{2.1}$$

Where each supernova was weighted by the volume it probes with a varying limiting absolute magnitude. However, a recent paper by [Lordet et al., 2026] showed that the rate of supernova is explosion might be more complex than anticipated and is much more dependent on its environment. Figure 2.3 highlights that the assumption that the rate is homogenous at low redshift isn't sufficient to describe supernova as we observe an excess at $z \sim 0.035$ while Figure 2.4 shows that the excess of supernova at that redshift seems to be located arond extremely dense regions.

At high redshift the rate becomes very dependent on the redshift. Its behaviour can be described by a power law with respect to $(1+z)$. Estimates by combining a multitude of probes in [Frohmaier et al., 2019] have shown that the rate can be described as:

$$R(z) = (2.27 \pm 0.19) \times 10^4 \times (1+z)^{(1.70 \pm 0.21)} \text{Gpc}^{-3}.\text{yr}^{-1} \tag{2.2}$$

1.4 Observations

Spectroscopic

Type Ia supernova spectra have a distinct spectra. Most of the light emitted from a type Ia is in the visible wavelengths with some parts of their spectra in the infrared and ultraviolet. During the initial phases of the explosion the external layers of the explosion are opaque. Near the maximum emission of light, we expect to see absoption lines of Silicium and Calcium caused by the fusion of Carbon and Oxygen. The days that follow, the opacity decreases and we start to see absoption lines from Nickel and Cobalt. Finally, in the following weeks the Nickel and Cobalt will disintegrate into Iron for which we see the absoption lines.

Photometric

When describing the photometric evolution of type Ia supernovae, we are describing the integrated flux in a observation band or filter that covers a specific wavelength. Generally, we use the standard filters U, B, V, R and I (also known as the Bessel filters [Bessell, 1990]) that cover the ultraviolet, visible and near-infrared range. We observe in all bands an increase the lightcurve that lasts approximately 20 days

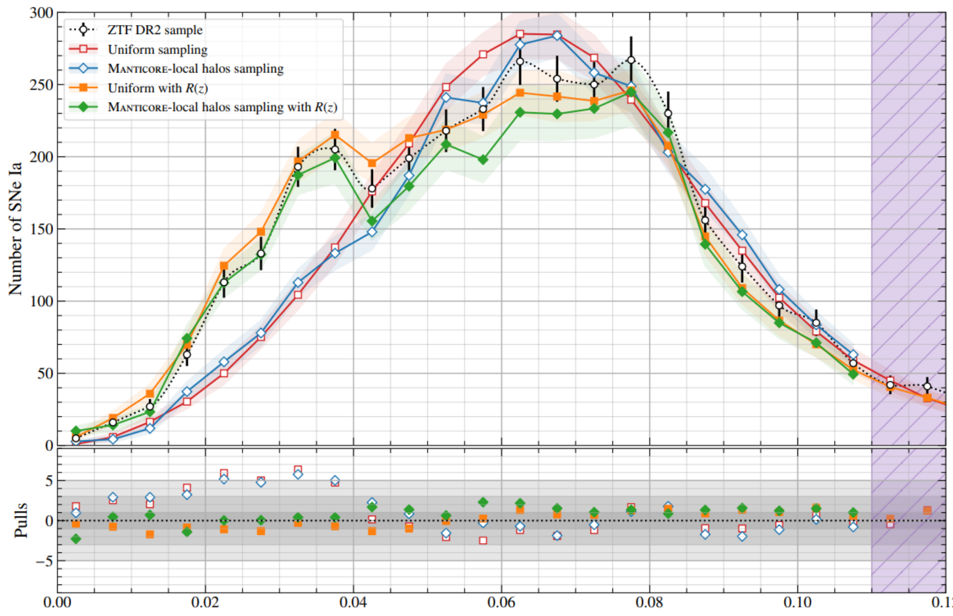


Fig. 2.3 Distribution of simulated supernova observed as a function of redshift z vs the ZTF DR2 sample of obsereved supernovae (circle with black uncertainty). The different simulations are done using different priors on their positions. Red assumes a uniform sampling like [Perley et al., 2020]. Blue uses a uniform sampling weighted by halo masses from Manticore-local. Oragne assumes a redshift dependent rate. Green assumes a redshift dependent rate and weights by the halo masses.

with an absolute magnitude in the B band of −19.5mag that occurs following the disintegration of Nickel. After that, the visible and ultraviolet bands measure an exponential decrease due to the disintegration into Iron. In the infrared bands, we observe a second peak somwhere between 20 to 25 days after the first peak.

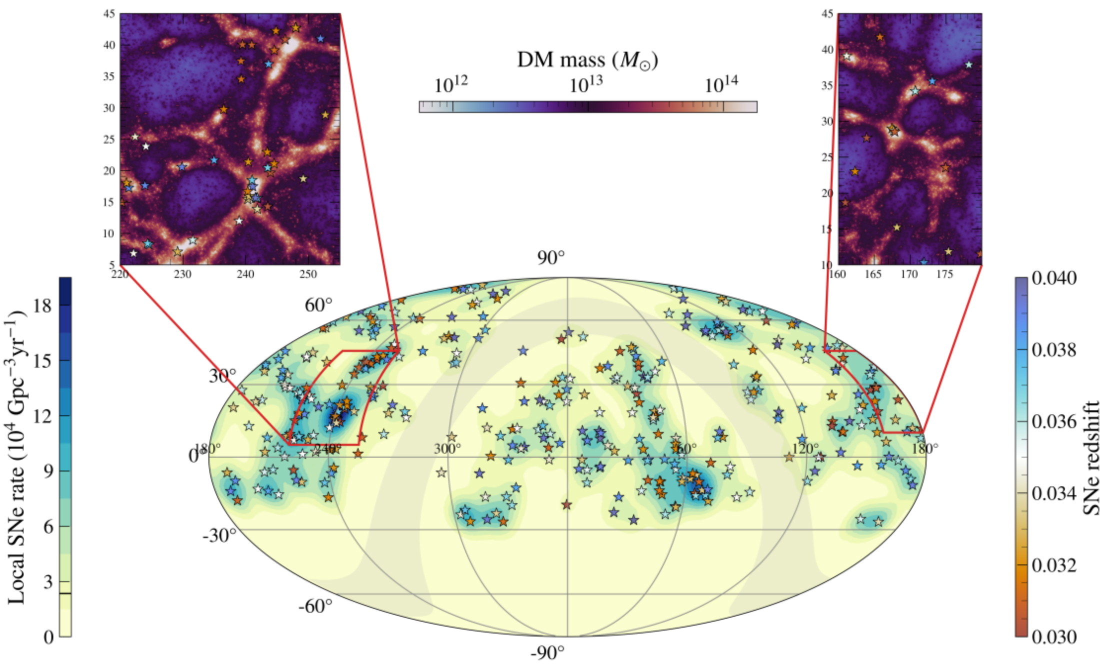


Fig. 2.4 Distribution of supernova obsereved vs the dark matter distribution posterior from Manticore-local as well as the local supernova rate. Top left corner we have a zoom in on the Hercules supercluster. Top right corner we have a zoom in on the Leo supercluster.

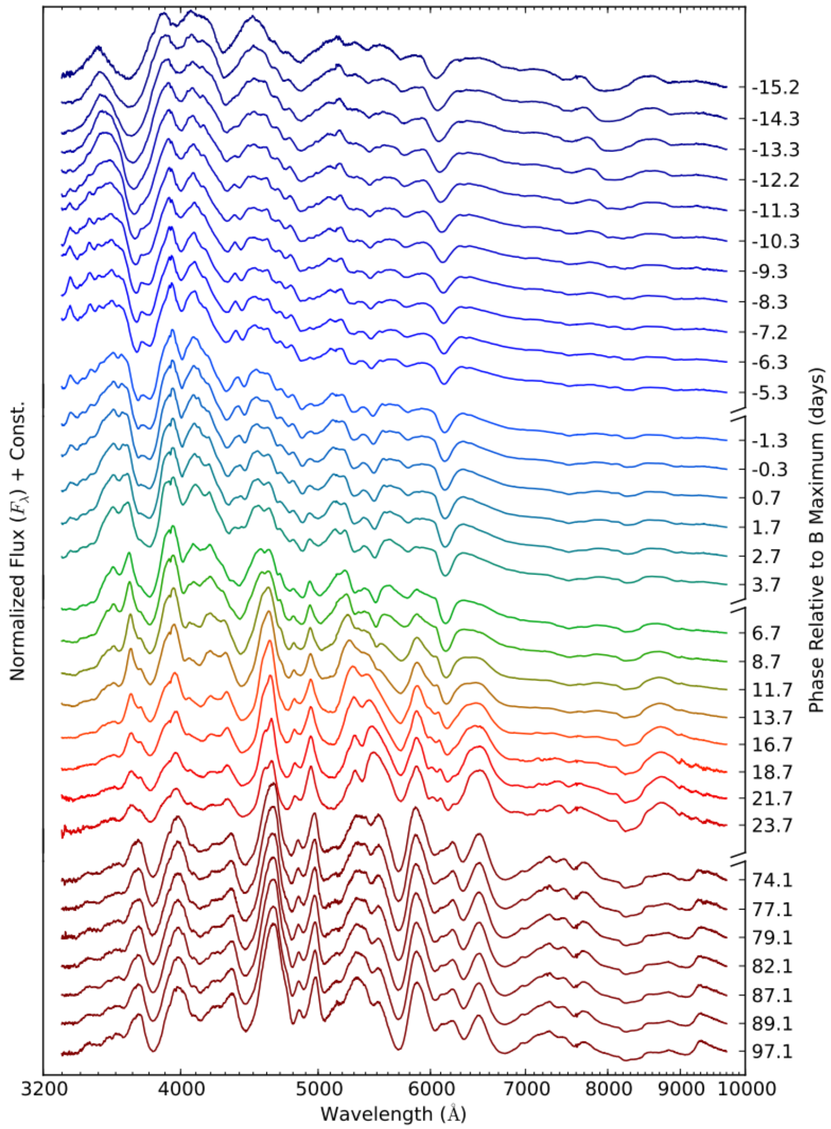


Fig. 2.5 Spectral series of SN 2011fe supernova taken by with the SNIFS instrument ([Pereira et al., 2013]). The spectra were captured betwenen -15 days to 50 days after the maximum emission data. They are ordered from top to bottom, or blue to red. The magnitude shown are relative to the maximum magnitude.

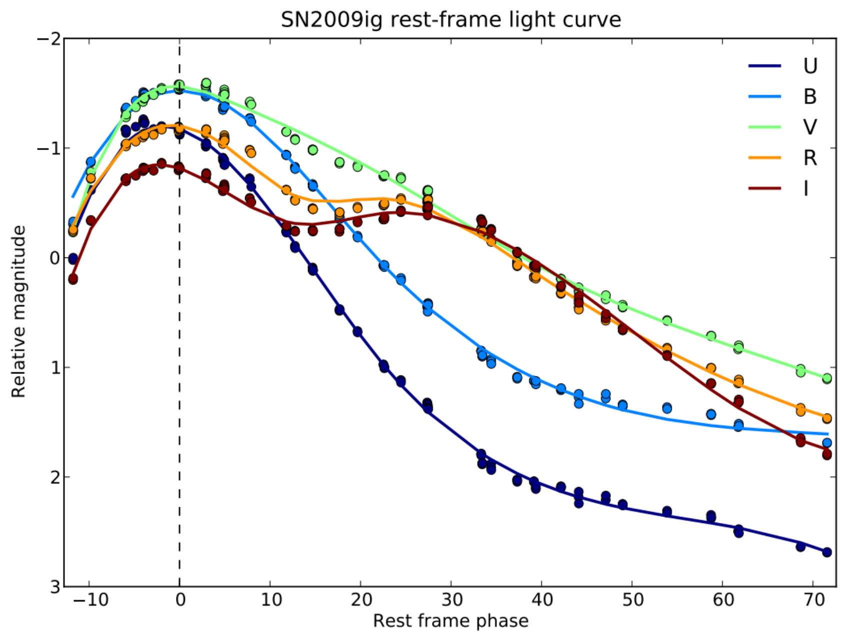


Fig. 2.6 Synthetic lightcurves of the SN 2009ig supernova in the UBVRI bands made for the SNfactory collaboration taken from [Chotard, 2011].

2 Cosmological distances

Observables such as the redshift or distances are used in order to constrain the matter content of the Universe as well as its expansion history and cosmological parameters. There are multiple ways one can define a distance in cosmology, each relying on a different set of parameters.

Physical distances

Under a FLRW metric, we can define the physical distance as the distance separating the observer and an event happening at χ:

$$d_p(\chi) = a_0 S_k(\chi) \tag{2.3}$$

with a_0 being the scale factor at t_0. It is more practical to express χ as a function of the redshift since it is not a direct observable. If we consider a photon emitted at a time t_e from the source object present at χ and received by the observer at t_0, then we get:

$$\chi = c \int_{t_e}^{t_0} \frac{dt}{a(t)} \tag{2.4}$$

By performing the following two variable changes:

$$H = \frac{\dot{a}}{a} = \frac{1}{a}\frac{da}{dt} \Longleftrightarrow dt = \frac{da}{aH} \tag{2.5}$$

$$1 + z = \frac{a_0}{a} \Longleftrightarrow dz = -\frac{a_0}{a^2} da \tag{2.6}$$

We get:

$$\chi = \frac{c}{a_0} \int_0^z \frac{dz'}{H(z')} \tag{2.7}$$

Which gives us the following equation for the physical distance:

$$\boxed{d_p(z) = a_0 S_k \left(\frac{c}{a_0} \int_0^z \frac{dz'}{H(z')} \right)} \tag{2.8}$$

In a flat Universe, this expression is reduced to:

$$d_p(z) = c \int_0^z \frac{dz'}{H(z')} \tag{2.9}$$

In practice, this distance isn't directly measurable as it requires to measure the values H at every redshift along the line of sight. However, other observable distances exist that we can relate to the physical distance.

Angular distances

We can firstly define the distance of an object based on its apparent size D and angle $\Delta\theta$. We call this the angular distance and is defined as:

$$\boxed{d_A = \frac{D}{\delta\theta} = a(t) S_k(\chi) = \frac{1}{1+z} a_0 S_k \left(\frac{c}{a_0} \int_0^z \frac{dz'}{H(z')} \right)} \tag{2.10}$$

Therefore knowing the redshift, the apparent angle and the size of an object gives us access to its distance.

Luminosity distances

We can also determine the distance of an object by linking its intrinsic luminosity to its emitted flux at the surface. This is defined as the luminosity distance d_L and is given by:

$$f = \frac{L}{4\pi d_L^2} \tag{2.11}$$

where f is the surface flux and L is the object's luminosity.

We need to account for the fact that the emitted photons are scattered on the surface S of the object, with $S = 4\pi a_0^2 S_k^2(\chi)$ as well as the fact that its energy $(\frac{hc}{\lambda})$ is is reduced by $1+z$ due to the cosmic expansion. The photons emitted during a duration of Δt will be received during a diluted duration $\Delta t(1+z)$. This means that the observed luminosity will differ from the emitted luminosity giving us:

$$L' = \frac{L}{(1+z)^2} \tag{2.12}$$

where L' is the luminosity seen by the observer. By definition, the surface that flux is seen at an observer frame is given by:

$$f = \frac{L'}{S} = \frac{L}{(1+z)^2 4\pi a_0^2 S_k^2(\chi)} \tag{2.13}$$

We can then identify a formula for the luminosity distance that is redshift dependent:

$$\boxed{d_L(z) = (1+z) a_0 S_k \left(\frac{c}{a_0} \int_0^z \frac{dz'}{H(z')} \right)} \tag{2.14}$$

We can see from the previous equations that:

$$d_L(z) = d_A(z)(1+z)^2 = d_p(z)(1+z) \tag{2.15}$$

At low redshift ($z \lesssim 0.15$) all three distance expressions become equivalent and we can approximate it to:

$$d_L(z) \approx \frac{cz}{H_0} \Leftrightarrow cz = v = H_0 d_L \tag{2.16}$$

where we can see that we recover the Hubble law. We compare in Figure 2.7 the evolution of the three methods to measure distances d_p, d_A and d_L.

3 Cosmology with type Ia supernova distances

3.1 Hubble diagram

The main tool used to constrain the expansion of the Universe with type Ia supernovae is the Hubble diagram. For supernovae, the Hubble diagram maps their luminosity distances to their redshifts. In practice what we attempt to do is relating the type Ia supernova's magnitude in the standard B band denoted as m_B. This magnitude is defined as:

$$m_B = -2.5 \log_{10} \left(\frac{f}{f_0} \right) \tag{2.17}$$

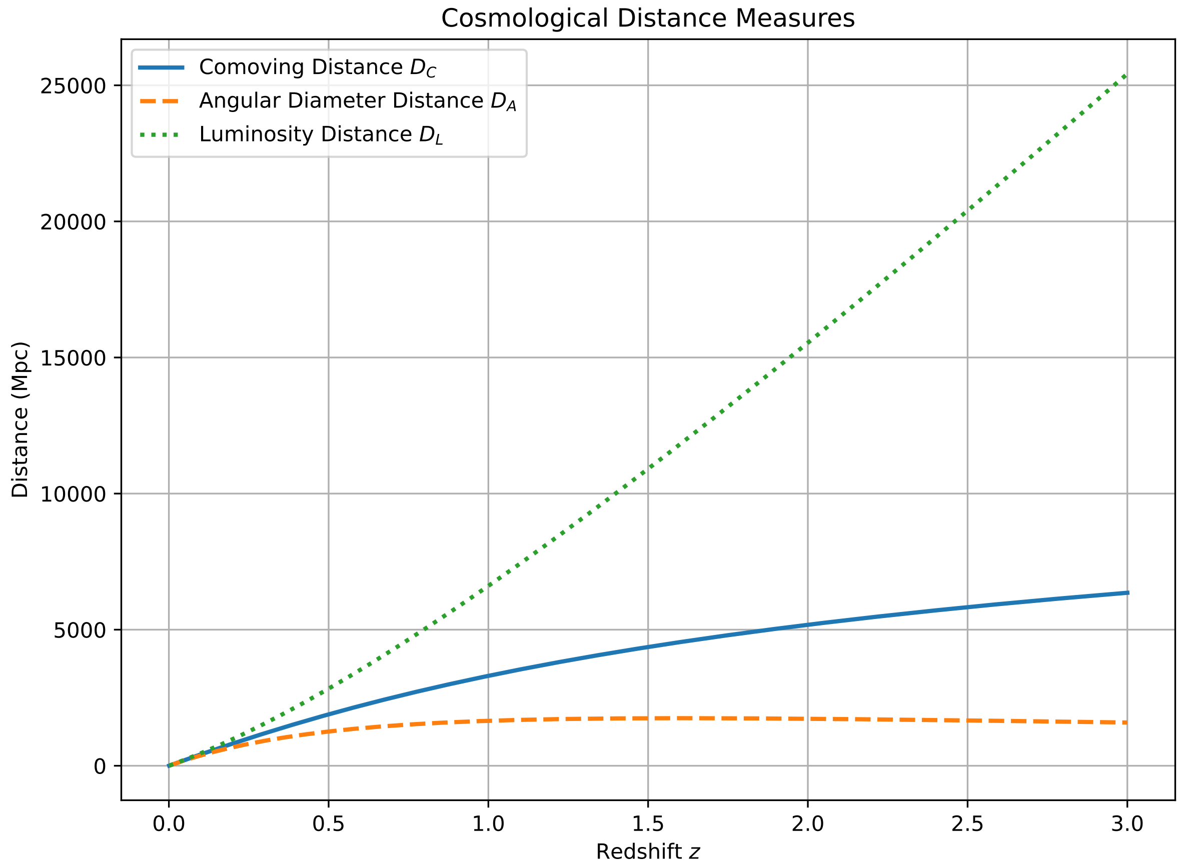


Fig. 2.7 Different distances in cosmology plotted as a function of redshift for a flat ΛCDM with $H_0 = 70\text{km.s}^{-1}\text{Mpc}^{-1}$ and $\Omega_m = 0.3$.

where f is the integrated flux in the observer frame B band and f_0 the flux of a reference star. We also define the absolute magnitude M of an object with an intrinsic luminosity L as the apparent magnitude if it was locatec at 10pc from the observer. Given the definition of the luminosity distance we can get:

$$M = -2.5\log_{10}\left(\frac{L}{f_0 4\pi(10\text{pc})^2}\right) \tag{2.18}$$

We can finally define the distance modulus of a type Ia supernova as:

$$\mu = m_B - M \tag{2.19}$$

and from Equation 2.11 we can identify:

$$\mu = 5\log_{10}\left(\frac{d_L}{10\text{pc}}\right) = 5\log_{10}(d_L) + 25 \tag{2.20}$$

where d_L is the luminosity distance measured in Mpc. If we manage to determine μ from observations, we can construct a Hubble diagram by plotting μ against z.

3.1.1 Measurement of $\Omega_m - w$

Once we have access to the supernova Hubble diagram, we can start deducing cosmological parameters. We recall that under a flat w-CDM cosmology, Equation 2.14 becomes:

$$d_L = \frac{c(1+z)}{H_0} \int_0^\infty \frac{dz'}{E(z')} \tag{2.21}$$

with $E(z) = \frac{H(z)}{H_0}$ being the dimensionaless Hubble parameter. Supernova measurements can go up to a redshift of 2, where the only contribution to the Hubble parameter are Ω_m and Ω_Λ. Given that $\Omega_m + \Omega_\Lambda = 1$ at that redshift, we can deduce that:

$$E(z) = \Omega_m (1+z)^3 + (1-\Omega_m)^{3(1+w)} \tag{2.22}$$

We can see that if we then adjust the Ω_m and w parameters and integrate Equation 2.21 we can compare our observations with the predicted values and constrain w and Ω_m. The value of H_0 for such analyses does not directly affect the values of w and Ω_m and it is usually absorbed in a generalized constant that contains M as well. Ω_m and w are also degenerated so with supernova alone we do a joint inference of the parameters. In order to break this degeneracy, combinations with other probes such as BAO measurements are done. Finally, doing this requires supernova accross a large range of redshift (a low-z sample at $z \lesssim 0.1$ and a high-z sample $0.1 \lesssim z \lesssim 2.0$) in order to observe the evolutions of Ω_m and Ω_Λ.

3.1.2 Measurement of H_0

We recall that for objects at a low redshift the distance luminosity is written as Equation 2.16. We can then express the magnitude as a function of the redshift with:

$$m \approx A - 2.5\log_{10}(LH_0^2) + 5\log_{10}(z) \tag{2.23}$$

where A is a constant. We can see that the realtionship between the magnitude and redshift naturally introduces the Hubble parameter H_0, however it is degenerated with the intrinsic luminosity of supernovae L. Breaking this degeneracy requires us to calibrate the intrinsic supernova luminosity. We commonly use the "distance ladder" method that uses Cepheid stars in the Milky-Way as distance calibrators.

Cepheids are a class stars that have variable periodic luminosities, which was firstly inferred by Henrietta Swan Leavitt thus establishing the Leavitt relation ([Leavitt, 1908]). Most recent confirmations of this relation were observed by [Breuval et al., 2025a]. Distances of Cepheids can be determined using "parallax": a method that measures the apparent displacement angle of an object with respect to a static background due to the motion of the observer. The application of this method is done by measuring the positions of the Cepheids with respect to a background of distant stars 6 months apart. The displacement angle of the observed Cepheid measured due to the half-rotation of the Earth around the Sun can be expressed as $\theta = \frac{r}{d}$ where r is the distance to the Cepheid and d the Earth-Sun distance. This means that the measurement of the distance luminosity of Cepheids can be calibrated using objects sufficiently close to the Milky-Way for their distance to be measured by parallax.

If we can identify nearby galaxies that contain both Cepheids and type Ia supernovae, we can use the Leavitt relation to measure the Cepheid distance modulus μ_{Cepheid}, which in turn calibrates the absolute magnitude M of the supernova:

$$M = m_B - \mu_{\text{Cepheid}} \tag{2.24}$$

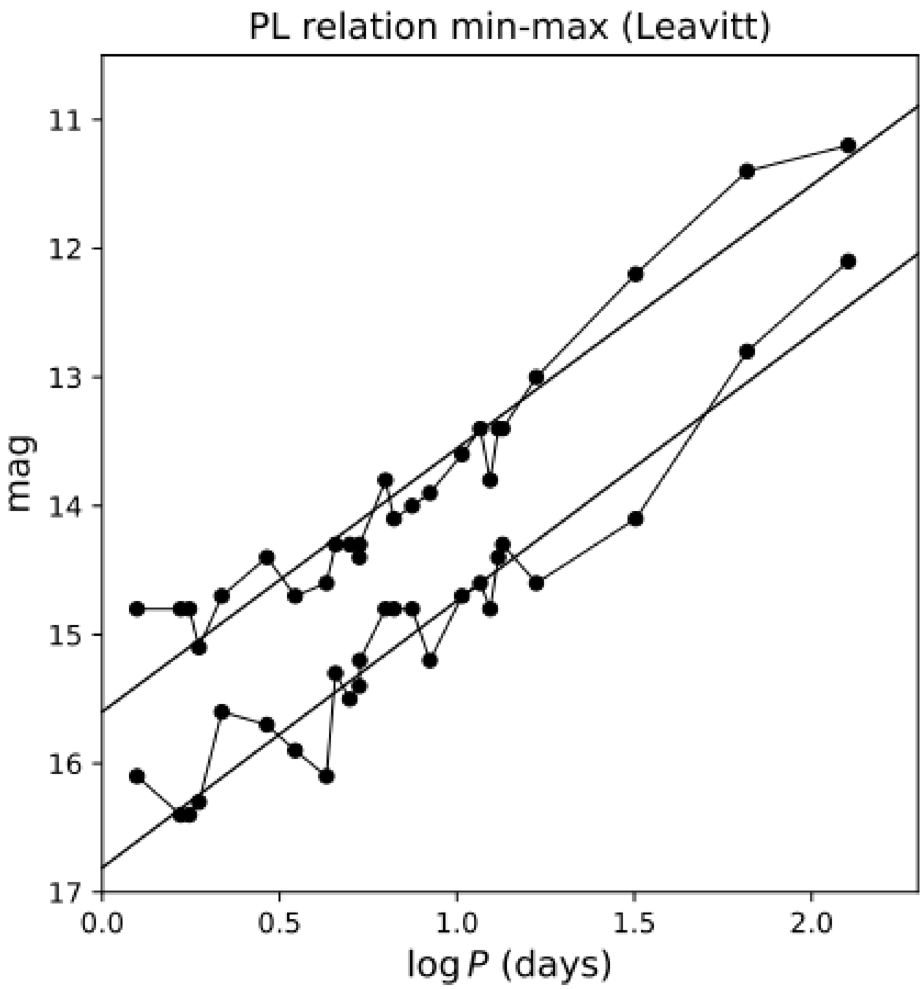


Fig. 2.8 Original measurements by Leavitt establishing the magnitude period relation adjusted in [Breuval et al., 2025b].

where m_B is the observed peak magnitude of the type Ia supernova in the rest-frame B band. This breaks the degeneracy between L and H_0, as M is now determined independently of the Hubble expansion. Applying this calibrated M to type Ia supernovae at low redshift, we can compute the luminosity distance d_L from the observed magnitude and directly infer H_0 from Equation 2.16, averaged over a large sample of supernovae. The most recent measurement using this method yields $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} ([Riess et al., 2022]).

3.2 Peculiar velocities

We define the peculiar velocity of a celestial object as the velocity due to its proper motion independent from the Hubble expansion. This is particularly important for cosmological analyses as it can be used as a test for GR. If we assume peculiar velocities originate from the infall into gravitational potential wells then the detection of correlations between peculiar velocities directly probes the "strength" of gravity at cosmological scales. The parameter probed here is the growth rate of structures $f(z)\sigma_8(z)$ which combines the growth rate function f as well as the matter fluctuations in a 8 Mpc box σ_8. A common parameterization of the growth rate of structures in ΛCDM is:

$$f(z)\sigma_8(z) = \Omega_m(z)^\gamma \sigma_8(z) \tag{2.25}$$

Where γ is the growth index. For GR γ is expected to be 0.55 therefore deviations from this value will indicate that GR is not the most suitable gravitational theory for the description of the large scale Universe.

3.2.1 Extracting peculiar velocities

Supernova peculiar velocity information is encoded in the observed redshift. If we define z_{pv} as the redshift due to the proper motion of the supernova and z_{cosmo} as the redshift due the cosmic expansion, we get the following formula for the observed redshit z_{obs} :

$$1 + z_{obs} = (1 + z_{cosmo})(1 + z_{pv}) = (1 + z_{cosmo})(1 + \frac{v}{c}) \tag{2.26}$$

Where v is the peculiar velocity. Measuring z_{pv} requires knowledge of z_{cosmo} which is not directly measurable. It can be deduced however once the expansion history has been determined. To put simply, if you know exactly the expansion history, then one can directly deduce the cosmological redshift of an object based on how far away it is. This is illustrated in Figure 2.9. This entails that it is crucial to be able to accurately measure the distances of supernovae. Moreover, high redshift supernovae redshifts are almost entirely dominated by the cosmological redshift so analyses on supernova peculiar velocities will be done on low redshift samples.

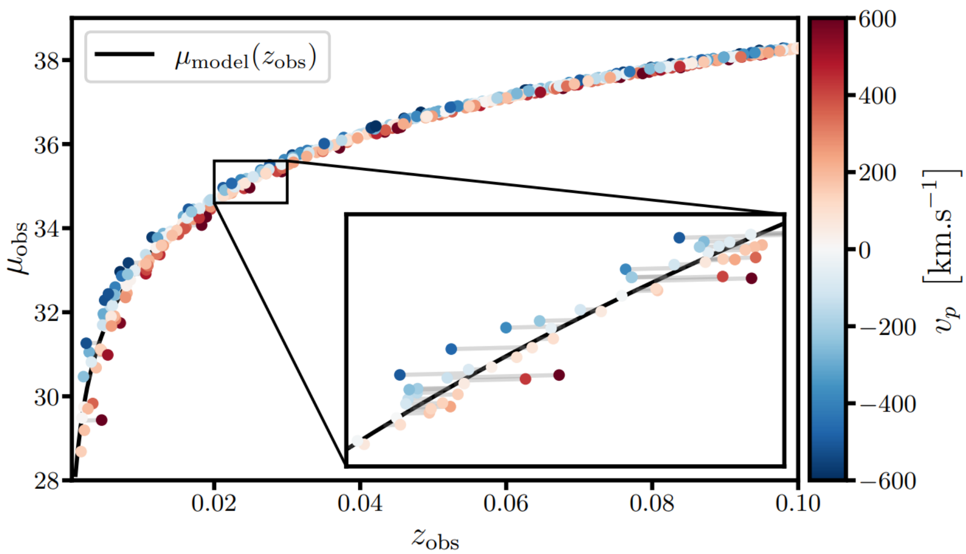


Fig. 2.9 Illustration of the impact of peculiar velocities on the distance modulus on a Hubble diagram from [Carreres et al., 2023]. The velocities were drawn randomly from a Gaussian distribution. The blue and red colors indicate the radial velocity of the supernova.

3.2.2 Measurement of $f\sigma_8$

Multiple methods in infering $f\sigma_8$ exist. We present here four different methods that exist.

Maximum likelihood estimation [Gordon et al., 2007, Carreres et al., 2023]

The maximum likelihood estimation (MLE) is a statistical method of estimating parameters given a model and a statistical distribution. The likelihood function measures how well the model explains the data by assimilating the statistical distribution. For MLE, the distribution is assumed to be Gaussian as the variable is the distance modulus of the supernova, a continuous parameter. The likelihood can be written as:

$$\mathcal{L} \propto |\mathbf{C}(\theta)|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\mathbf{v}(\theta)^T \mathbf{C}(\theta)^{-1}\mathbf{v}(\theta)\right) \tag{2.27}$$

where θ are the model parameters, $\mathbf{v}$ is the peculiar velocity calculated as the difference between the observed distance modulus and the predicted distance modulus defined by a model, $\mathbf{C}$ is the covariance matrix which can be divided into two parts $C = C_{\text{noise}} + C_{\text{PV}}$ where the second term directly depends on the value of $f\sigma_8$ and is left as a free parameter to be determined. The goal is to either maximize $\mathcal{L}$ or minimize it's logarithmic value $\log \mathcal{L}$ which can be numerically beneficial.

Compressed two-point statistics

The compressed two-point statistics is done by adjusting the amplitude of the theoretical 2 point velocity-velocity power spectrum to the data. If we would assume a ΛCDM cosmology, the power spectrum is given by:

$$\mathcal{P}_{vv}(k) \propto (f\sigma_8)^2 \int dk \mathcal{P}(k) j_1(kr)^2 \hat{\mathbf{r}}\hat{\mathbf{r}}' \tag{2.28}$$

where j_1 is the first order spherical Bessel function and $\mathcal{P}$ is the matter power spectrum. For a linear approximation we obtain:

$$\mathcal{P}_{vv}(k) \propto (f\sigma_8)^2 \frac{\mathcal{P}(k)}{k^2} \tag{2.29}$$

What remains is to then adjust the $f\sigma_8$ value to match the observed power spectrum. If the density field is indeed Gaussian, then the full power spectrum would contain all of the information on the distribution. However at small scales it is not the case.

Density field reconstruction

The density field reconstruction infers the velocity field of matter from a galaxy density field. If we call δ_g the galaxy density field and b the galaxy bias (term that models how well do galaxies trace matter) then the reconstructed velocity field becomes:

$$\mathbf{v} = \frac{aH\beta}{4\pi} \int d^3\mathbf{x}' \frac{\mathbf{x}' - \mathbf{x}}{|\mathbf{x}' - \mathbf{x}|} \delta_g(\mathbf{x}') \tag{2.30}$$

where $\beta = \frac{f}{b}$. Comparing this to the velocity field inferred from type Ia supernovae, one can determine β. If then this is combined with external measurements of $\sigma_{8,g} = b\sigma_8$, we can compute the growth rate of structures with $f\sigma_8 = \beta\sigma_{8,g}$. This has been done in [Lavaux and Hudson, 2011].

Field level forward modeling

The field level forward model treats the density and velocity field in itself as a free parameter. The idea is to evolve an inital simulated density field using a gravity model to obtain a final density field. The observables are modelled from the obtained field to compare to actual observations such as galaxies or supernovae.

$$\text{Initial field} \longrightarrow \text{Final field} \longrightarrow \text{Model observables} \Longleftrightarrow \text{Compare to observations}$$

Due to the nature of this method, the growth rate of structures can be directly determined via the final constrained simulation. The challenge of this method relies in multiple stages such as: determining an accurate gravity model while maintaining reasonable computational time and accurately modelling the relation between the observables and the density field.

4 Distance measurements from observed fluxes

The key information in doing a cosmological analysis with type Ia supernova is found in their restframe fluxes. The photometric fluxes as mentioned previously is a broadand flux, meaning it is integrated in a observation band. Due to the redshift of each supernova being different, these integrated zones will be different for an observer on Earth, which makes it difficult to compare the same physical quantities. This can be seen in Figure 2.10. The key is to be able to reconstruct the restframe flux from the obsereved flux in the same band. We use the restframe B band since it contains the maximum light emission from supernova. From now on we will call this band the B^* band. Once we have obtained the flux in the B^*, we can obtain the distance modulus.

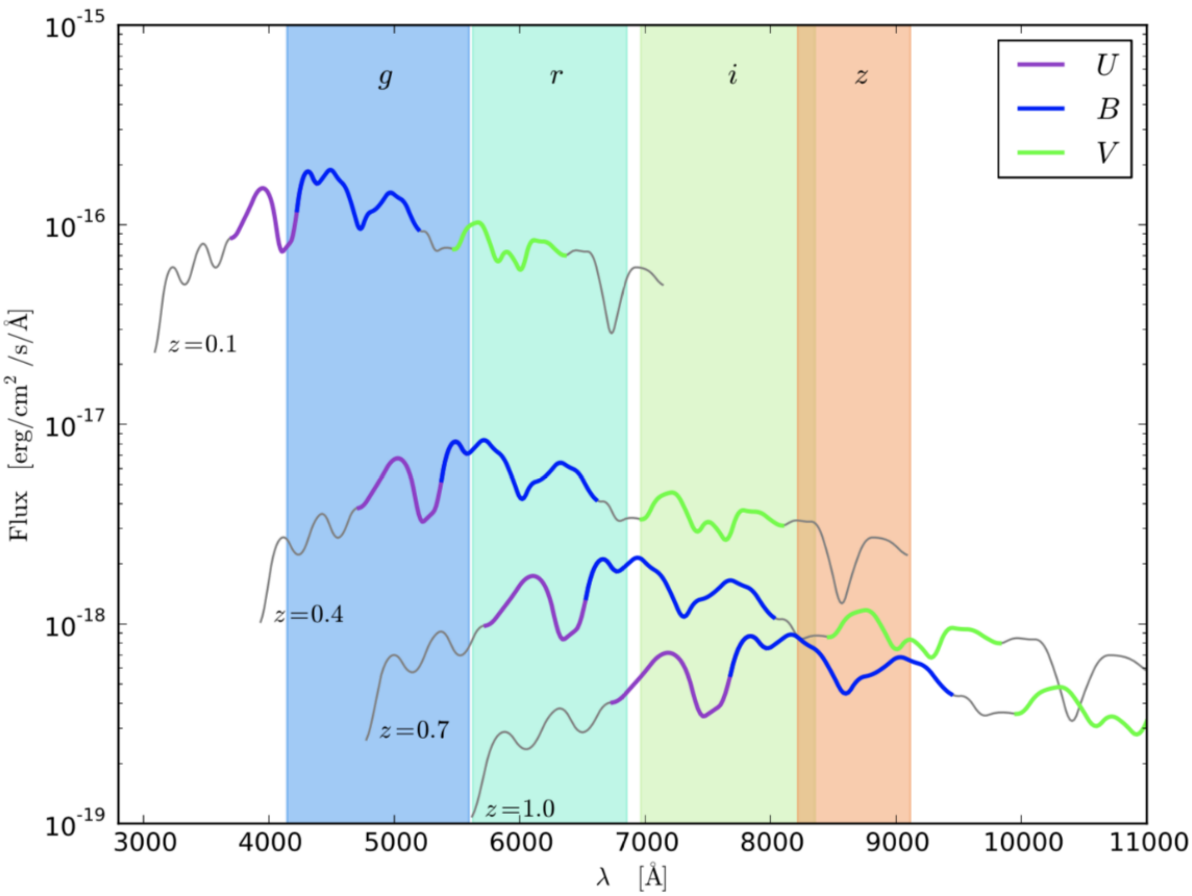


Fig. 2.10 Examples of four spectra at different redshifts 0.1, 0.4, 0.7 and 1.0. The highlighted parts of the spectra represent restframe band would measure. The highlighted areas on the graph are the regions where the observer bands measure.

Although supernova lightcurves look alike, there are still a couple of variations between them. This variability can be laregly accounted for using standardization parameters, hence type Ia supernovae are standardizable candles. Without taking those into effect, the dispersion that exists around peak magnitude in the standard B filter is around 40%. The most well known standardization relation is the Tripp relation ([Tripp and Branch, 1999]) which uses two standardization parameters named stretch and color. After standardization, the dispersion can go down to 15%. More complex standardization methods have been explored, however they necessitate spectral knowledge as well.

5 Standardization

5.1 Brighter-slower effect

The brighter-slower effect is an effect observed in type Ia supernovae where the lightcurves tend to have a slower decline the higher the magnitude of the supernova ([Phillips, 1993]). We define the decline parameter Δm_{15} as the difference of magnitude of the supernova in B band at the date of maximum emission and 15 days later:

$$\Delta m_{15} = M_B(t = t_{max}) - M_B(t = t_{max} + 15) \tag{2.31}$$

which quantifies that effect. We can also define a highly correlated parameter named the *stretch*, or X_1 ([Guy et al., 2007]), when modelling the spectrophotometric evolution with empirical models that measures the width of the lightcurve.

This effect has been revised in [Ginolin et al., 2025b] where this effect was measured on a sample of over 900 supernovae from the ZTF survey. It was demonstrated that this effect in fact presents two regimes that depend on the value of the stretch. In other words, the scaling of the brighter-slower effect varies based on the value of X_1. We refer to this variation as the "broken-alpha" effect.

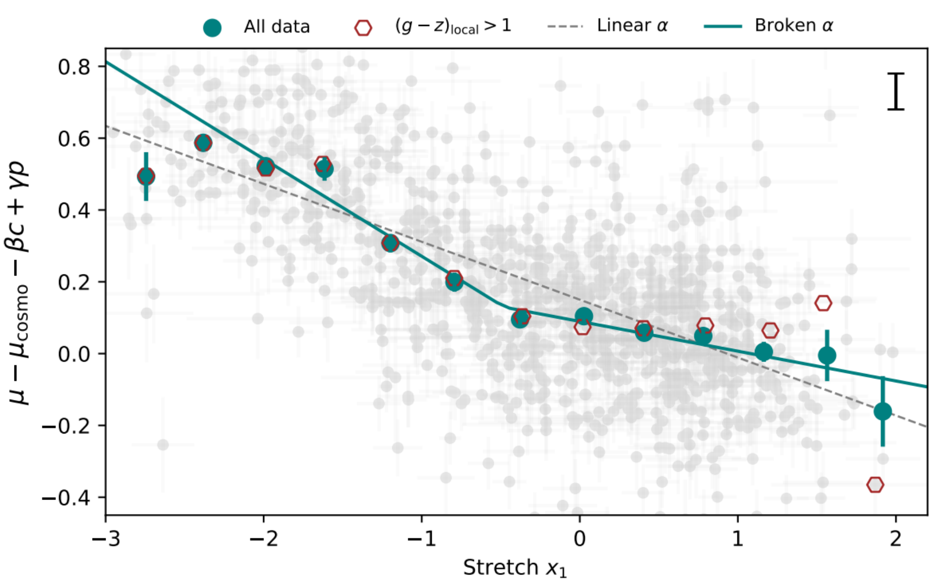


Fig. 2.11 Binned Hubble residuals as a function of the stretch. The data is fitted to highlight the two regimes of the broken alpha relation [Ginolin et al., 2025b].

5.2 Brighter-bluer effect

In a similar way, the brighter-bluer effect characterizes how bluer supernovae tend to have a higher magnitude ([Hamuy et al., 1995, Riess et al., 1996]). We can define an associated parameter, the *color* or c, as the deviation of the difference of magnitude at the time of maximum emission between the B and V from the average supernova. When c is negative we call the supernova blue, and if it is positive, red. Part of the reason this effect exist is due to the absorption of light by the host galaxy and the intergalactic dust in the line of sight of the observer. Another reason is due to intrinsic properties related to the physical properties of the progenitor.

This effect was confirmed in [Ginolin et al., 2025a] where it was studied on the ZTF DR2 sample of over 900 supernovae. A scale dependent effect was proposed in the paper but found no statistical significance for it.

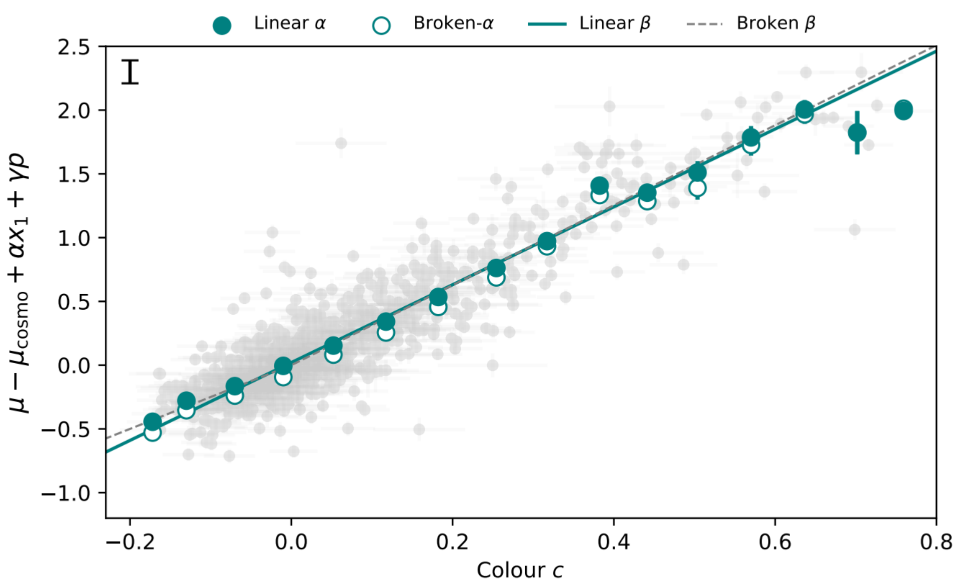


Fig. 2.12 Binned Hubble residuals as a function of color. The data is fitted to highlight the brighter-bluer effect ([Ginolin et al., 2025a]).

The Tripp relation combines both effects to standardize the observed magnitude with two parameters α and β:

$$m_{Tripp} = m_{obs} - \alpha X1 + \beta c \tag{2.32}$$

where m_{Tripp} is the standardized magnitude in the B band using the Tripp relation.

5.3 Environmental effects

We have already mentioned that galaxy properties can affect the luminosity of supernovae as seen in the brighter-blue effect. However, studies conducted on large samples of supernovae have shown that the standardized luminosity also depends multiple additional properties of the host galaxy. This is seen as a "step" in the Hubble residuals that depend on a certain property of the galaxy. One possible explaination would be the mass of the galaxies([Sullivan et al., 2010]). This dependancy presents itself as an offset in the standardized when the popultation of host galaxies' masses are divided into two categories: more or less than $10^{10} M_\odot$. The mass step has shown a 3.9σ significance afer correcting for stretch and color ([Sullivan et al., 2010]).

Another possible dependancy would be the star formation rate in those galaxies ([Rigault et al., 2020]). Galaxies can be divided into two subgroups: high star formation galaxies and low star formation galaxies. This step factor has shown a 2.6σ significance afer correcting for stretch and color.

The [Roman, M. et al., 2018] paper studied this dependancy on the supernovae local $U-V$ restframe colors (limited to a certain radius around the supernova). This step factor has shown a 7.0σ significance. When the same study was conducted on 927 ZTF supernova with the local colors $g-z$, the significance was found to be around 18.2σ significance.

We can then add an extra nuisance parameter to the Tripp relation γ that accounts for that offset:

$$m_{Tripp} = m_{obs} - \alpha X1 + \beta c + \gamma \tag{2.33}$$

where γ is fitted alongside the standardization parameters.

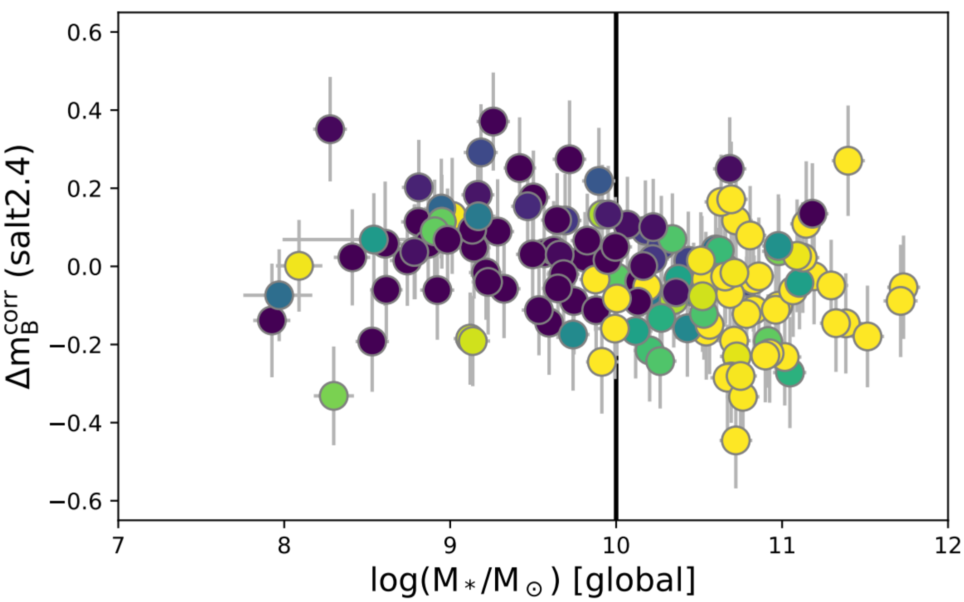


Fig. 2.13 Binned Hubble residuals as a function of the star formation rate ([Rigault et al., 2020]).

6 Spectrophotometric models of supernovae

Given that the B^* fluxes of supernovae are not directly observable, we rely on spectrophotometric models that allow us to reconstruct to emitted fluxes. Theoretical supernova explosion models haven't been able to replicate the obsereved fluxes with a good precision as seen in [Blinnikov et al., 2006], [Röpke and Sim, 2018] and [Pakmor et al., 2024]. Instead we rely on empirical models to describe the spectrophotometric data. Here we present the main models that exist.

6.1 SALT2/SALT3

The Spectral Adaptive Lightcurve Template ([Guy et al., 2007]) class of spectrophotometric models are the current state of the art for a cosmological analyses. The idea is to parameterize the model with two parameters that would represent the stretch and color in the Tripp relation. These models are particularly useful since they are determined and used on photometric data. Spectra are used to determine the spectral features of the model. Once the model has been obtained, one can directly use it on lightcurves to determine its parameters.

We define the restframe spectral flux density surface $\mathcal{S}(\lambda,p)$ the spectrophotometric evolution of type Ia supernova where p is the phase of the supernova, defined as the difference between the observation date and the date of maximum emission $p = t - t_{max}$ in days, and λ the wavelength in Å. The goal is to then obtain S.

We can define the spectral flux density as:

$$\phi_{spec}(\lambda,p) = \frac{1}{4\pi d_L^2(z)}\frac{1}{1+z}\mathcal{S}(\frac{\lambda}{1+z},\frac{p}{1+z}) \tag{2.34}$$

With that we can also deduce the broadband flux observed through a pass-band $T(\lambda)$ as:

$$\phi_{phot}(p) = \frac{1}{4\pi d_L^2(z)}\frac{1}{1+z}\int \mathcal{S}(\frac{\lambda}{1+z},\frac{p}{1+z})\frac{\lambda}{hc}T(\lambda)\mathrm{d}\lambda \tag{2.35}$$

Since the goal of doing supernova cosmology is reconstructing the restframe fluxes, we can rewrite the expression in the restframe of the supernova as:

$$\phi_{phot}(p) = \frac{1}{4\pi d_L^2(z)}(1+z)\int \mathcal{S}(\lambda,\frac{p}{1+z})\frac{\lambda}{hc}T((1+z)\lambda)\mathrm{d}\lambda \tag{2.36}$$

As mentioned previously, the luminosity distances of supernovae are defined using the emitted flux in the standard B band at the time of maximum emission, observed at 10pc from the supernova:

$$\Phi_B = \frac{1}{4\pi(10\mathrm{pc})^2}\int \mathcal{S}(\lambda,p=0)\frac{\lambda}{hc}B(\lambda)\mathrm{d}\lambda \tag{2.37}$$

By imposing the normalization of the model in the B^* band at phase 0:

$$\int \mathcal{S}(\lambda,p=0)\frac{\lambda}{hc}B(\lambda)\mathrm{d}\lambda = 1 \tag{2.38}$$

And defining the following parameter:

$$X_0 = \Phi_B\frac{(10\mathrm{pc})^2}{d_L^2(z)} \tag{2.39}$$

We obtain:

$$\phi_{spec}(\lambda, p) = X_0 \frac{1}{1+z} \mathcal{S}(\frac{\lambda}{1+z}, \frac{p}{1+z}) \tag{2.40}$$

$$\phi_{phot}(\lambda, p) = X_0(1+z) \int \mathcal{S}(\lambda, \frac{p}{1+z}) \frac{\lambda}{hc} T((1+z)\lambda) \mathrm{d}\lambda \tag{2.41}$$

The SALT parameterization of $\mathcal{S}$ is ([Guy et al., 2007]):

$$\boxed{\mathcal{S}(\lambda, p) = [M_0(\lambda, p) + X_1 M_1(\lambda, p)] 10^{0.4cCL(\lambda)}} \tag{2.42}$$

This parameterization is a classic principal component analysis, or PCA, with the addition of a color law. The M_0 and M_1 surfaces along with the CL color law are common to all supernovae. M_0 is the average spectral surface of a type Ia supernova, which represents what your average spectrophotometric evolution of a type Ia supernova looks like. M_1 is a surface representing the main variations to M_0. CL represents the color diversity of supernovae. In practice, the measured spectra obtained are only used to determine the spectral features of $\mathcal{S}$ as they are not as well calibrated as photometric data and are obtained from many different instruments for the same survey, making parameters such as X_0 determined primarily on the photometric data.

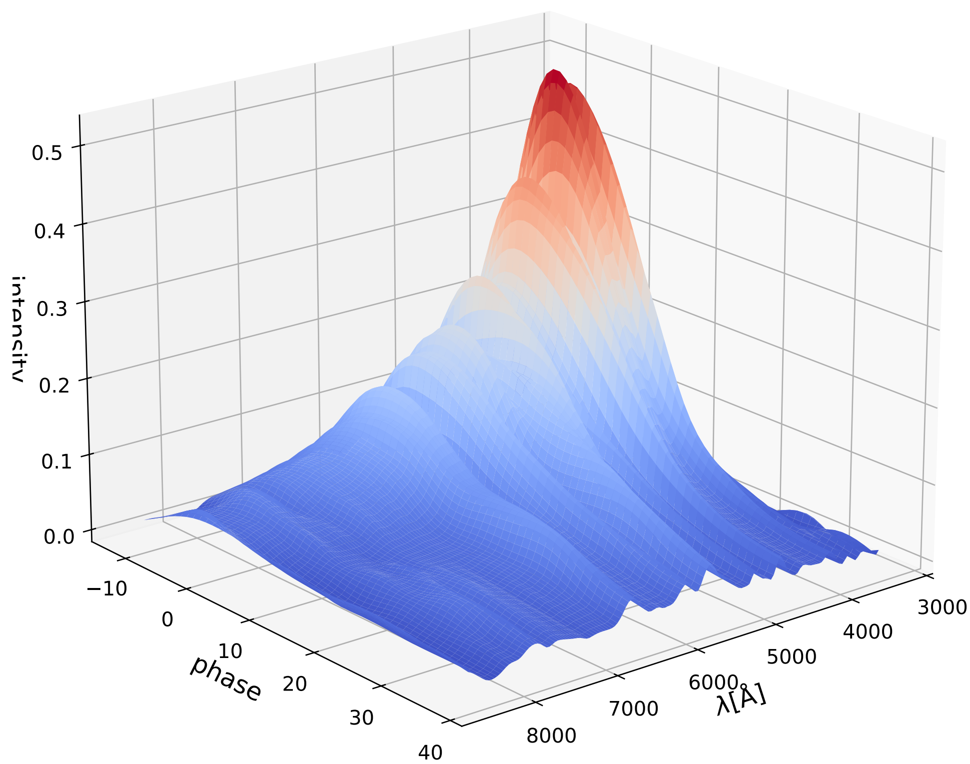


Fig. 2.14 The SALT2.4 M_0 surface with respect to the restframe wavelength and phase

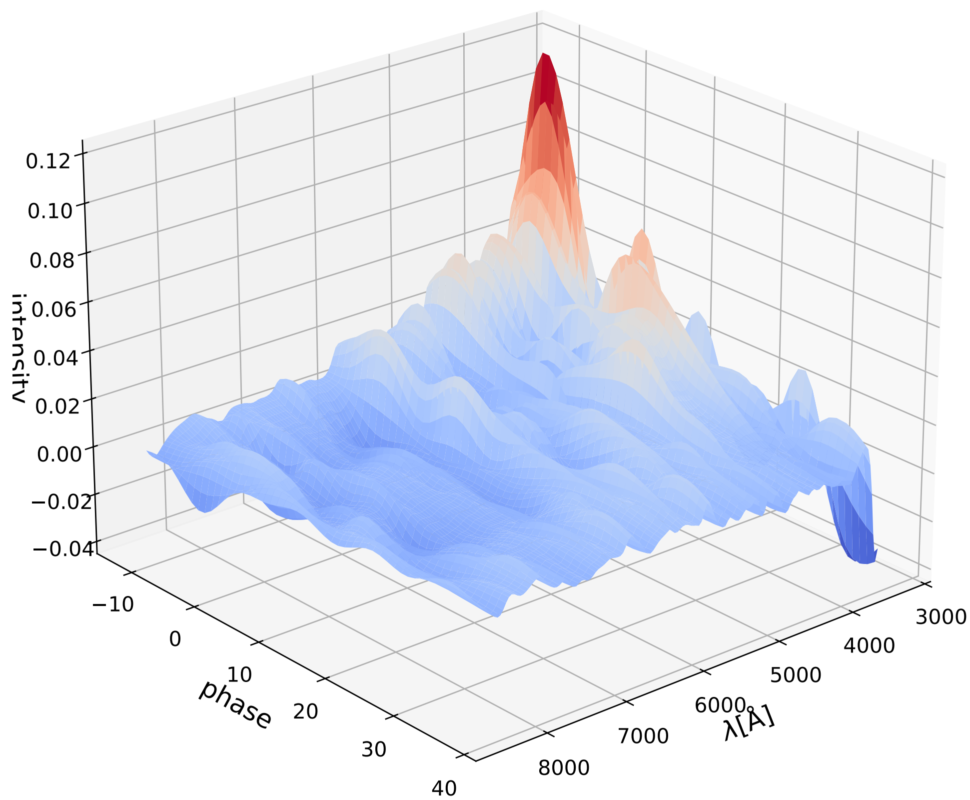


Fig. 2.15 The SALT2.4 M_1 surface with respect to the restframe wavelength and phase

Determining the global parameters is what we called "training the model" while determining the individual supernova parameters is called "fitting the lightcurve parameters". The model training is iteratively on a sample of supernovae, the training dataset. The date of maximum emission of light is the first thing that is determined (t_{max}). The global surfaces are then fitted and the residuals are analyzed. The variability not captured by the model is then accounted for using an error model. The last two steps are then repeated until the model converges. This includes accounting for calibration uncertainties, where for each passband in the dataset requires it's own model training in order which made the propagation of the calibration uncertainties a difficult task. Once the model has been trained, it is published. It is then fitted on a cosmological dataset, different from the one used in the training, to obtain the X_0, X_1 and c parameters.

The initial SALT2 model was trained on a dataset containing 52 low-z supernova from [Hamuy et al., 1996] and 121 high-z supernova from the first two years of observation from SNLS ()[Astier et al., 2006]). A later retraining of the model was done in [Guy et al., 2010] where 207 new supernovae were added from the SDSS-II sample, giving us the SALT2.4 model. This model was used in the [Betoule et al., 2014] JLA cosmological analysis.

The SALT3 model is the current state of the art iteration of the SALT models ([Kenworthy et al., 2021]).

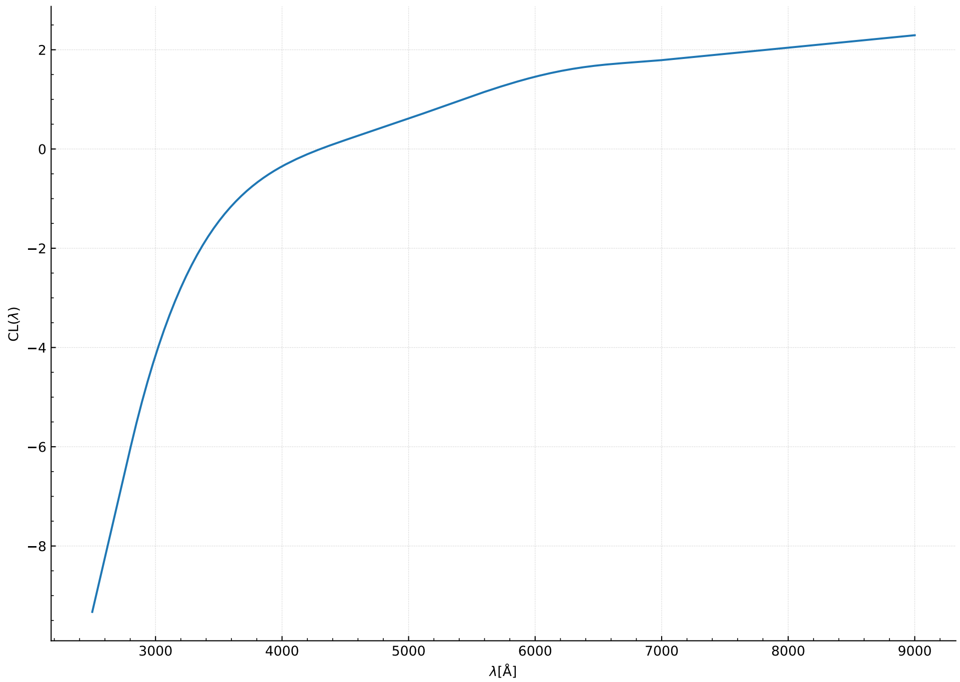


Fig. 2.16 The SALT2.4 color law CL surface with respect to the restframe wavelength

The entire training procedure was written in Python3 coding language and is publicly available for the community. The model was trained on a sample of 1083 supernovae, 2.5 times larger than SALT2. SALT3 improves the training by having consistent photometric handling between surveys and improved cross-calibration between surveys in order to reduce survey dependent systematics. The model was also trained on more high-z supernova adding data from PanSTARRS-1, DES and Pantheon, improving the handling of UV data regions. The color law was also readjusted to be a higher order polynomial than SALT2. The SALT3 was used for the DES cosmological analysis ([DES Collaboration et al., 2024]).

6.2 Models trained on the SNfactory dataset

The two-parameter standardization scheme introduced by [Tripp and Branch, 1999] manages to greatly reduce the dispersion between supernovae but still misses on some of the variability that is obsereved. In order to tackle this issue, spectrophotometric models have been developed with the aim of finding standardization parameters based on the spectroscopic features of supernovae. To do so, these models have been trained on the SNfactory dataset ([Aldering et al., 2002]). A survey that ran from 2004 to 2014 collecting a sequences of spectra of nearby supernovae at low-z (max $z \approx 0.11$) over a long period of time. This is the only dataset that contains a fine spectroscopic description in a large phase range. The well calibrated spectra in the SNfactory dataset will also mean that it doesn't need to be accompanied by photometric measurements to acquire distances.
These features makes this dataset unique and very interesting to use in order to study the properties of supernova.

6.2.1 SNEMO

SNEMO model (SuperNova Empirical MOdels, [Saunders et al., 2018]) generalises the PCA used in SALT2. In SNEMO, the $\mathcal{S}(\lambda,p)$ components are extracted using an Expectation Minimization for Factor Analysis or EMFA that takes into account the measurement uncertainties that affects the model. With SNEMO multiple model trainings can be done at different orders. The model is expressed at:

$$S_{SNEMO}(\lambda,p) = c_{SN,0}\left(e_0(\lambda,p) + \sum_{i=1}^{N} c'_{SN,i}e_i(\lambda,p)\right)10^{0.4A_sC(\Lambda)} \tag{2.43}$$

where c_{SN} are the individual supernvova parameters, e_i the model surfaces, A_s is the supernova extinction and C the extragalactic absorption law introduced by [Cardelli et al., 1989] (Cardelli extinction law). This method expands on the traditional SALT parameterization by adding as many extra surfaces as needed to fit the data. We can see that at second order the model becomes roughly the same as SALT2 with 2 parameters that represent the color and stretch of the supernova. Comparisons with the SALT2 color and stretch parameters have shown strong correlations. The [Saunders et al., 2018] paper shows that going up to the 7th order the supernovae variability decreases and at the 15th order the model fits the data best.

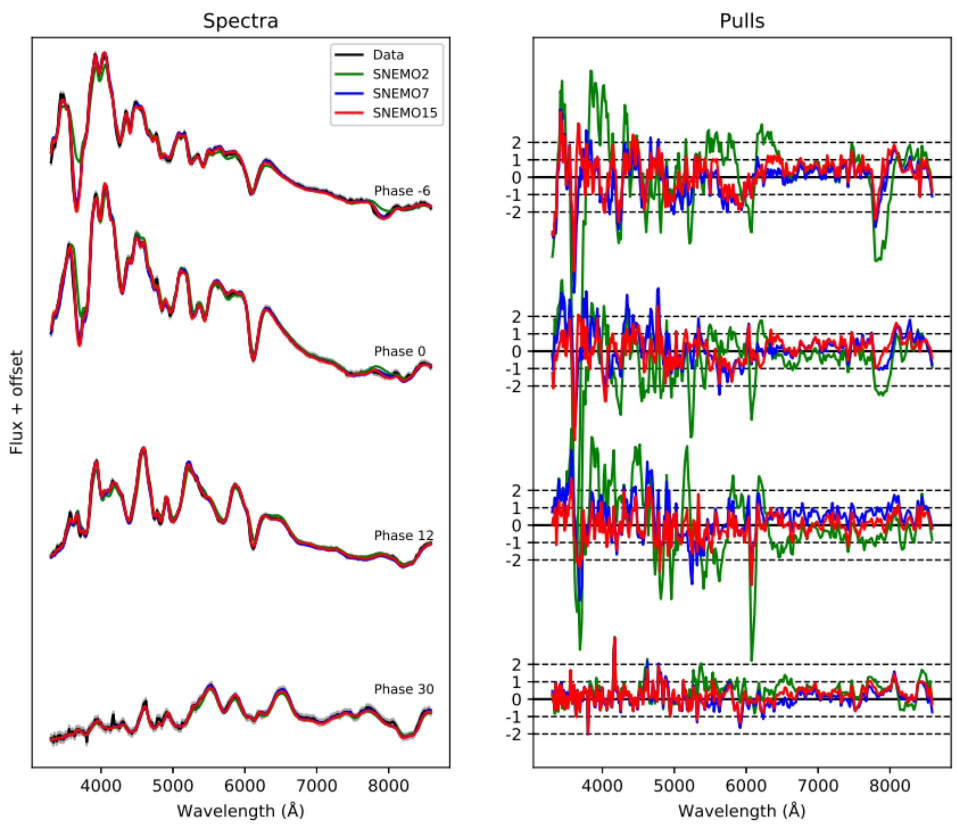


Fig. 2.17 Model reconstruction of SNEMO on SNfactory spectra ([Saunders et al., 2018]). On the left we see the fluxes of the spectra in comparison with 3 SNEMO models containing 2 (green), 7 (blue) and 15 (red) parameters. The right figures shows the weighted residuals of each model and spectrum.

6.2.2 SUGAR

The SUGAR (SUpernova Generator And Reconstructor, [Léget et al., 2020]) model uses EMFA like SNEMO but applied in a different space with the idea that supernova variability is encoded in their spectral data. Therefore, instead of adding surfaces that would explore the entire space for the standardization features, the model has a pre-selected set of spectral features to look out for when determining the model. The founders of SUGAR have chosen 13 spectral features to identify the supernova variability, for which they show accounts for its entirety. The model for each superniva i is expressed as:

$$M_i(\lambda, p) = M_0(\lambda, p) + \sum_{j=1}^{3} h_{i,j}\alpha_j(\lambda, p) + A_{V,i} f(\lambda, R_v) + \Delta M_{\mathrm{gray},i} \tag{2.44}$$

with $M_0(\lambda, p)$ being the magnitude of the aaverage supernova spectrum, $\alpha_j(\lambda, p)$ is the intrinsic variation related to the factor $h_{i,j}$, $A_{V,i}$ the extinction in the V band, $f(\lambda, R_v)$ is the Cardelli extinction law and $\Delta M_{\mathrm{gray},i}$ is a normalization factor. The parameters $h_{i,1}$ and $h_{i,3}$ are correlated with the SALT stretch while the $A_{V,i}$ parameter is correlated with the color. The work with SUGAR has essentially showed that the supernova variability can be captured by 3 parameters, one more than the traditional stretch and color parameters.

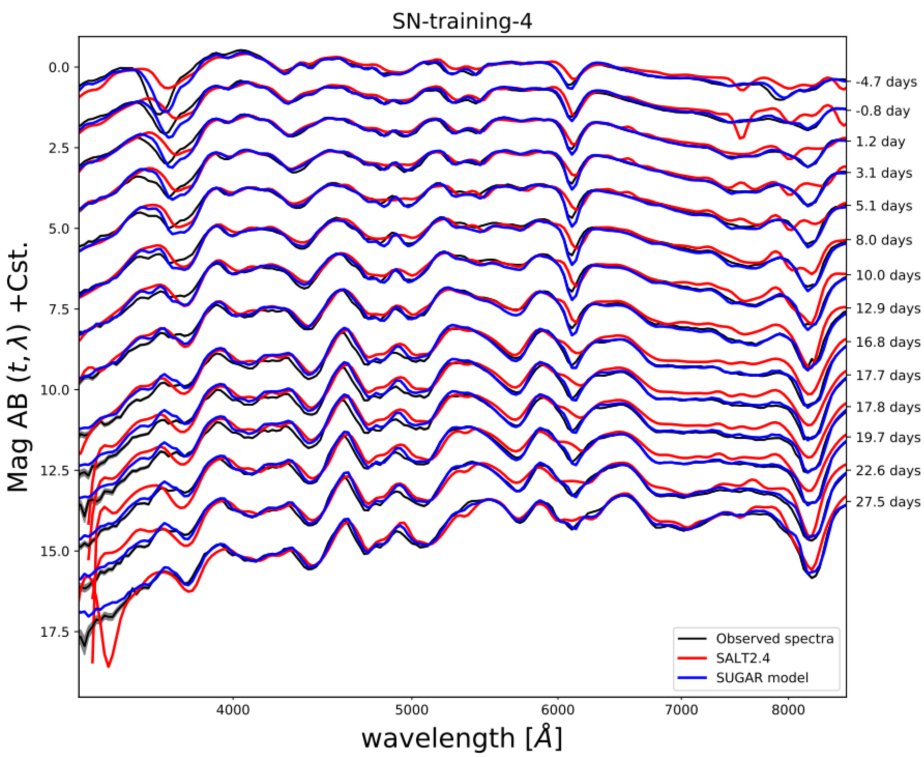


Fig. 2.18 SUGAR model (blue) vs SALT2.4 (red) on top of the SNfactory data (black) taken from [Léget et al., 2020].

6.2.3 Twins Embedding

The Twins Embedding model ([Boone et al., 2021]) uses a non-linear parameterization to describe the supernova diversity. It uses a new standardization method based on the spectral differences between supernova at specific regions. If the spectral differences between two supernovae is small enough they are considered as "Twins" (Figure 2.19). This is also called the spectral distance, denoted as γ_{ij}. The idea is to firstly determine the average spectrum near t_{max}, called f_{mean}, and account for any redshift uncertainties and galactic extinction using the Cardelli extinction law. The variability is then modeled by a non-linear model and adjusted using an algorithm that does a non-linear dimensionality reduction called Isomap which is a generalization of PCA while accounting for non-linearities. The spectral distance between two Twins i and j is written as:

$$\gamma_{ij} = \left[\sum_k \frac{f_i(\lambda_k) - f_j(\lambda_k)}{f_{mean}(\lambda_k)} \right]^{\frac{1}{2}} \tag{2.45}$$

with f_i and f_j the spectra of i and j. The model is described using three parameters. The first parameter ξ_1 highlights the the pseudo-equivalent widths (pEWs) of the Ca II features and it describes the most important variabilitis. THe second parameter ξ_2 affects the pEWs of the Si II lines and is correlated with the SALT2 stretch parameter X_1. The third parameter ξ_3 affects the ejecta velocities of the supernova.

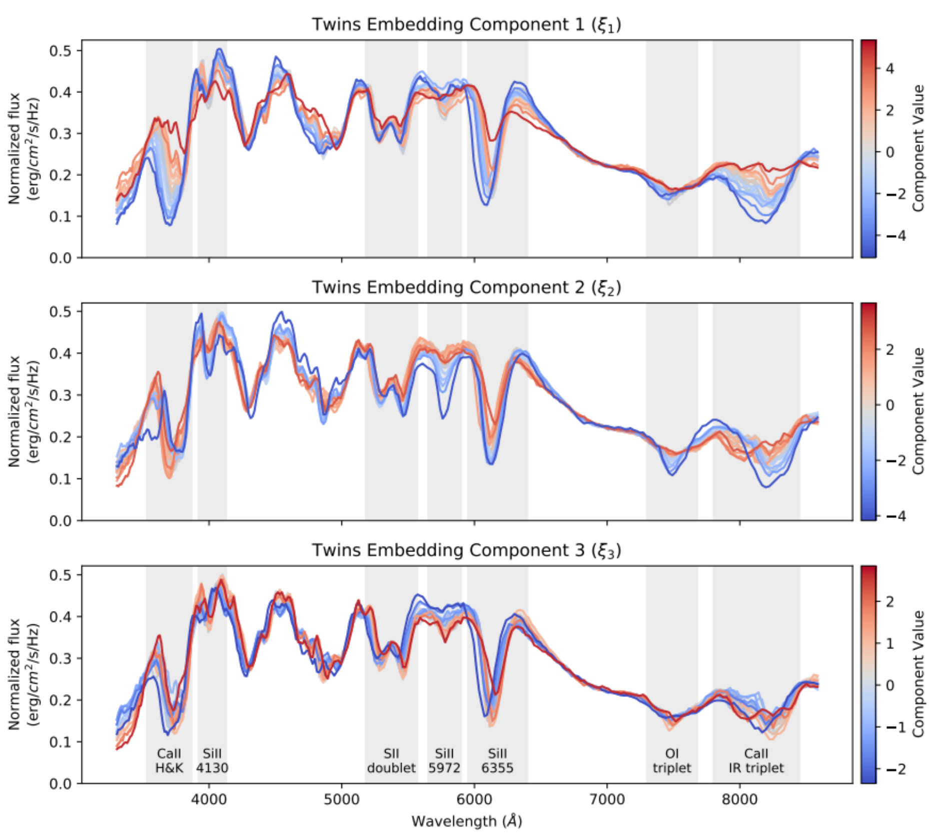


Fig. 2.19 Two examples of sets of supernova evenly spaced and considered as "Twins". Taken from [Boone et al., 2021].

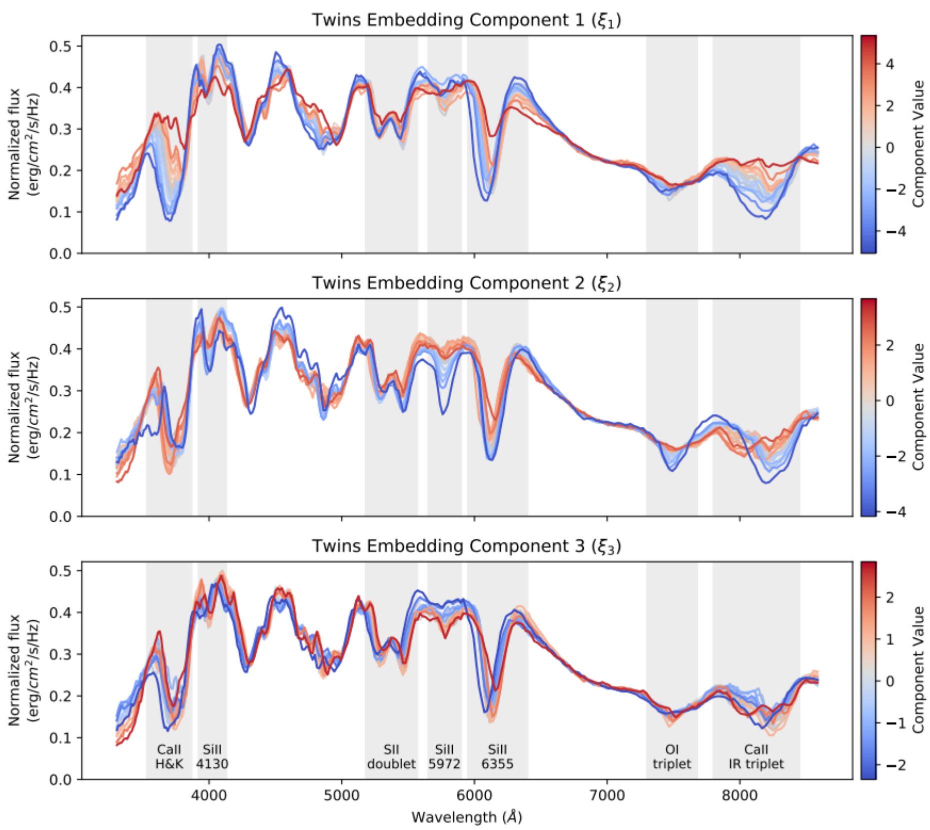


Fig. 2.20 Variations of the Twins Embedding model near maximum light when varying each one of its components ξ_1, ξ_2 and ξ_3. For each component, the spectrum shown is the median of 20 evenly populated bins of the coordinates of that component. Taken from [Boone et al., 2021].

6.2.4 SuPAErnova

The Super Probabilistic Autoencoder model (SuPAErnova) ([Stein et al., 2022]) uses a probabilistic autoencoder in order to assess the supernova diversity. The use of machine learning (ML) in this case leaves the model free to explore different ways to determine the necessary parameterization in order to describe the data. It Takes as input the spectra from SNfactory and encodes the parameterization of the model. The general definition of the model is:

$$F_{SN}(\lambda, p) = \mathbf{g}([z_1, ..., z_n], p) \times 10^{-0.4[CL(\lambda)\Delta A_V + \Delta M]} \tag{2.46}$$

where $\mathbf{g}$ is a decoder that learns to reconstruct the spectral series, z_i are the latent parameters of the model, CL is the color law, ΔA_V is the relative extrinsic extinction and ΔM the magnitude residuals. The results shown in the paper indicate that the with three latent parameters (z_1, z_2 and z_3) the reconstruction errors on the spectral series are minimal. The only setback is that it then becomes difficult to have a physical interpretation of the final parameters.

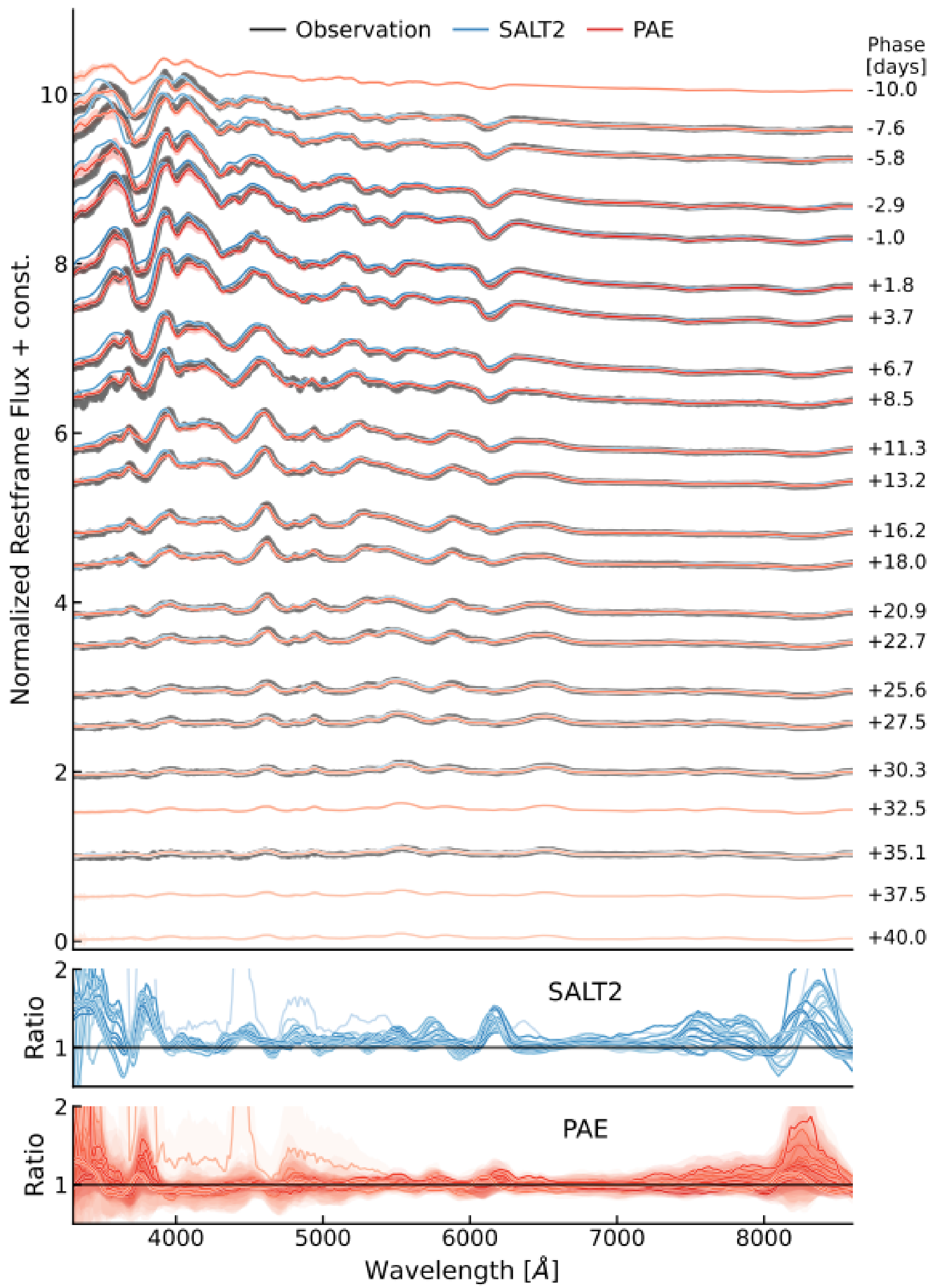


Fig. 2.21 Model comparisons from SuPAErnova (red) and SALT2 (blue) on the spectral series of the SN2010dt supernova observed by SNfactory (black) as well as ratios between model and data. Taken from [Stein et al., 2022]

7 Main goal of this PhD

We present here the overall goal of this PhD: the measurement of $f\sigma_8$ at $z \approx 0$ using low redshift supernovae and setup for a combined analysis using galaxy clustering. Two primary avenues in order to conduct such analysis were explored. The first involved the improvement of how we measure supernova distances. As we have mentioned before the measurements of supernova distances play a roll in multiple cosmological analyses (dark energy equation of state, H_0 and $f\sigma_8$), so one way we can improve the constraining power on the growth rate of structures is by improving our distance measurements. The second avenue was regarding the analysis method itself. Current measurements of $f\sigma_8$ using type Ia supernovae have been done using the Maximum likelihood method. This method falls short in reconstructing the non linearities in the velocity fields since it assumes a linear model. My goal will be to run a field level forwad model inference to extract the information on $f\sigma_8$ while accounting for non linearities. This method will also serve as a benchmark for a cross survey analysis which can be done naturally in this formalism.

CHAPTER 3

LEMAITRE project

Contents

1 LEMAITRE: Hubble diagram project overview and motivations

The LEMAITRE project aims at constructing the largest Hubble diagram by using an independent set of supernovae. One its objectives is to tackle the $(w_0 - w_a)$ tension observed by recent results from [Brout et al., 2022], [Rubin et al., 2023] and [DES Collaboration et al., 2024] supernova analyses as well as the combination with BAO results from [DESI Collaboration et al., 2025]. It is important to note that the low redshift supernova sample used in all three analyses is the same and that it seems to be the main motivating factor for the tension. By analyzing a completely new dataset for LEMAITRE, an independent assessment of this tension will be provided. Additionally with the LEMAITRE project comes a new analysis pipeline from end-to-end, starting from the lightcurve extraction and calibration up to the cosmological analysis. My work during the PhD involved the continued development and testing of a portion of the LEMAITRE pipeline named NaCl (see Chapter 3 and 4) which aims at training a spectrophotometric model.

2 Datasets: ZTF, SNLS, HSC

There are three datasets used in the LEMAITRE project: ZTF, SNLS and HSC. The three supernova datasets cover a large redshift from $z_{\mathrm{min}} \sim 0.02$ to $z_{\mathrm{max}} \sim 1.5$. They amount to roughly 3000 type Ia supernovae that can be used for a cosmological analysis most of which come from the ZTF survey at low redshift. All three surveys also observe the same portion of the sky which will allow for an improved handling of calibration systematics. In this section we present the three datasets in greater detail.

2.1 ZTF survey

We start off with the Zwicky Transient Facility survey as well as the the supernova sample that was primarily studied in this PhD the ZTF DR2. The ZTF camera is mounted on the Samuel Oschin 48-inch (1.2m) telescope in the Mount Palomar Obsrevatory, California, USA. The primary purpose of the survey is the study of nearby astrophysical transients, which makes it really good for studying supernovae in the local Universe.

Fig. 3.1 Mount Palomar Observatory. Credits: Caltech https://www.ztf.caltech.edu/

2.1.1 ZTF photometry

The ZTF camera is made up of 16 CCDs (model e2v CCD231-C6) each possessing 6144 $\times$ 6160 pixels. The Samuel Oschin telescope contains a 1.2m mirror (called the "primary mirror" M1) that redirects the photons to the focal plane of the camera. The mirror is a spherical pyrex-glass mirror coated with aluminium which allows for a large observation field of view. The telescope also contains an aspherical lens at the entrance to correct for geometric aberrations. It covers a solid angle of $\sim 47\text{deg}^2$ of the extra-galactical northern sky and manages to conduct a full sky coverage twice per night. The field of view can be seen in Figure 3.2 while the cadence is presented in Figure 3.3. Each image covers approximately 235 times the area of the full moon. The photometry is done using 3 filters g, r and i covering the ranges of $\approx 4000-5500$Å, $\approx 5500-7000$Å and $\approx 7000-9000$Å respectively. It is important to highlight that the transmissions of the filters are sensor-dependent and we display them by averaging over the sensor transmissions (Figure 3.4). The magnitude limits for each band are $m_g \sim 20.8-21.1$mag, $m_r \sim 20.6-20.9$mag, $m_i \sim 19.9-20.2$mag where the variations are laregly due to the phases of the moon. The full details on the ZTF photometry are in [Dekany et al., 2020].

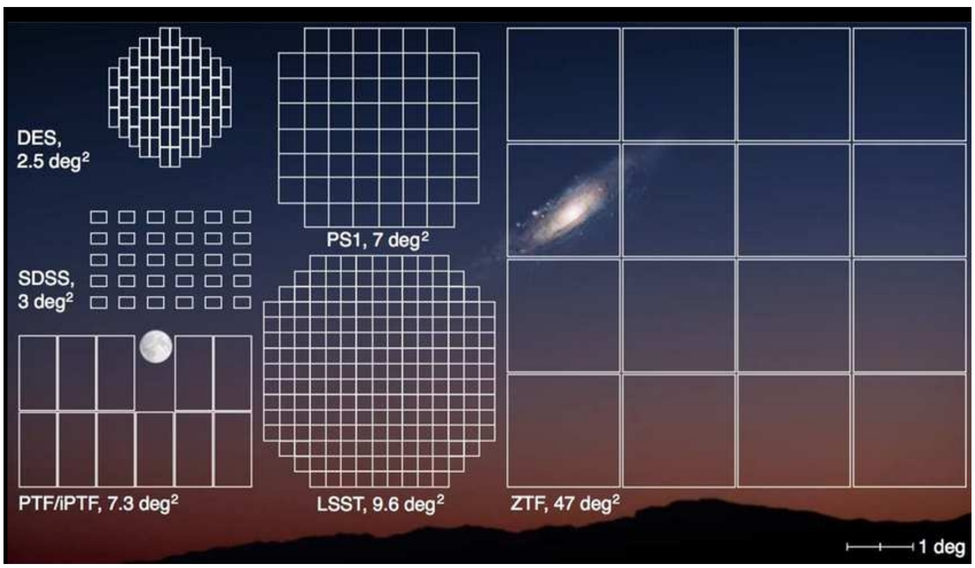


Fig. 3.2 Comparisons between the fields of view of ZTF, PS1, LSST, PTF, SDSS and DES. In the background we see in comparison the size of the moon and the Andromeda galaxy. Credits: Caltech `https://www.ztf.caltech.edu/`

2.1.2 Spectroscopic follow up

ZTF is accompagnied by a spectrograph named the Spectral Energy Distribution Machine (SEDM) located in the same observatory on the P60 telescope. The purpose of the object to conduct a follow up on the photometry in order to properly classify the transients obsereved. It's limit magnitude is 20.5mag. This follow up played an important role in acquiring the 60% spectra for the ZTF Bright Transient Survey (BTS) for which the magnitude limit was 18.5mag, a limited fraction of the ZTF total survey.

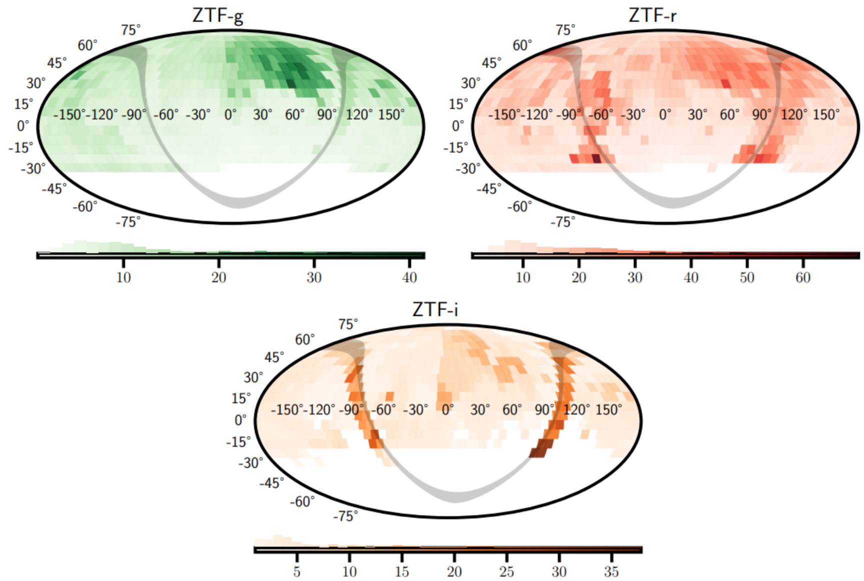


Fig. 3.3 ZTF cadence on the sky. Average number of visits per month for each field of ZTF in each band g, r and i. Taken from [Carreres, 2023]

2.1.3 ZTF DR2

We present in this subsection the type Ia supernova data release of ZTF named DR2. This encompasses data taken from March 2018 and December 2020. Figure 3.5 shows the image a supernova obsereved by ZTF. The sample contains 3628 spectroscopically identified supernovae up to a redshift of $z \approx 0.2$. Figure 3.6 shows examples of the sampling of the supernova obsereved by ZTF in DR2. We must note that not all supernova data is "good" enough to be used for a cosmological analysis. For such an analysis, quality cuts on sampling around the maximum emission of light, the color and the stretch must be applied. The details on the cosmology quality cuts can be found in [Rigault et al., 2020] and we remind them later in Chapter 4. After the cuts, the sample maintains 2667 supernovae that pass. An example of what the ZTF DR2 Hubble diagram looks like can be seen in Figure 3.7. A subset of the sample named the "volume-limited" sample contains all supernovae up to a redshift of $z = 0.06$ represents a complete sample of supernovae i.e. the sample is not influenced by any selection effects due to the magnitude limits of the telescope. This subset contains 994 supernovae.

2.2 SNLS survey

The second dataset is the SuperNova Legacy Survey (SNLS) (Astier et al., 2006), a type Ia supernova survey conducted with the MegaCam (Boulade et al., 2003) observing system. The camera is located on the Canada-France-Hawaii 3.6m telescope (CFHT) on top of the Maunakea volcano, Hawaii. The focal plane of the MegaCam camera is made up of 40 CCDs (model e2v CCD42-90) each having 2048 × 4612 pixels. The CFHT is constructed with two mirrors, the primary mirror has a parabolic shape and is made from a glass-ceramic material which makes it immune to thermal distortion. The camera is placed in the focal plane of the primary mirror. In addition, a Wide Field Corrector (WFC) is placed in front of the camera to guarantee a large field of view.

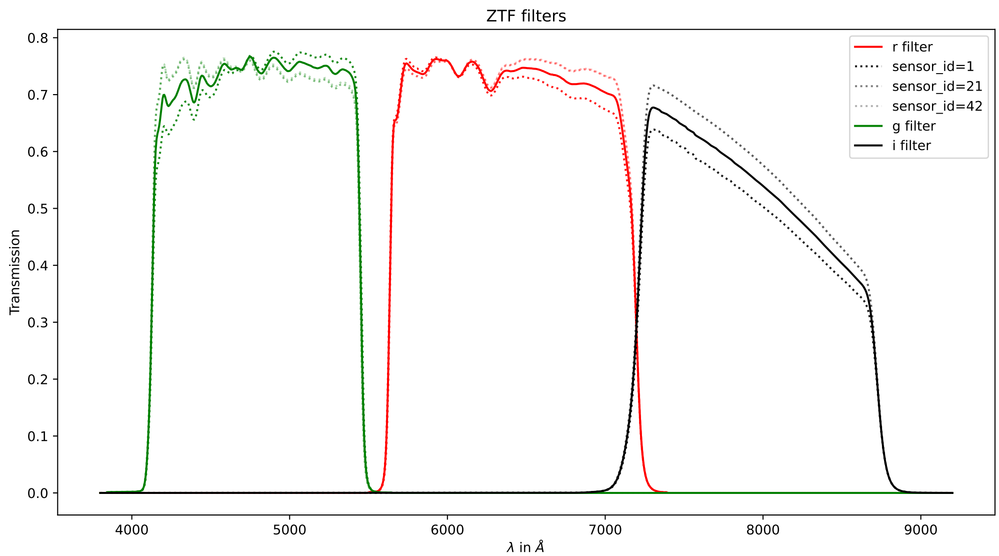


Fig. 3.4 ZTF filters. Dotted lines represent transmissions for different sensors. Solid lines represent the average transmissions over all sensors.

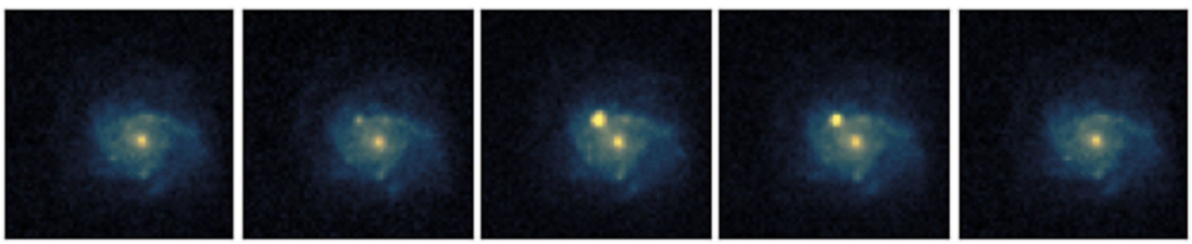


Fig. 3.5 Type Ia supernova detected by ZTF. Photos taken accross 2 months (from left to right 20th of Jan 2020, 25th of Jan 2020, 7th of Feb 2020, 20th of Feb 2020, 20th of March 2020). Credits: Mikaël Rigault.

The spectroscopic follow-up used to classify the supernovae and identify their redshifts are ensured by the European Southern Observatory Very Large Telescope (ESOVLT), the Gemini-North and South and the Keck-I and II instruments (Balland et al., 2009; Astier et al., 2006). The number of candidates for type Ia supernova obsereved photometrically exceed the number of events that can be followed up spectroscopically. This creates a need for a target selection process to try and eliminate transients that are not type Ia. The SNLS target selection is done by lightcurve fiting in real time to eliminate the majority of the type II supernovae. For the remaining candidates a limit magnitude is applied in the MegaCam i band at 24.5mag where supernovae are less likely to be spectroscopically obsereved.

The MegaCam footprint is a 0.90deg^2 rectangle, a much smaller footprint than ZTF that limits the survey from doing wide scans. Instead, deep observations are done in four fields equally distributed in right ascenstion up to a redshift of 1. (7ot figure). MegaCam contains 5 photometric filters used to observe transients: g, r, i, $i2$ (The i filter was broken during SNLS observation period and was replaced a similar filter) and z (figure). This covers the visible and near-infrared wavelength ranges. As seen in

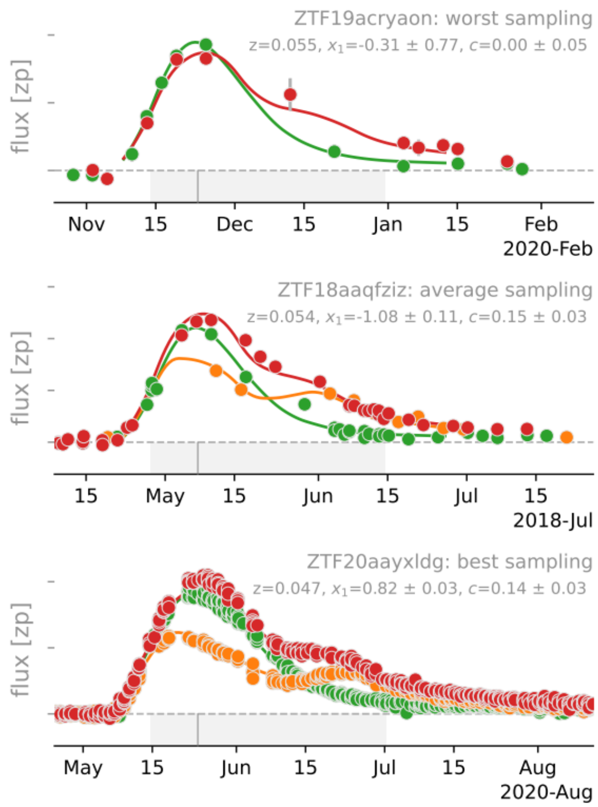


Fig. 3.6 Three supernova Ia lightcurves examples highlighting the worst (top), average (middle) and best (bottom) sampling in DR2. The colors indicate the band they were measured in and the solid lines are the best SALT2 fits. Taken from [Rigault et al., 2020]

Figure 3.4 for ZTF earlier, the sensor-dependent transmissions are also present in the MegaCam filters. The SNLS data used in the LEMAITRE project consists of five years of observation (SNLS 5y). The observations started in late 2003 and ended in 2009. The the 4th and 5th years have not yet been published and will be in (Betoule et al. in prep.). The number is expected to be around 300 supernovae from SNLS that will be used in LEMAITRE.

2.3 HSC survey

The final dataset comes from the Hyper Suprime-Cam Subaru Strategic Program (Aihara et al., 2022), also denoted HSCSSP. It is an astronomical observation program which dedicated some telescope time to the observation of type Ia supernovae at the end of 2010. It uses the Hyper Suprime Cam (HSC), a camera mounted on the Subaru 8.2m telescope. The Subaru telescope is also located at CFHT on top of the Maunakea volcano, Hawaii.

The redshifts of the HSC supernovae are determined by measuring the redshift of their host galaxies. They're collected by the AAT/AAOmega and DESI instruments and completed by observations from the Subaru/FOCAS and VLT instruments (Suzuki et al. in prep). The HSCSSP project also determines redshifts directly from photometry using several complementary methods. The Direct Empirical Photo-

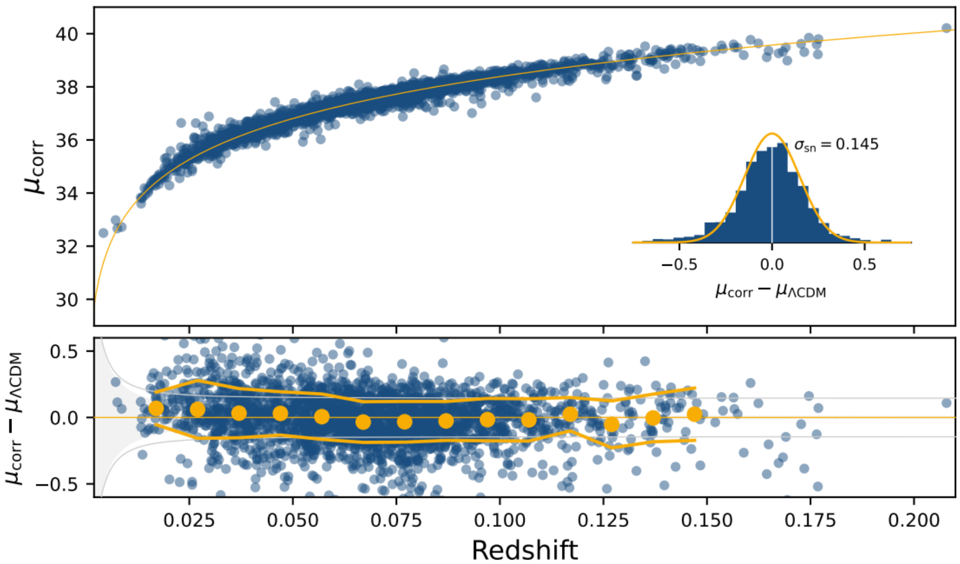


Fig. 3.7 Hubble diagram created with standardized ZTF DR2 type Ia supernovae compared to fiducial ΛCDM cosmology anchore on Planck results. Taken from [Rigault et al., 2020].

metric method (DEmP, [**?**]) and the TPZ method ([**?**]) both rely on machine learning to infer redshifts. DEmP does it by fitting a polynomial function relating redshift to magnitude, while TPZ is through random forest algorithms. Alternatively, the LePhare ([**?**, **?**]) and MIZUKI ([**?**]) methods adopt a template fitting approach with priors, where LePhare uses redshift distribution priors from the 30-band photometric redshifts of the COSMOS survey, while MIZUKI uses Bayesian priors.

The Hyper Suprime-Cam camera footprint is a 1.8deg^2 disc. The main focus of HSCSSP is to build a galaxy catalog for weak lensing analysis, however the observation cadence of its deep fields were adapted for supernovae observation. The observation strategy of the HSC supernova survey is restricted to four deep fields (XMM-LSS, E-COSMOS, ELAIS-N1, DEEP2-F3) and two ultra-deep fields (SXDSand COSMOS) partially overlapping with the SNLS D1 and D2 fields, which go up to a redshift of $z \sim 1.6$.

HSC observation are done in 7 photometric filters: g, r, $r2$, i, $i2$, z and y spanning the visible and near-infrared wavelength ranges. The transmissions are sensor-dependent like the ZTF and SNLS filters. The data used in the LEMAITRE porject consists of observations from a first session in 2017 up to observations from a second session in 2020. The number is expected to be around 300 supernovae from HSC that will be used in LEMAITRE.

3 Pipeline

3.1 Pipeline overview

3.2 Cosmological analyses with LEMAITRE

3.3 Uncertainties in SN measurements?

4 LEMAÎTRE Data Challenges

In this section we will be detailing the LEMAÎTRE Data Challenges, DCs from now on. The goal of these challenges is to validate the new analysis pipeline that is part of the LEMAÎTRE project, which is an extremely important key test before running the full analysis chain on the real data.

4.1 General description of the DCs

The LEMAÎTRE analysis pipeline is brand new from the point where the lightcurves are extracted from the ZTF, SNLS and HSC surveys up until comsological parameters are determined. The pipeline can be divided into 2 main parts : the lightcurve extraction and calibration pipeline, and the Georges pipline (or the cosmological analysis pipeline). Determining whether this new pipeline can accurately predict cosmological parameters without any biases is crucial and must be investigated thoroughly, especially since many of the elements of the pipeline depend on each other.
We will be focused here on the validity of the cosmological analysis pipeline, Georges. For that, the pipeline will face a series of challenges, each with an increasing level of complexity representing a challenge we are expected to face in the final analysis on the LEMAÎTRE dataset. Those challenges are the DCs and they are constituted of realistic simulations of ZTF, SNLS and HSC lightcurves and spectra. We will firstly describe the different elements of the Georges pipeline and then the different elements of the DCs. Each DC is constituted of 100 realizations of the simulations, essentially containing the exact same supernova but by applying the noise on the data differently. This will allow us to perform a robust coverage test for each DC.

4.2 Georges analysis pipeline

The Georges pipeline consists of mainly four modules : PeTS, NaCl, EDRIS, cosmologix.

4.2.1 PeTS & miniPeTS

PeTS

The PeTS module aims at "cleaning" the training dataset we will ll be using in the cosmological analysis. The reason we need to do that is because some of the supernova lightcurves can contain issues such as few points, very noisy data or a mix of both that can introduce issues when trying to accurately model the supernova with NaCl.

In order to avoid that, PeTS applies a set of data quality cuts that will ensure that we select a subset on which we will be able to properly measure all of the supernova parameters.
The selection will be applied on supernovae that meets the following criteria:

- The date of maximum emission (t_{max}) needs to be properly defined
- The color and the stretch parameters need to be accurately measured
- The supernova must contain at least 5 points with an SNR superior to 3, present in 2 bands and present within 50 days of the initial guess of t_{max}

In order to do that, PeTS initially fits all of the lightcurves with the SALT2.4 model using sncosmo ([Barbary et al., 2021]). **The sncosmo fit needs to converge in order to select a supernova in our final dataset**. That initial fit allows us to have an initial guess on the parameters c, X_1 and t_{max}.

The next thing is to create a log-likelihood profile while varying the value of t_{max}. The supernova is then selected if the following criteria are met on the parameter t_{max}:

- The uncertainty on t_{max} derived from the log-likelihood profile must be inferior to 1 day
- The log-likelihood profile must be symmetric. The condition chosen for that in PeTS is :

$$|3\sigma(t_{max})_{Left} - 3\sigma(t_{max})_{Right}| < 0.3$$

- The log likelihood profile must present only 1 minimum within 3σ

Lastly, the uncertainties on c and X_1 need to be inferior to 0.3 and 1.

If all of these conditions are met, the supernova will be present in the final dataset. For the DCs, we want to apply PeTS on only 1 realization since the quality cuts shouldn't vary between realizations. Running PeTS, however, presents a difficulty. Creating the log-likelihood profile for each supernova takes a lot of computational time. It would take us a really long to do so if we wanted to perform multiple tests on any DC. In order to avoid that and gain a bit of time, we introduced miniPeTS.

miniPeTS

With miniPeTS the goal is to achieve similar quality cuts to PeTS but much faster. This means that we don't expect to necessarily obtain the best quality dataset, but good enough to test the pipeline on the DCs. We apply similar cuts to the ZTF DR2 cuts (ref). We can summarize the quality cuts done with miniPeTS in the following list:

- The supernova needs to be measured in at least 2 bands
- The supernova needs to have at least 5 total measurements
- The supernova needs to have at least 2 measurement points before t_{max}
- The supernova needs to have at least 2 measurement points after t_{max}
- $-0.2 < c < 0.8$
- $|X_1| < 3$
- SNLS and HSC supernova need to be measured at $z > 0.2$
- $E(B-V)_{MW} < 0.25$

4.2.2 Data compression

Before we can start training a model with NaCl on the data, we decided to accelerate the training by compressing the data on the basis of the model.

Lightcurve compression

The way we compress the lightcurves data is by averaging the fluxes measured per night. We find that doing so reduces the size of lightcurves files by almost half.

Spectra compression

We have also noticed that the resolution of the spectra is much higher than the resolution of our model. The spatial dimension of our model in NaCl is defined by a B-spline basis of order 4. Projecting the spectra on that basis can be written as

$$\sum_i p_i \mathcal{B}_i(\lambda)$$

The idea is retain the p_i parameters alongside their uncertainties. The size of the spectral data is then reduced by a factor of 10.

4.2.3 NaCl

The next step in the Georges analysis pipeline is the model training with NaCl. The X_0, X_1 and c along with the marginalized covariance matrix will be the input for EDRIS in order to estimate the supernova distances.

4.2.4 EDRIS

The observed distance modulus of type Ia supernova contains a natural dispersion with respect the Hubble law of around ~ 0.4mag. This can be reduced by standardizing the distance moudli. We commonly use the Tripp relation (ref) which reduces this dispersion to ~ 0.15mag. It is expressed as follows :

$$\mu = m_B - (M_B - \alpha X_1 + \beta c) \tag{3.1}$$

$$m_B = -2.5 \log_{10}(X_0) \tag{3.2}$$

with m_B being the observed magnitude, M_B the absolute magnitude and (α, β) are the standardization parameters also known as the brighter slower and brighter bluer parameters. This is due to the fact that type Ia supernova with a larger stretch parameter tend to be brighter as well as supernova that are bluer.
The remaining 0.15mag dispersion are due an intrinsic variability between supernovae that isn't modeled by the Tripp relation.

Determining the standardized parameters can be difficult. Since telescopes can only observe objects up to a certain magnitude we expect a cutoff in magnitude for each survey. Coupled with the intrinsic dispersion of supernova, this means that the population of supernova observed by a survey will skewed, favoring redder and faster supernova. We call this **selection effects**. This makes a direct determination of (α, β) biased and will bias any cosmological analysis.

The goal of EDRIS is to accurately estimate unbiased standardized distances for complete and incomplete supernova Ia samples. EDRIS takes as input the parameters from NaCl alongside their covariance matrix, and will use that to infer standardization parameters (α, β), M_B, selection effects and the intrinsic dispersion denoted σ_{int}.

4.2.5 Cosmologix

The final step of the pipeline is to infer the cosmological parameters from the distances and we do that with cosmologix [1] (Betoule et al in review), a fast JAX-coded software designed to accurately infer cos-

[1] https://lemaitre.pages.in2p3.fr/cosmologix/home.html

mological parameters. It is also possible to combine the supernova results with other probes such as BAO and CMB. This will be useful in the analysis of the real data.

4.3 DC1

The first data challenge will be a consistency test for the full chain. It's the first time we've tested the full pipeline together on the same dataset. This will also serve as a stepping stone for the analysis of the future DCs since they will be incrementing layers of complexity on top of what was done in DC1. We will be including simulated lightcurves and spectra from ZTF and SNLS as well as lightcurves from HSC.

The simulated data will include measurement uncertainties, an error snake and an intrinsic dispersion. We will not be including any selection effects so EDRIS will need to find the standardization parameters. The main challenge will be for NaCl to ensure that training the model with our current model constraints and regularization on a LEMAÎTRE like dataset does not bias our cosmological analysis.

The metric used to determine if we have passed the DC or not will be to check the coverage for the cosmological parameters after 100 realizations and ensuring it stays within 1σ.

4.4 DC2

The second data challenge serves as a test to the model selection effects in EDRIS. The simulations will be identical to DC1 with the addition of a selection function for each survey.

The metric used to determine if we have passed the DC or not will be the same as DC1.

4.5 DC3

In the third data challenge, calibration uncertainties and supernova outliers will be added the simulation. This serves as a main test to NaCl's error model to see how well it can identify outliers as well as determining the impact of calibration uncertainties on the reconstructed cosmology.

The metric used to determine if we have passed the DC or not will be the same as DC1 and DC2.

4.6 DC4

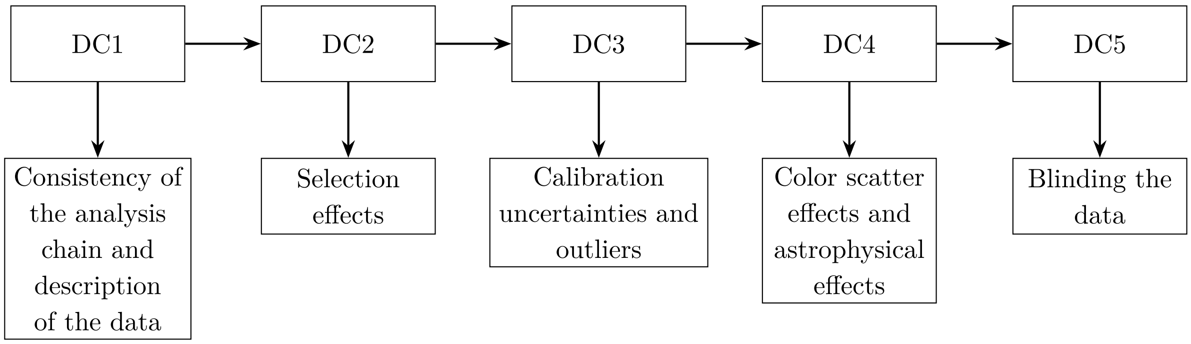
In the fourth data challenge, the simulations will include as many realistic effects as possible adding all sources of uncertainties and systematics that we know can affect our analysis. This includes realistic distributions of the stretch and color (ref madeleine), non type Ia contamination, astrophysical effects and color scatter effects. This will be a challenge for both NaCl and EDRIS.

The metric we will use here to determine if we have passed the DC or not will be the same as before. Since astrophysical effects aren't directly taken into account during the modeling of type Ia supernova, we will quantifying its effects on the reconstructed cosmology and see how we can minimize it.

4.7 DC5

In the fifth and final DC we will be placing ourselves in the same conditions of the realistic simulations by blinding the data. The party that will be blinding the data will not be the same as the one running the pipeline simulation. The goal is to obtain the correct cosmological parameters after unblinding the data.

We can summarize the DCs in the following diagram :



CHAPTER 4

NaCl

Contents

1 NaCl framework

The first part of my work for this thesis was the continuation of the development of a new framework designed to train spectrophotometric models of type Ia supernova called NaCl ("Nouveaux Algorithmes de Courbes de Lumière") which was initially started by G. Augarde during his PhD, defended in September of 2022 ([Augarde, 2022]).

1.1 The idea behind the framework

The framework is meant to be able to easily train a spectrophotometric model all while being able to accurately propagate all known sources of uncertainties. It is meant to be easy to use and manipulate in order to account for any new developments in how spectrophotmetric models are parameterized. The idea is to then have a framework that is fast and capable of conducting model training easily every time one can do so on a cosmological dataset of type Ia supernova. We will see how this will allow us to overcome issues previous spectrophotometric models have encountered, such as calibration issues. The entirety of the framework has been coded in Python3.

1.2 Differences with the SALT models

This parameterization, like SALT2, is a classic principal component analysis, or PCA, with the addition of a color law. The main differences with SALT models is in the training procedure. NaCl training and lightcurve fitting is done simultaneously on the same dataset used for the cosmological analysis. This reduces the model systematics obtained from SALT models due to the change of datasets when conducting a cosmological analysis. It also means we can directly obtain a covariance matrix containing the correlations between supernova lightcurve parameters and also model parameters. This directly affects the distance modulus.

In NaCl the date of maximum emission t_{max} is fitted during the training of NaCl. This is a big difference from what is done for the SALT models where t_{max} is fitted beforehand. The reason why in NaCl we do that is to account for the possible changes of t_{max} that may arise due to the changes in the spectrophotmetric model during the training.

Calibration uncertainties are modeled as nuisance parameters in NaCl that are fitted during the training, held by a covariance matrix allowing us to account for the correlations between filters. This also ensure we have an exact propagation of the calibration systematics in our training and prevents us from doing multiple trainings.

1.2.1 Practical implementation of the SALT2 model

In order to be able to determine the global model surfaces (M_0 and M_1), we will be describing them with the help of B-splines. This is a particular set of polynomials of order $m-1$, called splines of degree m, that can form a basis. Their smoothness will allow our model surfaces to have complex forms. We use splines of degree 4 in NaCl. M_0 and M_1 are then defined as:

$$M_{0|1} = \sum_{k=1}^{n_p} \sum_{l=1}^{n_\lambda} \theta_{kl}^{[0|1]} \mathcal{B}_k(p) \mathcal{B}_l(\lambda) \tag{4.1}$$

where $\mathcal{B}$ are the splines, θ are the model parameters that we will determine. Similarly, the evaluation of the filters is also done on the same spline basis of the model. For the color law $CL(\lambda)$ it is implemented as a degree 5 polynomial.

Since we do not determine the magnitude of the supernova on the spectra, we add an extra polynomial that for each spectra in our dataset. The polynomial $SpecRecal(\lambda)$ can be then added to the spectroscopic element of the model such that:

$$\phi_{spec}(\lambda, p) = \frac{SpecRecal(\lambda)}{1+z}\mathcal{S}(\frac{\lambda}{1+z}, \frac{p}{1+z}) \tag{4.2}$$

This entails that there will be a degeneracy between the color law CL and the recalibration polynomials, therefore during the fitting of the spectra we do not determine CL (essentially it is not treated as a parameter that can influence the overall shape of the spectral features of the model).

The model parameters are then determined by minimizing a log-likelihood which takes the form of:

$$-2\ln(\mathcal{L}) = N\ln(2\pi) + \ln(|\mathbf{V}|) + \mathbf{R}^T\mathbf{V}^{-1}\mathbf{R} \tag{4.3}$$

where $\mathbf{R}$ are the model residuals and $\mathbf{V}$ the covariance matrix independent of the parameters. If $\mathbf{V}$ contains only measurement uncertainties, the problem then becomes minimizing the following quantity:

$$\chi^2 = \mathbf{R}^T\mathbf{V}^{-1}\mathbf{R} \tag{4.4}$$

which will be the case for us during this PhD.

The minimization of the log-likelihood is via the Fisher scoring algorithm. This method allows us to obtain the desired model parameters that best describe the data using a second order descent of χ^2 (slope and curvature information). If we denote $\mathbf{M}$ the model, β the model parameters, $\mathbf{J}$ the model Jacobian and $\mathbf{H}_\chi$ the Hessian (second order derivatives matrix) of the χ^2 then it possible to show using the Fisher scoring algorithm that for a set of parameters β_i at an instance i to be updated at an instance $i+1$ with $\beta_{i+1} = \beta_i + \delta\beta$ we have:

$$\delta\beta = -(\mathbb{E}[\mathbf{H}_\chi])^{-1}(\beta_i)\nabla\chi^2(\beta_i) \tag{4.5}$$

where $\mathbb{E}$ is the expected value. It is possible to relate the the χ^2 Hessian with the model Hessian in order to simplify this expression. Will be omitting this demonstration but we get $\mathbb{E}[\mathbf{H}_\chi] = 2\mathbf{J}^T\mathbf{V}^{-1}\mathbf{J}$ for an unbiased model and this relation is true at the minimum value of the log-likelihood. Additionally, the gradient of χ^2 can be written as $\nabla\chi^2 = -2\mathbf{J}^T\mathbf{V}^{-1}\mathbf{R}$ where $\mathbf{R}$ is the model residuals vector. Finally, we get the following expression for $\delta\beta$:

$$\delta\beta = \left(\mathbf{J}^T\mathbf{V}^{-1}\mathbf{J}\right)^{-1}\mathbf{J}^T\mathbf{V}^{-1}\mathbf{R} \tag{4.6}$$

This algorithm requires us to invert the Hessian matrix $\mathbf{H} = 2\mathbf{J}^T\mathbf{V}^{-1}\mathbf{J}$ at each step, which is always positive definite by construction, allowing us to use efficient factorisation methods such as the Cholesky decomposition. In practice, minor degeneracies are handled by augmenting the diagonal of the Hessian with a small damping term following the Levenberg-Marquardt approach, and by enforcing parameter constraints through Lagrange multipliers, both of which improve the numerical stability of the inversion.

1.3 Constraints

Since this model is built empirically, we can see that there will be a lot of degeneracies between the individual and model parameters. To break those degeneracies we will be adding a set of constraints.

When NaCl was first developed, the constraints used were the same as the ones for SALT2 and SALT3 ([Kenworthy et al., 2021]) models. Although those constraints break the degeneracies that exist between parameters, we've decided to include a new set of constraints that would address two things : the first that one of the constraints is non linear which made the model training take a longer time, the other was that some constraints depended on the supernovae used in the training sample. An example such a constraint was $< X_1 >= 0$ which defined the average stretch of a supernovae. This is isn't an issue if we are comparing the stretch of supernovae that were fitted using the same model. However since NaCl will be used to train different models on different datasets, it'll be more difficult to directly compare the parameters of the supernovae.

During this PhD, the following sets of constraints were developed. I will be detailing what they do and why they were developed.

Model normalization:

The model normalization is degenerated with the value of X_0. We impose that :

$$\int M_0(\lambda, p=0)\frac{\lambda}{hc}B(\lambda)d\lambda = 1 \tag{4.7}$$

$$\int M_1(\lambda, p=0)\frac{\lambda}{hc}B(\lambda)d\lambda = 0 \tag{4.8}$$

where $B(\lambda)$ is the restframe standard B-band.

Date of maximum emission:

There exists a degeneracy between the model and the date of maximum emission t_{max}. It's possible to shift the entire model in order to compensate for a change in the date t_{max}. For that we impose that at $p=0$, the model presents a maxima in the standard B-band :

$$\frac{d}{dp}\int M_0(\lambda, p=0)\frac{\lambda}{hc}B(\lambda)d\lambda\Big|_{p=0} = 0 \tag{4.9}$$

$$\frac{d}{dp}\int M_1(\lambda, p=0)\frac{\lambda}{hc}B(\lambda)d\lambda\Big|_{p=0} = 0 \tag{4.10}$$

Color Law:

We also need to account for the degeneracy that exists between the spectral shape of the model and CL. Tilting the M_0 and M_1 can be completely corrected for by changing CL. We impose then that $CL(B) = 0$ and $CL(V) = 1$ with B and V being the central wavelengths of the restframe standard B-band and V-band.

Average color:

Another degeneracy exists between the individual supernova color c and the M_0. Changing the spectral features of the model can be completely compensated by the value of c. For that we need to define the "average" supernova color. We impose that the the M_0 surface would have its $B-V$ color equal to 0 at peak luminosity and that c accounts for the relative variations in color with respect to that. The constraint now becomes :

$$\int [M_0(\lambda, p=0)B(\lambda) - M_0(\lambda, p=0)V(\lambda)] \frac{\lambda}{hc} d\lambda = 0 \tag{4.11}$$

Average stretch:

A similar degeneracy also exists with X_1. A global change of the X_1's can be absorbed by changing the shape of the M_0 surface along the phase axis. We have decided to fix the value of $X_1 = 1$ such that it would be equivalent to having the same X_1 value using SALT2. For that we fix the difference between the model evaluated for $X_1 = 0$ and $X_1 = 1$ at a phase of 15 :

$$\int [M_0(\lambda, p=15) + M_1(\lambda, p=15)] \frac{\lambda}{hc} B(\lambda) d\lambda - \alpha \int M_0(\lambda, p=15) \frac{\lambda}{hc} B(\lambda) d\lambda = 0 \tag{4.12}$$

with $\alpha = 10^{-0.4 \times 0.162}$ the flux difference we have found in SALT2.

Average width:

The last unwanted degree of freedom is regarding the width of the model. We can see that the definition of X_1 relies on a difference between the model with different stretch parameters but it doesn't define what the width of the model is. We can always make the model larger or smaller while maintaining the constraints on all of the other parameters. One way to bypass that issue is to constrain the model at a certain phase. We choose to fix the model at a phase $p = 15$, also known as Δm_{15} as :

$$\int M_0(\lambda, p=15) \frac{\lambda}{hc} d\lambda = 10^{-0.4} \tag{4.13}$$

In practice, we write down the constraints as a function $C_k(\beta) = \alpha_k$ where $C_k(\beta)$ is a function evaluating the k-th constraint of value α_k. This is then added as penalty that takes the form $\mu_{cons} \sum_k (C_k(\beta) - \alpha_k)^2$ where μ_{cons} is hyperparameter that determines the strength of the penalty. We can write this in a matrix $\mathbf{C}(\beta)$ and we obtain :

$$\boxed{\chi^2_{cons} = \mu_{cons} \mathbf{C}(\beta)^T \mathbf{C}(\beta)} \tag{4.14}$$

1.4 Regularization

Because our training dataset contains a finite number of data points, it's impossible that the entirety of the global parameters would be constrained by data. To deal with that we add regularization to the model during the fit, which is a penalty added to the model in the areas where we would have extremely large variations due to poor constraining power of the data. This is specifically really important for NaCl since we are fitting the date of maximum emission there are data points that could go outside of the model range.

The regularization penalty that was initially developed was expressed as :

$$\mu_{reg} \sum_{kl} \left(\theta_{k,l}^2 + \left(\theta_{k+1,l} - \theta_{k,l} \right)^2 \right) \tag{4.15}$$

where θ_{kl} are the model parameters as seen in equation 4.1. The first sum is a penalty that makes the θ's tend to zero when no data is available. The second term is a penalty of the phase derivatives of the surfaces that limits the sharp variations of the model observed when no data is available, or at poorly constrained regions of the model. Finally, μ_{reg} is a hyperparameter that determines the strength of the penalty. Writing this in a matrix form gives us :

$$\chi^2_{reg} = \mu_{reg} \beta^T \mathbf{P} \beta \tag{4.16}$$

where $\mathbf{P}$ is what we call the regularization matrix, that allows us to obtain the expression in equation 4.15, and β our set of parameters.

1.5 Error models

Dealing with an empirical model means that we expect that our model would not perfectly describe the data. More specifically, the residuals of the model would not be explained solely by the measurement uncertainties on the data. This is a variability that exists between supernovae, unable to be captured by the SALT2 parameterization.
It's also important to note that all of the photometric measurements are affected by uncertainties of the photometric instruments. This can play a crucial role in our measurements of cosmological parameters as it has been shown in the most recent analyses to be one of the leading systematics ([Rubin et al., 2024], [Vincenzi et al., 2024]).
We can account for both the supernova variability and calibration uncertainties with the help of error models.

1.5.1 Error Snake

The first error model is the Error Snake, which is supposed to capture errors directly due to variability in the flux. If we call V the variance of the data points, then the error model is implemented as :

$$V = \sigma^2_{meas} + \sigma^2_{snake} \tag{4.17}$$

where σ_{meas} are the measurement uncertainties and σ_{snake} is the additional variance added that is our Error snake model.

The general expression of this term is :

$$\sigma_{snake} = G(\lambda, p) \times \phi(\lambda, p) \tag{4.18}$$

where ϕ is the concatenation of both the measured photometric and spectroscopic fluxes used in our training dataset. $G(\lambda, p)$ can take multiple forms depending on how we would like to model our Error snake.

The simplest way to model it would be as a constant. The expression becomes :

$$G(\lambda,p) = g \tag{4.19}$$

Physically this means that the variability of the supernovae in our dataset are independent of phase and wavelength.

Alternatively, we can imagine something more complex by modeling G as a surface like M_0 and M_1 that is to be determined during the training. The expression then becomes:

$$G(\lambda,p) = g(\lambda,p) \tag{4.20}$$

Here, the variability does depend on phase and wavelength.

Lastly, we thought of adding a bit more complexity in our Error snake since we know that it's possible that our dataset could be contaminated by very peculiar supernovae or by objects that are hard to distinguish from type Ia supernova such as type Ib and type Ic supernovae or . We decided to also include an extra term in G that takes up one value for each supernova in our dataset, we call it γ_{SN}. We expect that the γ_{SN} value of a peculiar supernova would be larger than the others. The expression then becomes :

$$G(\lambda,p) = \gamma_{SN} \times g(\lambda,p) \tag{4.21}$$

which is what we would like to use in the final analysis of the LEMAÎTRE dataset. However it is important to note that for the data challenges, specifically DC1 (see Chapter 4), we will be using the simplest form of the Error snake.

1.5.2 Color Scatter

COMMENT : Waiting for the final implementation

1.5.3 Calibration Uncertainties

The calibration uncertainties are taken into account directly in the model. One parameter η is added for each transmission band in our dataset held by a prior that is known beforehand. For the LEMAÎTRE project we have 12 bands, so we'll have 12 η parameters. The expression of the photometric model becomes :

$$\phi_{phot}(\lambda,p) = X_0(1+\eta)(1+z)\int S(\lambda,p)T(\lambda)\frac{\lambda}{hc}d\lambda \tag{4.22}$$

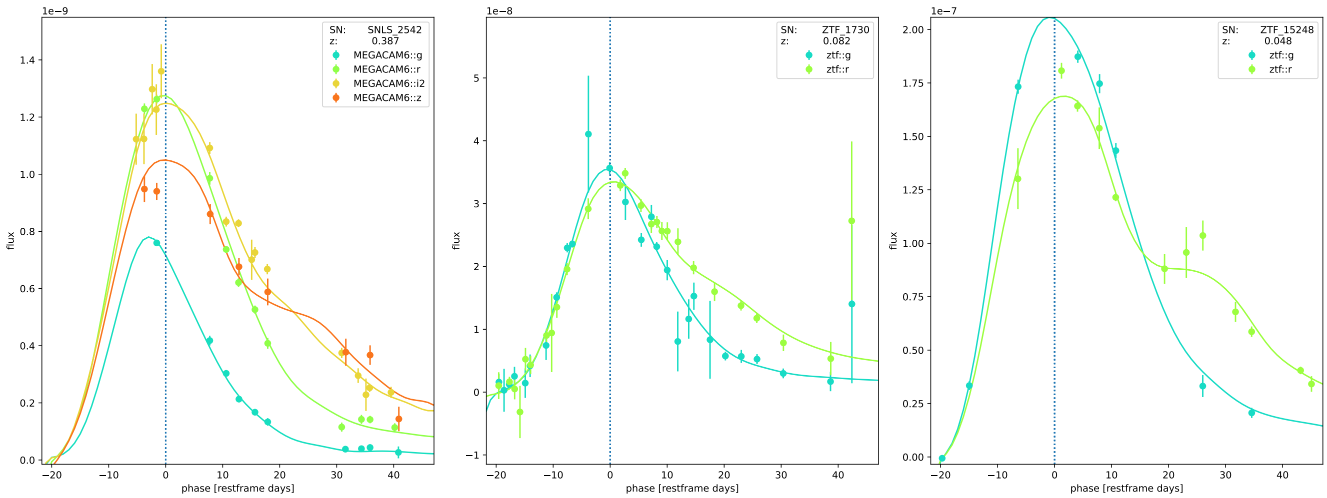
With an additional penalty to the model :

$$\chi^2_{calib} = \eta^T {\mathbf{V}_\eta}^{-1}\eta \tag{4.23}$$

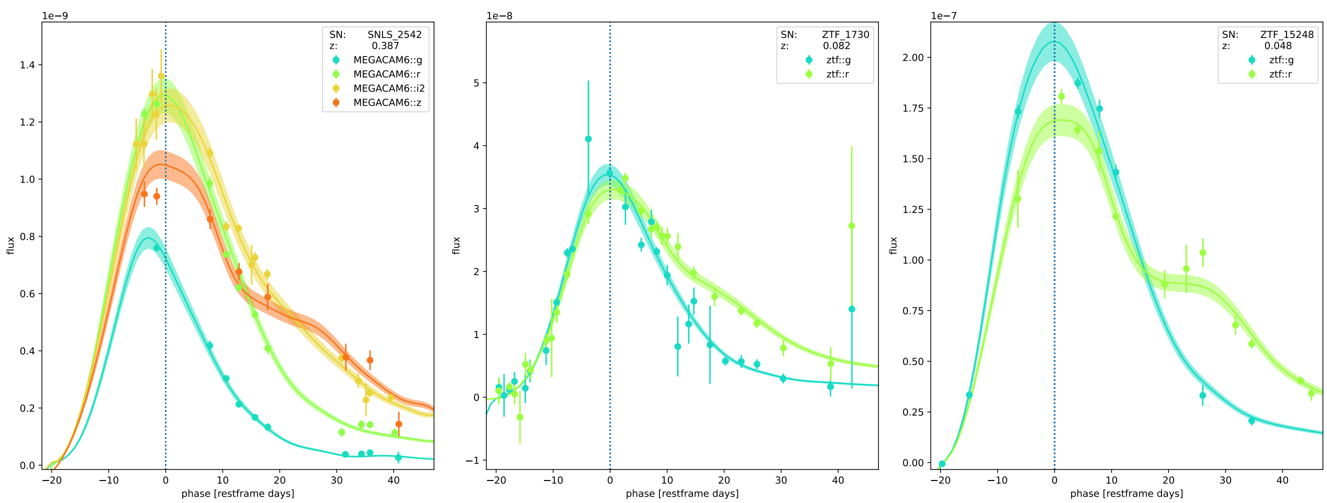
$\mathbf{V}_\eta$ being the η covariance matrix that is already known.

1.6 NaCl output

At the end of a training, we get a vector containing all of the model and individual supernova parameters and the full covariance matrix calculated from the Hessian of the model : $\mathbf{V} = 2\mathbf{H}^{-1}$. For the cosmological analysis, we will be marginalizing over all but the X_0, X_1 and c parameters.



(a)



(b)

Fig. 4.1 Visual representation of the error snake on simulated lightcurves from DC1 (see section (reference)). In 4.1a we see how the model represents the data without an error snake model. In 4.1b we see a highlighted section for each lightcurves which are the variations modeled by the error snake.

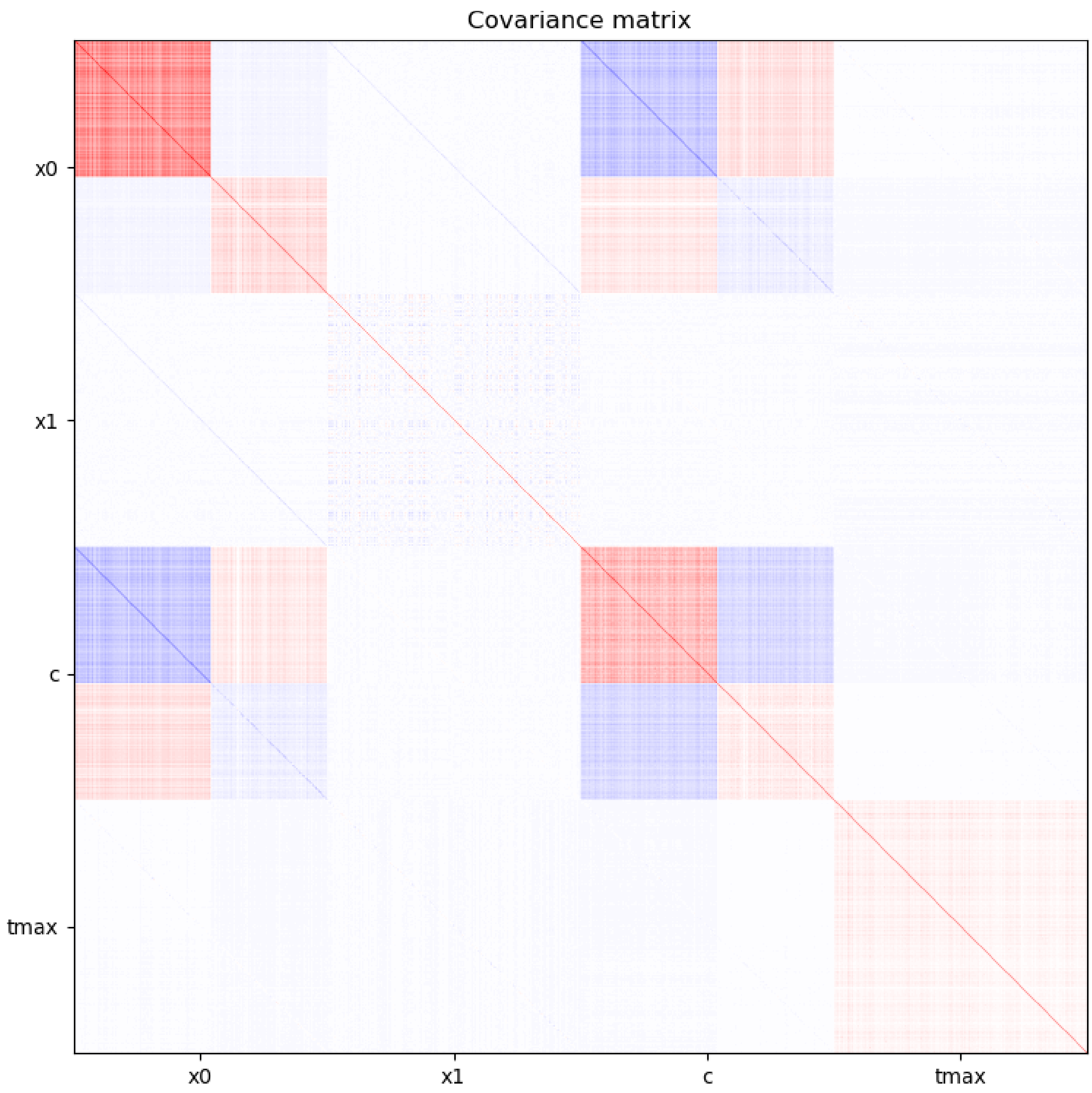


Fig. 4.2 Example of the NaCl covariance matrix after an output, marginalized over all but the X_0, X_1 and c parameters.

2 Improving NaCl

The foundational framework of `NaCl` was established by Guy Augarde ([Augarde, 2022]), whose work defined its structure and validated it on LSST-like simulations. However, several improvements were identified as necessary to handle realistic data. As part of my PhD, I contributed to developing these improvements, which are presented in the following sections.

2.1 Implementing a model for spectrophotometric data

Training a spectrophotometric model is done on two types of dataset. Classical models used in cosmological analyses like SALT2 or SALT3 rely on photometric and spectroscopic data obtained separately. Models that aim at refinined the standardization scheme of supernovae rely on spectrophotometric data like the SNfactory data ([Aldering et al., 2002]). The SNIFS instrument used for SNfactory allows for us to obtain a series of well calibrated spectral sequences for each observed supernova making the need for separate photometric data obselete. Being able to use such datasets in NaCl will then require us to add supplementary information in the model to account for that.

If we recall in Chapter 2, the SALT parameterization is expressed as:

$$\mathcal{S}(\lambda, p) = [M_0(\lambda, p) + X_1 M_1(\lambda, p)] 10^{0.4cCL(\lambda)} \tag{4.24}$$

used to describe the following spectroscopic and photometric fluxes:

$$\phi_{spec}(\lambda, p) = \frac{1}{1+z} \mathcal{S}(\frac{\lambda}{1+z}, \frac{p}{1+z}) \tag{4.25}$$

$$\phi_{phot}(\lambda, p) = X_0(1+z) \int \mathcal{S}(\lambda, \frac{p}{1+z}) \frac{\lambda}{hc} T((1+z)\lambda) \mathrm{d}\lambda \tag{4.26}$$

For spectrophotometric data, the goal is to then describe one data-type that follows:

$$\phi_{specphot}(\lambda, p) = X_0 \frac{1}{1+z} \mathcal{S}(\frac{\lambda}{1+z}, \frac{p}{1+z}) \tag{4.27}$$

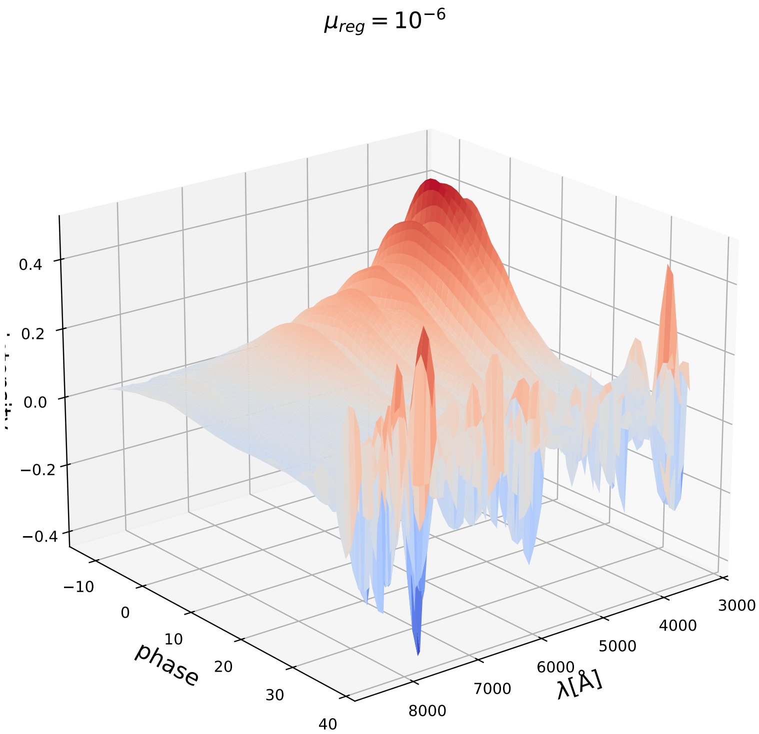
thus removing the need for a recalibration polynomial *SpecRecal* and reintroducing the color law CL which will determined on the spectra. My work was to code this model in NaCl which lead to the first use of NaCl on a real dataset where the results are presented in Section 3.

2.2 Regularization

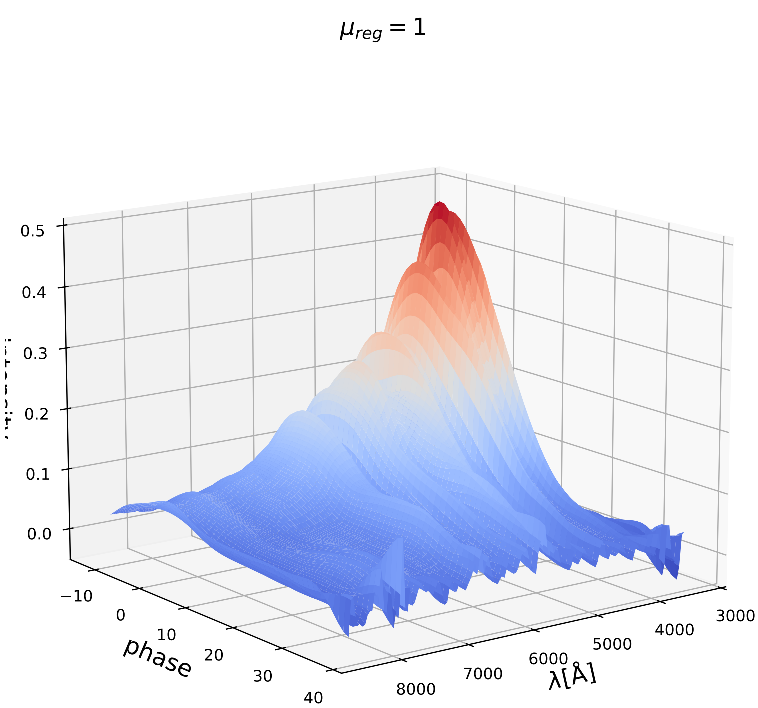
An issue that we've had with the previously mentioned form of regularization is that it is very sensitive to the value of μ_{reg}. If it's too small the model continues to present very strong variations. If it's too strong the model becomes attenuated and this biases our parameters, which means the cosmological parameters calculated with the output of NaCl would be biased. We show the effects of the strength of the regularization on the model in Figure 4.3

To try and overcome this issue, I've developed during this PhD a regularization scheme that adapts the force of the regularization based on the number of points seen by each model parameter. This regularization scheme is inspired by [Lenz et al., 2022]. We can separate $\mathbf{P}$ matrix as follows:

$$\mathbf{P} = \mathbf{P}^{data} + \mathbf{P}^{smooth} \tag{4.28}$$



(a)



(b)

Fig. 4.3 Effects of the strength of the regularization on the shape of the M_0. In 4.3a we see how the model behaves with a $\mu_{reg} = 10^{-6}$. In 4.3b we see how the model behaves with a $\mu_{reg} = 1$.

with $\mathbf{P}^{data}$ and $\mathbf{P}^{smooth}$ the matrices determining the strength based on the number of points and to smoothen the model where there is no data as much as possible. In order to determine $\mathbf{P}^{data}$, we define $\mathcal{N}_i$ as the number of data points seen by each θ_i parameters (θ_{kl} written as a vector). We then introduce for each $\mathcal{N}_i$ a parameter μ_i which will act as a gate function of $\mathcal{N}_i$. This will determine the force of our regularization. The idea is that if $\mathcal{N}_i = 0$ we regularize model strongly, otherwise we don't.

COMMENT : This specific version of the regularization wasn't implemented in the final version of NaCl since we still had to test other things in between. I will be rewriting this part of the manuscript again.

μ_i is then defined as:

$$\mu_i^{data} = \max(\mu_{reg}^{data} - \mathcal{N}_i, 0) \tag{4.29}$$

We can use that to define a diagonal matrix $\mathbf{M}^{data}$ where $\mathbf{M}_{ii}^{data} = \mu_i^{data}$. It is then clear that the simplest way to define $\mathbf{P}^{data}$ will be to write it as:

$$\mathbf{P}^{data} = \mathbf{M}^{data} \times \mathbb{1} \tag{4.30}$$

We then define $\mathbf{P}^{smooth}$ as a penalty on the first order derivative of the model in phase, like in equation 4.15. We apply the same adaptive method mentioned previously and we get:

$$\mathbf{P}^{smooth} = \mathbf{M}^{smooth} \times \mathbb{P} \tag{4.31}$$

with $\mathbb{P}$ being :

$$\mathbb{P} = \begin{bmatrix} 2 & -1 & 0 & & & \dots \\ -1 & 1 & -1 & 0 & \dots & \\ 0 & & & \dots & & \\ & & \dots & & & 0 \\ & \dots & 0 & -1 & 1 & -1 \\ \dots & & & 0 & -1 & 2 \end{bmatrix} \tag{4.32}$$

with μ_{reg}^{smooth} would be the hyperparameter determining the strength of the regularization on the derivatives.

Since the hyperparameters are now included in the $\mathbf{P}$ matrix, the penalty now becomes :

$$\boxed{\chi_{reg}^2 = \beta^T \mathbf{P} \beta} \tag{4.33}$$

2.3 NaCl training recipe

During the tests of NaCl on simulations of LEMAITRE-like supernova, we have noticed that fitting all of the parameters can cause issues. If the model parameters are not close to the truth we found cases where the error model would dominate. This is not problematic for the minimizer as it sees this as a way to describe the data but physically none of the parameters are being fitted. Another issue encountered was the possibility of converging the minimizer at a local minima rather than a global minima. We've developed during the PhD a training recipe that allows for having good initial guesses of the supernova and model parameters. This helps accelerate the fit and makes it more stable. The current recipe is the following:

1. Lightcruve fit with the initial NaCl model. In this part we are only fitting X_0, X_1, c and t_{max} for all supernova lightcurves.

2. *SpecRecal* fit with the initial NaCl model. Here we are estimating the recalibration polynomial coefficients. This part of the fit is near instantaneous due to the linearity of the polynomial.

3. Initial model guess. In this part we aim to obtain relatively close initial guesses for M_0, M_1 and CL to the truth. This also ensures that the model respects the constraints.

4. Initial error model guess. Here we have an initial guess for the error model for a fixed model.

5. Full training. All parameters are fitted.

The first two steps of the recipe do not take a lof time to complete. The Initial model guess step can take some time to converge due to large number of parameters of the M_0 and M_1 surface, however the number of steps is limited to 15 in that step since the goal is to simply get an inital guess. The error model fit is also near instantaneous. Finally, freeing all of the parameters in step 5 takes the most amount of time.

3 NaCl on real data

There have been models that were empirically constructed in similar ways as the SALT models with SNfactory data, namely SNEMO ([Saunders et al., 2018]), SUGAR ([Léget et al., 2020]), Twins Embedding ([Boone et al., 2021]) and SuPAErnova ([Stein et al., 2022]) as presented earlier. These models usually incorporate different parameterizations than SALT2 in order to try and fully understand the spectrophotometric evolution of supernovae. One thing to note is that it is easy to implement different models in NaCl. So one could modify the current implementation of the SALT2 model and add extra parameters and test if it improves the description of supernovae.
For now, we present the results obtained when training NaCl on the SNfactory data with a SALT2 parameterization.

3.1 NaCl training results

3.1.1 Data coverage

Training a SALT2-like model with NaCl on the SNfactory was the first time NaCl was used on real data and was part of the PhD to test the framework. The SNfactory dataset used here contains 224 supernova and 3135 spectra. We can see in the coverage of the data on the model's $\lambda - p$ space in Figure 4.4 that we have a very rich dataset that covers a very large part of the model's domain of definition.

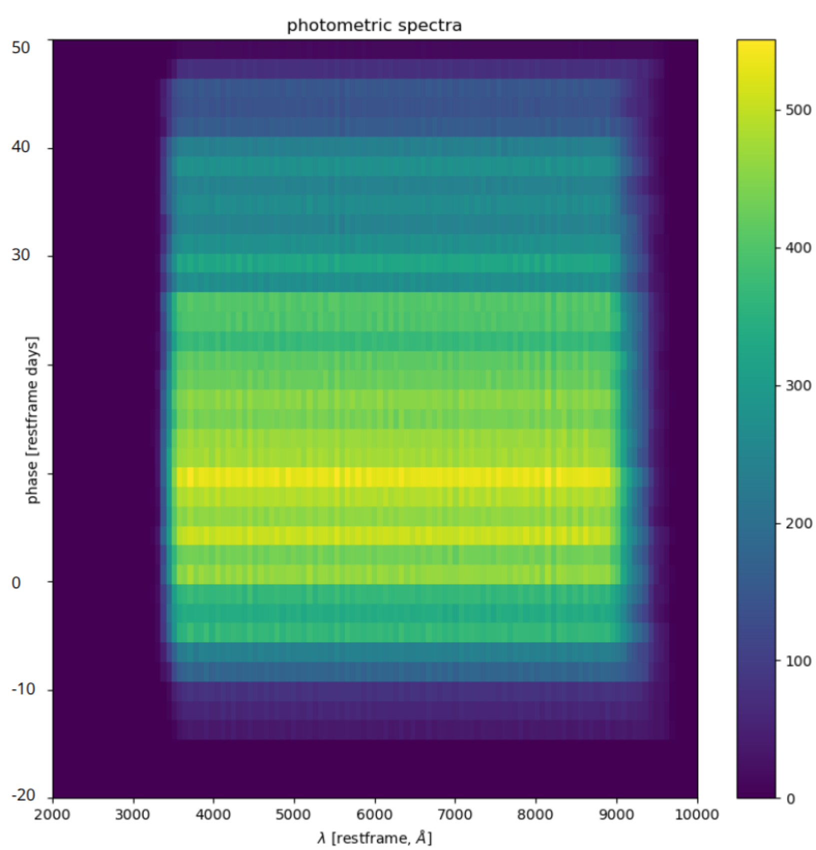


Fig. 4.4 Coverage of the SNfactory data in the $\lambda - p$ space.

3.1.2 Fitting the spectra

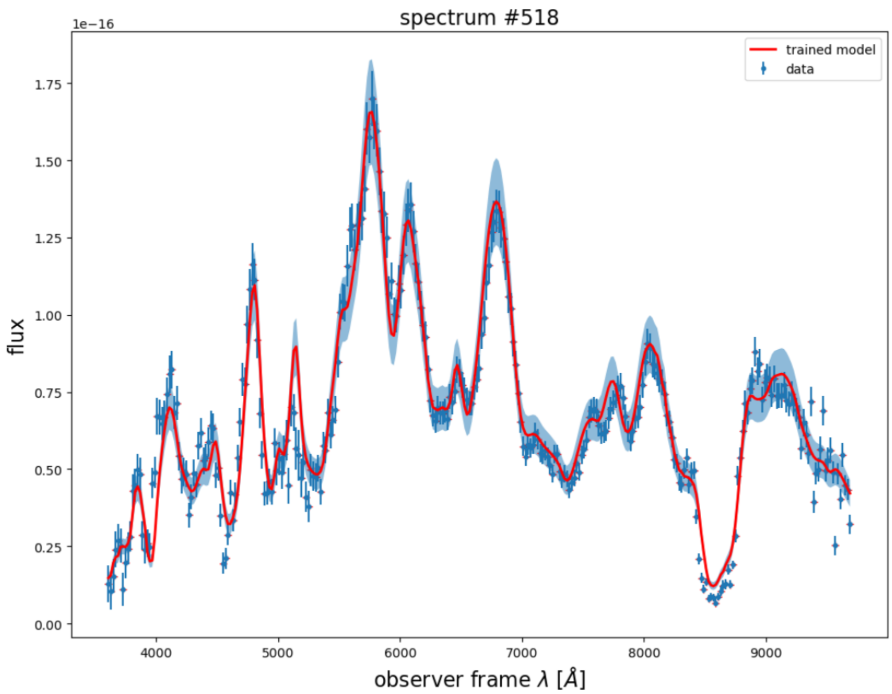
We present in Figure 4.5 examples of spectra that were fitted with NaCl. These are 2 spectra randomly picked out that belong to two different supernova and are taken at different phases. What we see is that the trained model seems to accurately describe the evolution of those spectra. The errorsnake model employed shows an intrinsic dispersion of the fluxes at an order of about 10%. We present in the next subsection the results of full dataset.

The entirety of the model training takes around 30 minutes on a Dell Precision 3581 laptop equipped with an Intel(R) Core(TM) i7-13700H, 14 cores, up to 5.0 GHz, 32 GB RAM and a NVIDIA RTX A1000 6GB GPU.

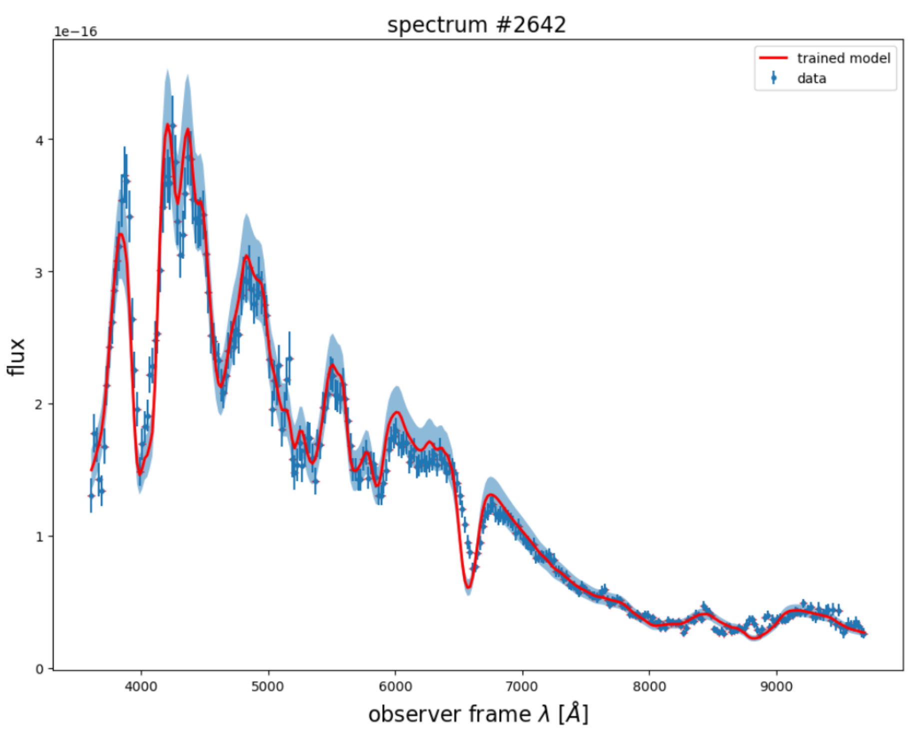
3.1.3 Training residuals

We would like to examine how well a SALT2 like parameterized model is able to describe the spectrophotometric evolution of supernovae. In order to do that, we will be looking at the weighted residuals of the model on the entirety of the data and check if there are any particular features that stand out. This is presented in Figure 4.6 where we have calculated the weighted residuals which represent by how many σ our model differs from the data, also known as pulls. We do that for each data point and stack them on top of each other if they have been observed at the same time. So for example if we have 2 spectra measured at phase of $p = -1$ with data points at 5000Å, it'll appear as one point on the Figure and its value will be the average of the 2 pulls. We then get a Figure that represents the accuracy of the model in the $\lambda - p$ space. This also means that each horizontal line represents on average how well a spectrum is modeled at that particular restframe phases, while the vertical lines are for specific restframe wavelengths.

We see in the Figure 4.6 that for most of the $\lambda - p$ space the residuals show good agreement between the model and the data. But what also stands out is that we can identify three distinct wavelength regions where the model is unable to capture the variability of the data. These three regions are found at ~ 3500Å, ~ 6000Å and ~ 8000Å, which correspond to UV, Si II and Ca II. These are known to have very diverse forms ([Wang et al., 2012], [Parrent et al., 2016]) and we can see here that they are not captured by a SALT2 parameterization. It is important to mention that the models mentioned in Chapter 2 that go beyond the SALT2 parameterization can capture those variabilities with the help of extra parameters.



(a)



(b)

Fig. 4.5 Two spectra with the train NaCl model. The blue points are the data, the red line is the trained model and the highlighted area is the error model.

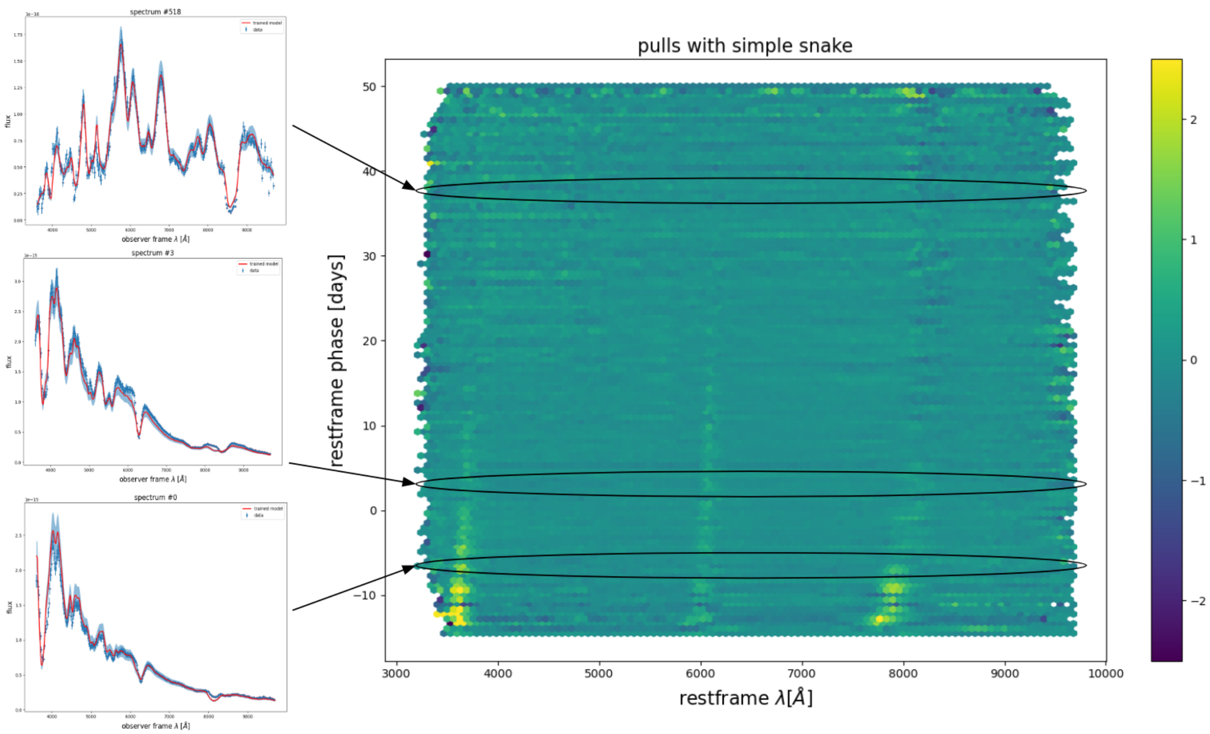


Fig. 4.6 Weighted residuals at the end of the training done with NaCl on the SNfactory data. The color bar represents by how many σ our model differs from the data. We also use as examples 3 spectra at 3 different phases and show where they would be present on the plot.

4 pre-DC1

Part of my PhD work, I took charge of the analysis of pre-DC1, a slightly different version of the DC1 where we use a SALT model that respects the NaCl constraints to generate the supernova lightcurves. This allows us to remain coherent in assessing whether the pipeline introduces internal biases at the level of the model training or not.

4.1 Data generation : simulation

All of the elements concerning the generation of the pre-DC1 supernovae can be summarized in the following list :

- All of the supernova data was generated with SkySurvey [1].
- The cosmology that was used is Planck18 found in astropy.cosmology ($\Omega_m = 0.30966$) (ref).
- The color and stretch of the supernovae were drawn from a normal distribution centered around 0, with $\sigma(c) = 0.1$ and $\sigma(X_1) = 1$.
- There are no redshift errors added to any supernovae.
- A spectrum is generated for each supernova from ZTF and SNLS drawn from a phase when the SN was observed.
- Magnitude cuts were done before the application of any noise and intrinsic dispersion. For each survey the magnitude cuts are written in table 4.1. This ensures that there are no selection effects added to the simulations.
- The rate of the supernovae were chosen to be Frohmaier+19 (ref).
- The milky way dust extinction was drawn out from the Planck dust map (ref) and the $E(B-V)_{MW}$ values follow the CM89 dust law (ref).
- The addition of the intrinsic dispersion, σ_{int}, is applied on the observed magnitudes of each supernova.
- Lightcurve measurement uncertainties were drawn from sncosmo [2] as a function of the skynoise taken from the real observational logs.
- An error snake is also added to the lightcurves drawn from a normal distribution with a dispersion of 5% of the flux. The error snake term is the same for a given band and given night for a given supernova.

ZTF	18.3
SNLS	24.2
HSC	24.6

Tab. 4.1 List of the magnitude cuts on DC1 simulations

[1] https://github.com/MickaelRigault/skysurvey
[2] https://sncosmo.readthedocs.io/en/stable/

4.2 Goals

The goal behind pre-DC1 is to check the consistency of the Georges pipeline in reconstructing the input distances and cosmology. The final results of the analysis chain will be determined after running the Georges pipeline on 100 realizations of the pre-DC1 dataset.

We define the following set of goals to achieve at the end of pre-DC1 :

1. That the reconstructed supernova parameters X_0, X_1, c and t_{max} are unbiased
2. That the reconstructed distance moduli μ are unbiased
3. That the reconstructed parameter uncertainties extracted from the inverse of the Fisher matrix (σ_{Fisher}) are equivalent to the parameter uncertainties extracted from the empirical covariance matrix constructed after the 100 realizations (σ_{Emp})
4. That the reconstructed cosmological parameters Ω_m and w are unbiased in the context of a ΛCDM cosmology with a constant w.

5 Results : Potential biases & systematics

5.1 Model & data agreement

The results we present here are obtained on 1 realization of the pre-DC1 simulations. The average time it takes to train one pre-DC1 sample of supernova is X. This is done on the IN2P3 Computing Cluster[3].

In Figures 4.7 and 4.8 we are plotting the photometric and spectroscopic residuals weighted by the measurement uncertainties. We see that after the model training, we are able to accurately describe the data with NaCl within the model's definition domain. This is also extended in Figure 4.9 when we look at the weighted residuals for each measurement obtained with a specific filter.

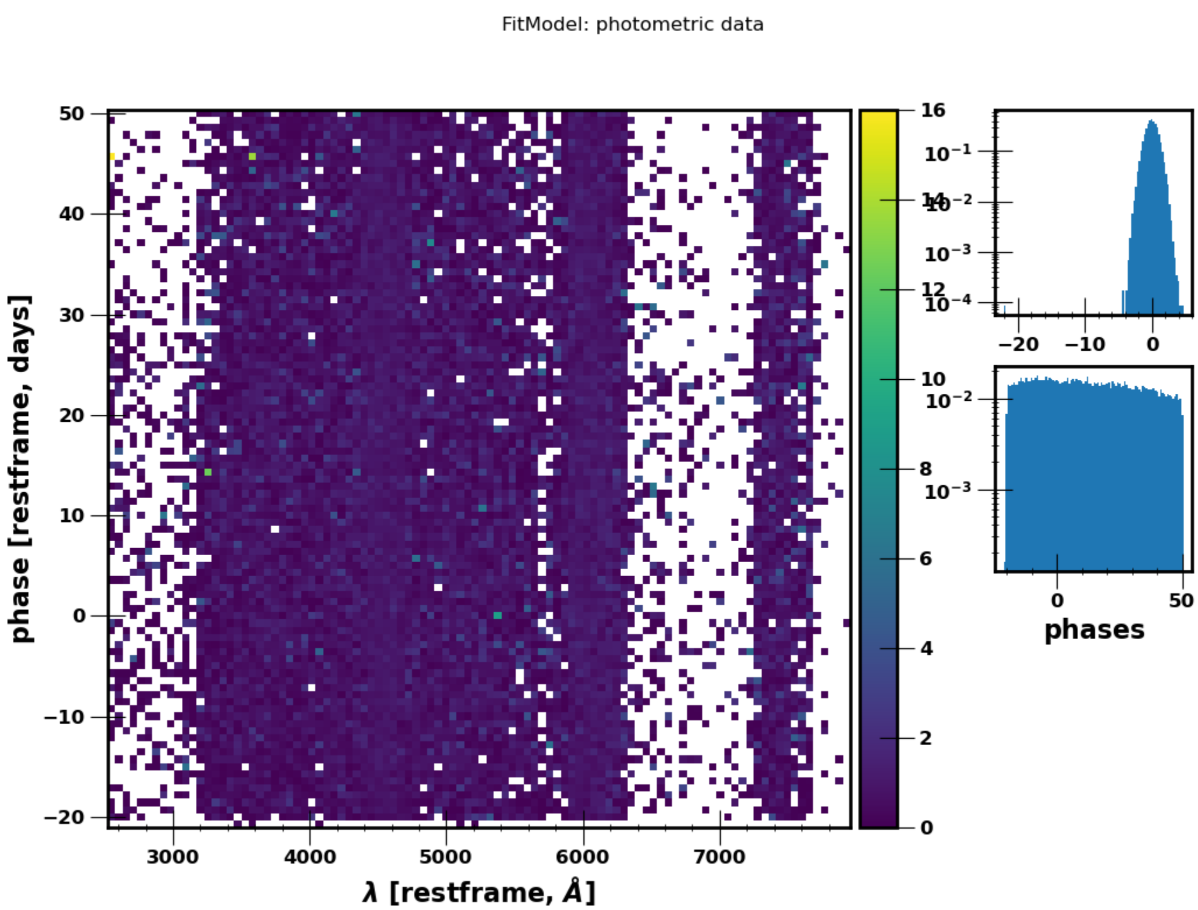


Fig. 4.7 Photometric residuals for 1 realization of pre-DC1

5.2 Parameter reconstruction and covariance matrix

Since the pre-DC1 dataset is generated with a model that respects the NaCl constraints, we expect to reconstruct the exact same parameters used in the generation. In Figure 4.10 we compare the generated and reconstructed parameters averaged over all 100 realizations of pre-DC1. We see that the reconstructed

[3] https://cc.in2p3.fr/en/

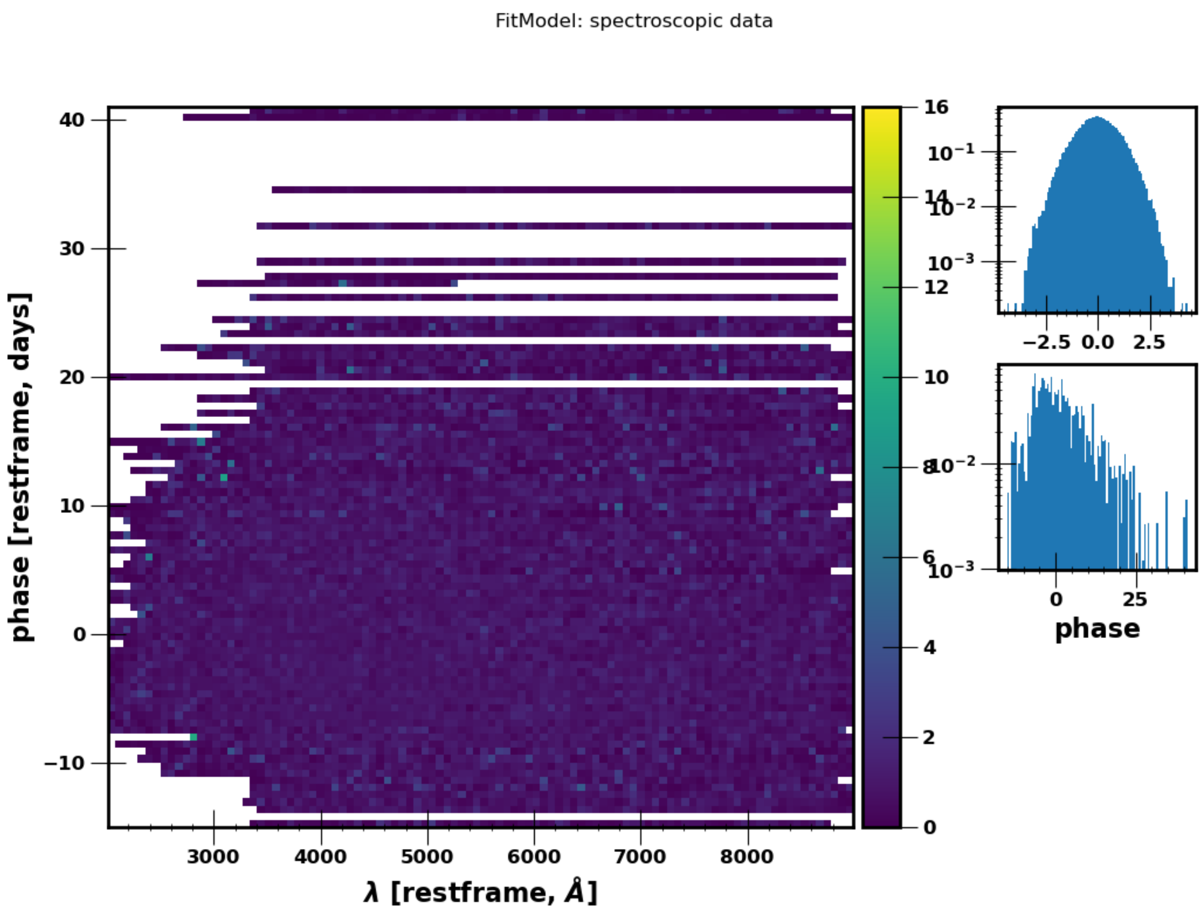


Fig. 4.8 Spectroscopic residuals for 1 realization of pre-DC1

parameters match the input parameters within the statistical uncertainties of the reconstruction.

We are also interested in comparing the constructed empirical covariance matrix and the Fisher covariance matrix from NaCl. In Figure (ref), we compare the output uncertainties (σ_{Fisher}) from 1 pre-DC1 realization with the uncertainties extracted from the empirical covariance matrix (σ_{Emp}) constructed from the run on the 100 pre-DC1 realizations. We see that for each block of parameters X_0, X_1 and c, σ_{Fisher} is equivalent to σ_{Emp} and well within the statistical uncertainties of σ_{Emp}. This is also highlighted when we look at their ratios where they are centered around 1. We are now convinced that the NaCl can produce unbiased parameters along their uncertainties in a LEMAITRE-like analysis.

5.3 Distance modulus and cosmological parameters reconstruction

We use the EDRIS module to determine the distance modulus of the supernovae at the end of the lightcurve fit and model training with NaCl. It's important to note that we are not assessing the efficiency of EDRIS in reconstructing distances as this has been already established in D. Kuhn's PhD thesis[4] where it was shown that EDRIS does not bias supernova distances in our setup. If we do find

[4] https://theses.hal.science/tel-05451069

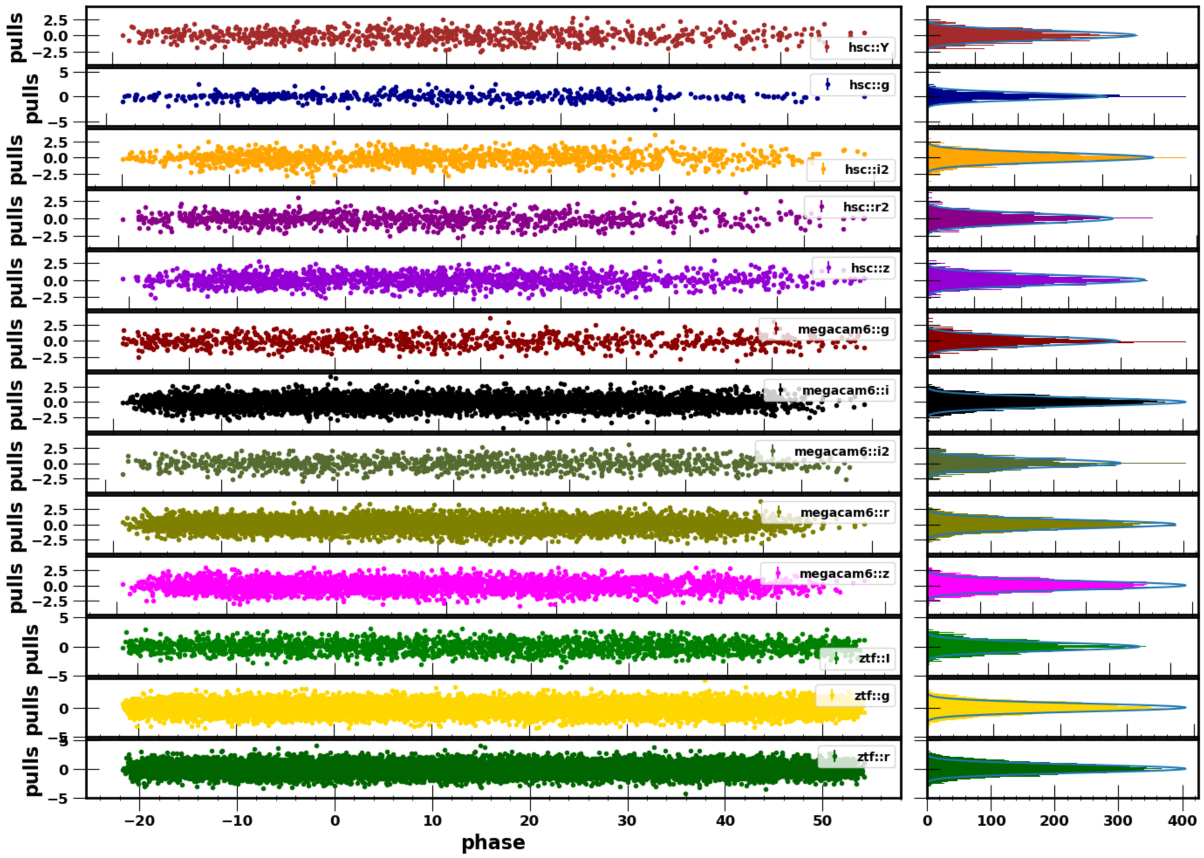


Fig. 4.9 Photometric residuals per band for 1 realization of pre-DC1

biased distances, then it would come from NaCl.
The results in Figure 4.11 show the mean binned distance residuals from 100 realizations of pre-DC1 as well as a comparison between between the empirical uncertainties on the binned distances and the Fisher uncertainties from EDRIS. We see that NaCl does not introduce biases on the supernova distances. We can now move on to determining the cosmological parameters with cosmologix.

In Figures 4.12 and ref(om and w coverage plots) we show the results from 1 realization of pre-DC1 and from the 100 realizations of pre-DC1. We are plotting the contours for all the Ω_m and w parameters obtained. What we see is that we recover unbiased results within 1σ ensuring that our pipeline works.

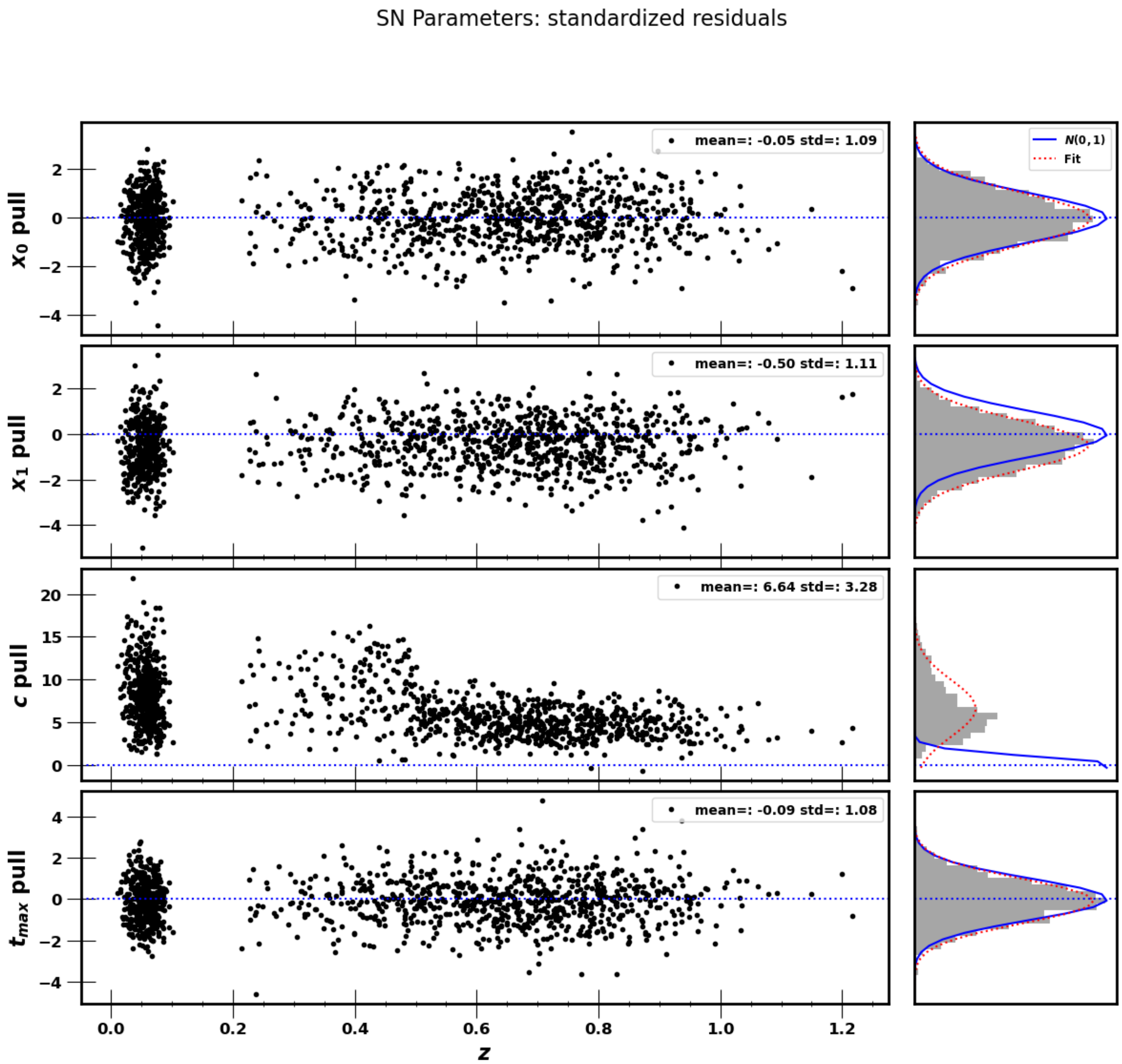


Fig. 4.10 Weighted residuals of each supernova parameter for 100 realization of pre-DC1

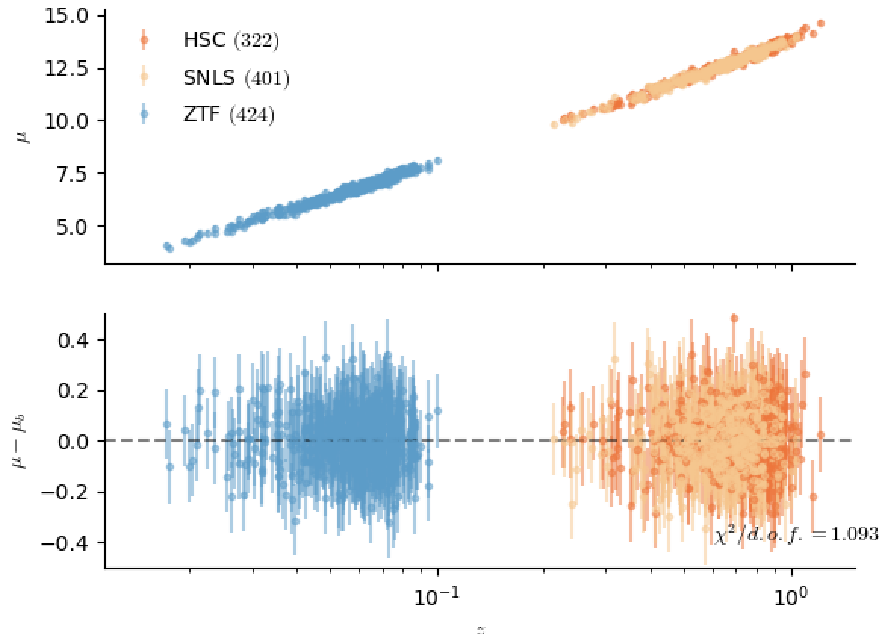


Fig. 4.11 Hubble diagram for 1 realization of pre-DC1

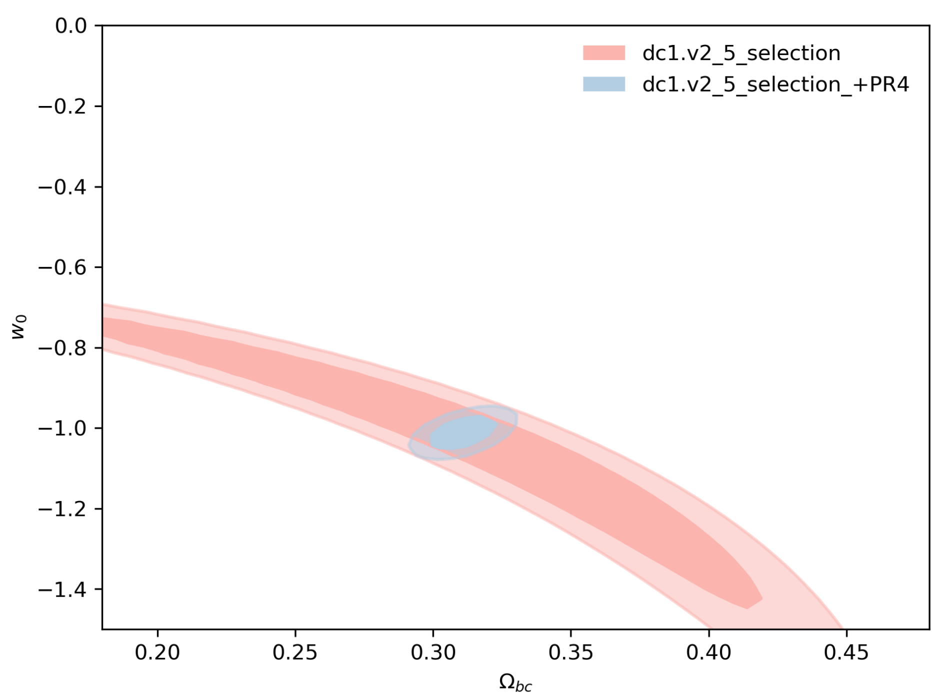


Fig. 4.12 Ω_m and w contours for 1 realization of pre-DC1

CHAPTER 5

Field level inference for cosmological analyses

Contents

1 Field level-inferences

1.1 Definition from [Leclercq, 2025]

A field level inference generally refers to methods and techniques used to infer physical fields alongside a set of parameters from observations, without relying on compressed data from the corresponding field. As a result, a natural byproduct in cosmology is a cosmological map. These inferences are very high dimensional and require advanced statistical techniques. There are motives behind doing such analyses as they include more complexities than traditional summary statistics which could lead to less biasing results. For this PhD, we will be discussing field-level inferences specifically of matter densities and velocity fields.

1.2 Motivations

Unlike compressed statistics such as the matter power spectrum ($\langle\delta(\mathbf{k_1})\delta(\mathbf{k_2})\rangle$), a field-level inference preserves all statistical information such as higher-order correlations. The structures observed today are non-linear, meaning correlations of higher order such as the bispectrum ($\langle\delta(\mathbf{k_1})\delta(\mathbf{k_2})\delta(\mathbf{k_3})\rangle$) or trispectrum ($\langle\delta(\mathbf{k_1})\delta(\mathbf{k_2})\delta(\mathbf{k_3})\delta(\mathbf{k_4})\rangle$) are non-zero and must be calculated to extract maximal information on cosmological parameters. Since the field-level inference aims at reconstructing the entire underlying field as well as its structures, these correlations will always be accounted for at all orders. Moreover it preserves phase information. The compressed statistics mainly calculates the amplitudes of the correlations that exist while missing out on the phase for which these amplitudes are possible. We can assume that the contribution to the correlation function of two over dense regions would be the same had they been present anywhere in the Universe as long their separating distance remains the same.

Another motivation for field-level inference is reducing the uncertainties that arise due to cosmic variance at the local-Universe. Traditional summary statistics assume ergodicity i.e. averaged measurements over a sufficiently large volume of the Universe approximates the averages over many hypothetical realizations, which becomes problematic at the local Universe due to the limited volume. With field-level inference the goal is reconstruct the specific realization that produces our observations, allowing us to the properly adjust our model parameters to that realization.

The outcome of field-level inferences done on observations of our Universe is what we call a Digital Twin.

avoid compression

BORG vs other methods

1.3 Field-level vs compressed summary statistics

results from Ngyuen et al 2025 and Porqueres et al 2023

2 The forward model

2.1 The BORG algorithm

BORG stands for *Bayesian Origin Reconstruction from Galaxies* ([Jasche and Wandelt, 2013]) - a Bayesian inference framework that uses a forward model to tackle the inverse problem of how structures are formed in at the large scales of the Universe. The way BORG can be applied on type Ia Supernova is summarized in Figure 5.1.

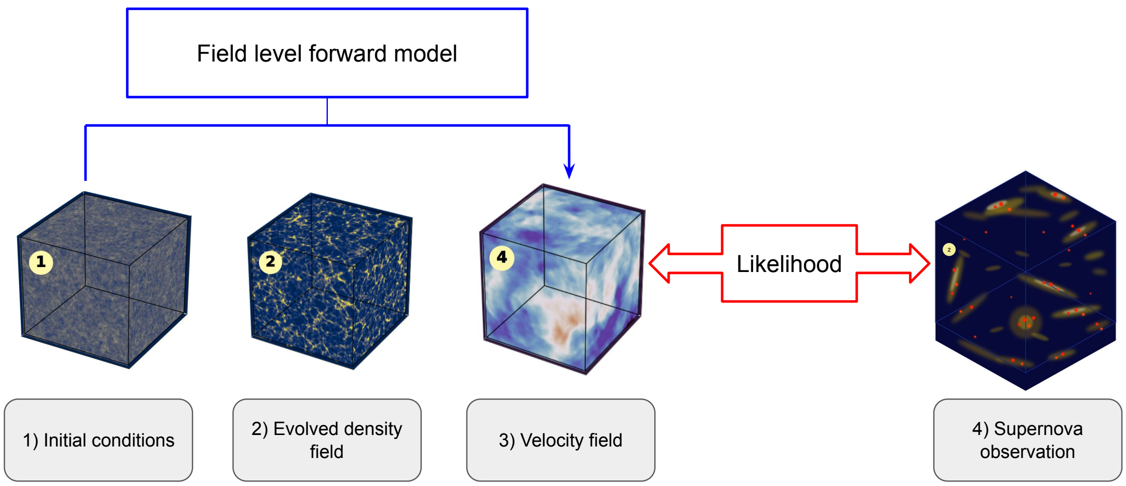


Fig. 5.1 Schematic summarizing what BORG intends to do when faced with type Ia Supernova data.

It is a field level approach designed to infer an initial density field δ_{IC}, which would be our initial conditions and forward model its evolution to obtain a final density field δ_f. The Bayesian logic behind BORG makes it that we are exploring a large scale structure posterior by infering a Gaussian prior for the initial density field δ_{IC}. Modeling the evolving density field from δ_{IC} to δ_f requires a gravity model that can capture the non linear effects of structure formation. I will be explaining the details of the different parts of BORG in the next subsections. In a sense, we are generating, or simulating, a numerical "twin" of our Universe by sampling the posterior of the initial conditions using a Bayesian approach.

2.1.1 BORG's Bayesian inference chain

The logic behind BORG relies on Bayesian inference, as stated in the name of the algorithm, and this method is very well adapted for the problem we are trying to solve. The Bayesian framework works in the following way : Based on the Bayes theorem, the probability $\mathcal{P}$ of a specific realization of an event or data d being described by a set of parameters θ can be written as :

$$\mathcal{P}(\theta|d) = \frac{\mathcal{P}(d|\theta)\mathcal{P}(\theta)}{\mathcal{P}(d)} \tag{5.1}$$

where $\mathcal{P}(d|\theta)$ is the probability that the data is known given a model which is also called the likelihood, $\mathcal{P}(\theta)$ the probability of the model parameters in the absence of any data, also known as the prior, and $\mathcal{P}(d)$ the probability of the data before it is known, also called the evidence. The probability $\mathcal{P}(\theta|d)$ is called the posterior.

In the context of our problem, determining a prior on the evolved density field is a challenge due to the non-linear gravitational nature of structure formation ([Fis, 1995], [Jasche and Kitaura, 2010]). So even if one assumes a Gaussian or a log-normal for the δ_f they will be missing the description of fine filamentary features of the observed cosmic web ([Peebles, 1980]). Instead what is done in B0RG is treat the problem as a Bayesian inference on the inital conditions (δ_{IC}). The biggest gain out of that, is that the prior is physically motiviated from results obtained from Planck CMB data ([Ade et al., 2016], [Collaboration et al., 2019]) and inflation which is a Gaussian prior. Combining that with a gravity model to evolve the density field and by marginalizing over the final density fields, we can then constrain our inital conditions.

From [Jasche and Lavaux, 2019] we can obtain the final posterior as :

$$\mathcal{P}(\{\delta_{IC}^{(i)}\}|\{N^{(i)}\}) = \frac{\mathcal{P}(\{\delta_{IC}^{(i)}\})\mathcal{P}(\{N^{(i)}\}|\{G^{(i)}(\{\delta_{IC}^{(i)}\})\})}{\mathcal{P}(\{N^{(i)}\})} \tag{5.2}$$

i being the index of the voxel we are evaluating with $N^{(i)}$ the number of objects within that voxel and $G^{(i)}(\{\delta_{IC}^{(i)}\})$ is the gravity model or the structure formation model defined such that

$$G^{(i)}(\{\delta_{IC}^{(i)}\}) = \delta_f^{(i)}$$

Given the dimensionality of the problem at hand, directly integrating this equation is practically unfeasible due to the computational power required. The solution to that is to then explore the posterior distribution of initial conditions to find the best configuration set that matches our observations. This will require us to use a set of Markov Chain Monte Carlo (MCMC) sampling algorithms.
A Markov Chain is a set of sequences $\{\theta_0, \theta_1, \theta_2, ..., \theta_n, ...\}$ where the conditional distribution of θ_{n+1} given the previous elements only depends on θ_n. If this distribution does not depend on n we call it a stationary transition probability, which is what we are interested in. To guarantee that distribution, the following condition must be satisfied :

$$\mathcal{T}(\theta|\theta')\mathcal{P}(\theta') = \mathcal{T}(\theta'|\theta)\mathcal{P}(\theta) \tag{5.3}$$

where $\mathcal{T}$ is the transition probability distribution. A property of that condition is that the probability flux from $\theta \rightarrow \theta'$ is equal to $\theta' \rightarrow \theta$ with respect to the target distribution. This is also known as the "balance flow".

Metropolis-Hastings algorithm

A very common example of an MCMC algorithm is the Metropolis-Hastings algorithm ([Metropolis et al., 1953], [HASTINGS, 1970]). To summarize how it works : given a state θ_n, a new state is drawn from a proposal distribution which is often Gaussian q such that :

$$\theta' \sim q(\theta', \theta_n) \tag{5.4}$$

With a Gaussian distribution we get $\theta' = \theta_n + \epsilon; \quad \epsilon \sim \mathcal{N}(0,\sigma)$ with ϵ being a random perturbation drawn from the normal distribution where σ controls the step size. The next step is to then compute an acceptance ratio :

$$\alpha = \min\left(1, \frac{\mathcal{P}(\theta'|d)q(\theta_n|\theta')}{\mathcal{P}(\theta_n|d)q(\theta'|\theta_n)}\right) \tag{5.5}$$

The state then becomes :

$$\theta_{n+1} = \begin{cases} \theta' & \text{With probability } \alpha \\ \theta_n & \text{With probability } 1-\alpha \end{cases} \tag{5.6}$$

This guarantees that better proposals are more likely to be accepted while worse proposal have a probability proportional to how much worse they are. The problem BORG might have with this algorithm is the step size and the computation time since the complexity of the algorithm is $\mathcal{O}(D)$ where D is the dimension of the parameter space. We expect BORG to have around 10^7 parameters to fit so if the step size that is chosen is too small the chain will only make tiny steps and take a very long time to reach a converged state, while if the step size is too large most proposals will land in low-probability regions and get rejected, also making the computational time too long and inefficient.

HMC algorithm

The Hamiltonian Monte-Carlo algorithm (HMC, [Brooks et al., 2011]) is much more suitable for our problem and is currently what is implemented in BORG. The general idea of HMC is to use concepts from Hamiltonian classical mechanics on high-dimensional problems. It uses gradient information to make significant proposals with a high acceptance rate. Let us define Θ as the position vector containing the a set of parameters to solve, $\mathbf{p}$ the momentum variables and $\mathcal{H}$ the Hamiltonian, the Hamilton equations give us :

$$\frac{\mathrm{d}\Theta}{\mathrm{d}t} = \frac{\partial \mathcal{H}}{\partial \mathbf{p}} \tag{5.7}$$

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = -\frac{\partial \mathcal{H}}{\partial \Theta} \tag{5.8}$$

with the Hamiltonian being constant in time. Let $\mathbf{M}$ be the mass matrix which is symmetric and positive definite, then the Hamiltonian is :

$$\mathcal{H} = \frac{1}{2}\mathbf{p}^{\mathrm{T}}\mathbf{M}^{-1}\mathbf{p} + U(\Theta) \tag{5.9}$$

where $U(\Theta)$ is the potential energy and $\frac{1}{2}\mathbf{p}^{\mathrm{T}}\mathbf{M}^{-1}\mathbf{p}$ the kinetic energy. It's important to note that the choice of the mass matrix can strongly impact the performance of the sampler.
The link between Hamiltonian dynamics and probabilities can be established with the concept of canonical distributions from statistical mechanics. The probability distribution over the states and momenta are written as :

$$\mathcal{P}(\Theta, \mathbf{p}) = \frac{1}{Z}\exp\left(\frac{-\mathcal{H}}{k_B T}\right) \tag{5.10}$$

where k_B is the Boltzmann constant, T the temperature of the system and Z is the partition function. We can impose $k_B T = 1$ for simplicity and absorbing the Z partition function into the normalization which leads us to:

$$-\ln\mathcal{P}(\Theta, \mathbf{p}) = \mathcal{H}(\Theta, \mathbf{p}) = \frac{1}{2}\mathbf{p}^{\mathrm{T}}\mathbf{M}^{-1}\mathbf{p} + U(\Theta) \tag{5.11}$$

We can see that the Hamiltonian formalism naturally allows the parameters Θ to be displaced along trajectories of a constant $\mathcal{H}$, guided by the gradient of the potential energy $U(\Theta) = -\ln\mathcal{P}(\Theta, \mathbf{p})$ towards regions of higher posterior probability. This means that the joint distribution is :

$$\mathcal{P}(\Theta,\mathbf{p}) \propto \exp(-\mathcal{H}(\Theta,\mathbf{p})) \tag{5.12}$$

We can then the same acceptance condition as the Metropolis-Hastings algorithm where we have :

$$\alpha(\Theta',\mathbf{p}',\Theta,\mathbf{p}) = \min\left(1, \frac{\mathcal{P}(\Theta',\mathbf{p}')}{\mathcal{P}(\Theta,\mathbf{p})}\right) = \min\left(1, \exp(-\mathcal{H}(\Theta',\mathbf{p}') + \mathcal{H}(\Theta,\mathbf{p}))\right) \tag{5.13}$$

In exact Hamiltonian dynamics, $\mathcal{H}$ is conserved so the expected acceptance rate is unity but due to numerical errors we expect to see a lower acceptance rate. Lastly, we marginalize over the momenta to obtain the final posterior on our parameters, for us that will be the initial conditions.

$$\mathcal{P}(\Theta) = \int \mathcal{P}(\Theta,\mathbf{p})\mathrm{d}\mathbf{p} \tag{5.14}$$

2.1.2 Particle mass assignment

The inital density field δ_{IC} is drawn from a Gaussian prior and represented as a set of particles distributed on a grid. The space we are working in is assigned on a grid so it is important to be able to assign the particles to that grid. BORG addresses by doing a Cloud In Cell (CIC) mass assignment on the grid. This mass assignment works by assigning a "cloud" to each particle and distributing its mass to the neighbouring grid points according to the overlap between the cloud and these points.
The assignment is done in the following way : Each particle is treated as a uniform-density cube having a mass of unity. Each particle overlaps with up to 8 neighbouring grid cells. The fraction of the particle's mass assigned to each cell is proportional to the volume overlap between the particle cube and the grid cell. If we consider a 3D grid with a cell size of Δx and a particle at the position $X_p = (x_p, y_p, z_p)$ lives inside a cell whose lower-left corner is a grid point (i,j,k) where

$$i = \left\lfloor \frac{x_p}{\Delta x} \right\rfloor, \quad j = \left\lfloor \frac{y_p}{\Delta x} \right\rfloor, \quad k = \left\lfloor \frac{z_p}{\Delta x} \right\rfloor$$

We define the fractional distance as

$$d_x = \frac{x_p}{\Delta x} - i, \quad d_y = \frac{y_p}{\Delta x} - j, \quad d_z = \frac{z_p}{\Delta x} - k$$

So that $d_x,;d_y,;d_z \in [0,1[$.
The particle's mass m_p is distributed to the 8 surrounding nodes $(i+a, j+b, k+c)$ with $a,b,c \in 0,1$:

$$\rho_{i+a,j+b,k+c} += \frac{m_p}{\Delta x^3}.T_x^{(a)}.T_y^{(b)}.T_z^{(c)} \tag{5.15}$$

with $T^0_{x,y,z} = 1 - d_{x,y,z}$ and $T^1_{x,y,z} = d_{x,y,z}$.

2.1.3 Gravity Model

Simply evolving the δ_{IC} field using the equations of motion would be too computationally expensive since the number of particles in our field can go up to 10^9 particle. Developments in the implmentation of a gravity model that is accurate and fast have been a very important topic in the development of BORG. I will be presenting the main models that are used in BORG. Additionally, the movements of the particles in the density field directly gives us access to the underlying velocity field.

LPT

The first gravity model is the Lagrangian Perturbation Theory or LPT, it can also be known as the Zeldovich approximation ([Zel'dovich, 1970a], [Crocce and Scoccimarro, 2006]). The idea is to expand the density field pertubatively $\delta(x,t) = \delta^{(1)}(x,t) + \delta^{(2)}(x,t) + ...$, which is the Eulerian expansion, as well as the displacement field $\Psi(q,t) = \Psi^{(1)}(q,t) + \Psi^{(2)}(q,t) + ...$ which maps the initial positions q (Lagrangian coordinates) to the final positions x (Eulerian coordinates). In comoving coordinates, the equations of motion for dark matter particles are :

$$\ddot{x} + 2H\dot{x} = -\frac{1}{a^2}\nabla_x\phi \tag{5.16}$$

where $H = \dot{a}/a$ is the Hubble parameter, a the scale factor and ϕ is the gravitational potential satisfying the Poisson equation :

$$\nabla_x^2\phi = 4\pi G\bar{\rho}a^2\delta(x,t) \tag{5.17}$$

With the position being:

$$x(q,t) = q + \Psi(q,t) \tag{5.18}$$

This means we can rewrite the equations of motion as :

$$\ddot{\Psi} + 2H\dot{\Psi} = -\frac{1}{a^2}\nabla_x\phi \tag{5.19}$$

To ensure the mass conservation, the continuity equation in the Lagrangian formalism gives us :

$$1 + \delta(x,t) = \frac{1}{|\det(J)|} \tag{5.20}$$

With J being the Jacobian that maps q to x.

$$J_{ij} = \frac{\partial x_i}{\partial q_j} = \delta_{ij} + \frac{\partial \Psi_i}{\partial q_j} \tag{5.21}$$

At first order, we can expand the determinant $\det(J) \simeq 1 + \nabla_q.\Psi^{(1)}$ giving the first term in the expansion of the density field as :

$$\delta^{(1)}(x,t) \simeq -\nabla_q.\Psi^{(1)} \tag{5.22}$$

If we continue the demonstration, we can obtain what the first order approximation is for the position of the density field particles at a time t and its velocity as:

$$\boxed{x(q,t) = q - D_1(t)\nabla_q\phi^{(1)}(q)} \tag{5.23}$$

$$\boxed{v(q,t) = -aHfD_1(t)\nabla_q\phi^{(1)}(q)} \tag{5.24}$$

Where D_1 is the linear growth factor and $f = d\ln D_1/d\ln a$ the linear growth rate of structures.
It's important to note that this approximation breaks down when dense structures start to form such as halos, therefore this approximation is a good description of structure formation at large scales. Going to higher orders such as 2LPT (2nd order approximation) delays this breakdown but does not prevent it. Physically, it can be shown that this occurs around $k \approx 10^{-1}h\text{Mpc}^{-1}$ at $z = 0$

2LPT

We can also use the second order approximation of the Lagrangian Perturbation Theory (2LPT) as our gravity solver. That means the displacement field is written as $\Psi(q,t) = \Psi^{(1)}(q,t) + \Psi^{(2)}(q,t)$ and one can demonstrate that the second order term of the displacement is :

$$\Psi^{(2)}(q,t) = -D_2(t)\nabla_q \phi^{(2)}(q) \tag{5.25}$$

where $D_2(t)$ is the second order growth factor. We also get the following Poisson equation for the gravitational potential:

$$\nabla_q^2 \phi^{(2)}(q) = \sum_{i>j} \left(\frac{\partial^2 \phi^{(1)}}{\partial^2 x_i} \frac{\partial^2 \phi^{(1)}}{\partial^2 x_j} - \left[\frac{\partial^2 \phi^{(1)}}{\partial x_i x_j} \right]^2 \right) \tag{5.26}$$

The final position and velocity of the density field in 2LPT are :

$$\boxed{x(q,t) = q - D_1(t)\nabla_q \phi^{(1)} - D_2(t)\nabla_q \phi^{(2)}} \tag{5.27}$$

$$\boxed{v(q,t) = -aH \left[f_1 D_1 \nabla_q \phi^{(1)} + f_2 D_2 \nabla_q \phi^{(2)} \right]} \tag{5.28}$$

where f_1 is the linear growth factor $f_1 = d\ln D_1/d\ln a$ and $f_2 = d\ln D_2/d\ln a$.

PM

The goal of the Particle Mesh (PM) gravity model is to evolve the particles in the simulation after having done a mass assignment, like CIC. The equations of motion and the Poisson are adapted to be solved on the BORG grid chosen, they are given by Equation 5.18 and $\frac{\mathrm{d}^2\Psi(q,a)}{\mathrm{d}a^2} = -\nabla_x \phi(x,a)$. I will explaining the steps it takes to evolve the field from a time t_i to a time t_{i+1} without going into the derivation of the equations at each step.

- Mass assignment, like CIC, to obtain the density field δ.
- Transform to Fourier space $\delta(x) \to \tilde{\delta}(k)$.
- Solve the Poisson equation in Fourier space.
- Compute the force field, or the gravitational acceleration, in Fourier space $\tilde{g}(k)$ then transform it to cartesian space to get $g(x)$.
- Interpolate the forces back to particles.
- Time integration to get the next position and momentum of the particles.

The PM gravity model is accurate up to the resolution of the grid, which is the most accurate gravity model we can use in BORG at the current moment. It allows us to have very accurate density and velocity fields but getting there requires having to repeat the full PM cycle at least 100 times ([Tassev et al., 2013]) which will produce very long computational time. If we decide to reduce the steps in PM we would lose accuracy on all scales.

COLA

The last model I will be presenting is COmoving Lagrangian Acceleration or COLA ([Tassev et al., 2013]). The goal of COLA is to be able to solve the equations of motion and the Poisson equation for particles

that undergo trajectories in LPT. This method allows us to trade accuracy at small scales to gain computational speed while maintaining accuracy at large scales.
This is done by decomposing the displacement field into two components :

$$\Psi(q,a) = \Psi^{LPT}(q,a) + \Psi^{COLA}_{res}(q,a) \tag{5.29}$$

where $\Psi^{LPT}(q,a)$ is the LPT displacement field and $\Psi^{COLA}_{res}(q,a)$ the residual displacement of each particle as observed from a frame comoving with with an "LPT observer" whose trajectory is defined by Ψ^{LPT}. If we temporarily omitt the Hubble expansion, we get the following equations of motion :

$$\frac{\mathrm{d}^2\Psi(q,a)}{\mathrm{d}a^2} = -\nabla_x \phi(x,a) \tag{5.30}$$

This can be rewritten in the comoving frame of the "LPT" observer as :

$$\boxed{\frac{\mathrm{d}^2\Psi^{COLA}_{res}(q,a)}{\mathrm{d}a^2} = -\nabla_x \phi(x,a) - \frac{\mathrm{d}^2\Psi^{LPT}(q,a)}{\mathrm{d}a^2}} \tag{5.31}$$

We can think of the LPT as an additional force acting on the particles due to the use of a non-inertial frame of reference. LPT also experiences lower typical accelerations compared to the natural cosmological frame. Numerically, this produces equivalent accuracy to solving the full equations of motion while using fewer steps ([Tassev et al., 2013]). The expected number of STEPS in COLA is around 10 unlike PM. A visual representation of what COLA does can be seen in Figure 5.2.

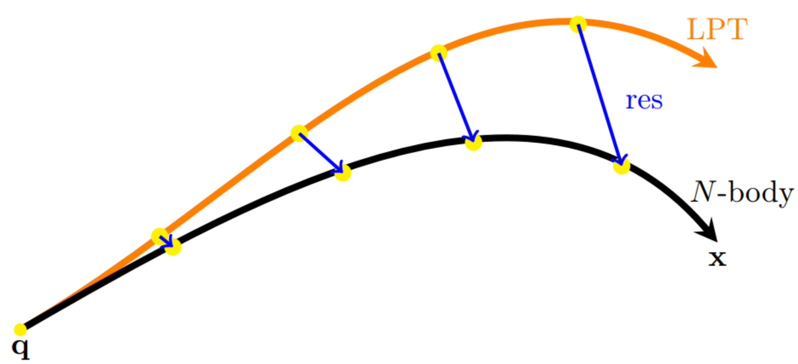


Fig. 5.2 Schematic representing how COLA works found in [Bartlett et al., 2025]. The orange line represents the motion of the particle from the LPT frame of reference and the blue lines are the residuals COLA calculates from the true position to correct for it.

If we denote $x(a) = x_{LPT}(a) + x_{COLA}(a)$ and the momentum $p = p_{LPT} + p_{COLA}$ the equations to solve then become :

$$\frac{\mathrm{d}x}{\mathrm{d}a} = \mathcal{D}(a) p_{COLA} - \frac{\mathrm{d}D_1}{\mathrm{d}\Psi^{(1)}} + \frac{\mathrm{d}D_2}{\mathrm{d}\Psi^{(2)}} \tag{5.32}$$

$$\frac{\mathrm{d}p_{COLA}}{\mathrm{d}a} = \mathcal{K}(a)\left[\nabla(\Delta^{-1}\delta) - \mathcal{V}[D_1](a)\Psi^{(1)} + \mathcal{V}[D_2](a)\Psi^{(2)}\right] \tag{5.33}$$

with $\mathcal{D}(a) = \frac{f(a)}{a^2}$, also called the "drift" prefactor, and $\mathcal{K}(a) = -\frac{3}{2}\Omega_m H_0^2 f(a)/a$, also called the "kick" prefactor. These terms come from the kick-drift-kick (KDK) algorithm that is used to update the particle positions. The kick phase boosts the momentum of the particles while the drift phase changes their positions. The KDK scheme goes as follows : At an instant n, positions and momenta at an instant $n+1$ are determined in the following way :

- Kick $p^{COLA}_{n+1/2} = p^{COLA}_{n} + F^{COLA}_{n}.\frac{\Delta a}{2}$
- Drift $x_{n+1} = x_n + p^{COLA}_{n+1/2}.\delta a$
- Kick $p^{COLA}_{n+1} = p^{COLA}_{n+1/2} + F^{COLA}_{n+1}.\frac{\Delta a}{2}$

where at each step, the force F^{COLA} is computed using a PM solver from above.

We can look at a visual difference between the 2LPT gravity solver, COLA and what a "true" density field like which is a N-body simulation named GADGET ([Springel, 2005], [Springel et al., 2021]) in Figure 5.3. The 2LPT results are obtained in 3 timesteps, the COLA results in 10 timesteps and GADGET results are obtained in 200 times. As expected, the COLA results manage to capture a lot of the details found on small scales as opposed to the 2LPT results while requiring far fewer timesteps than a full N-body simulation.

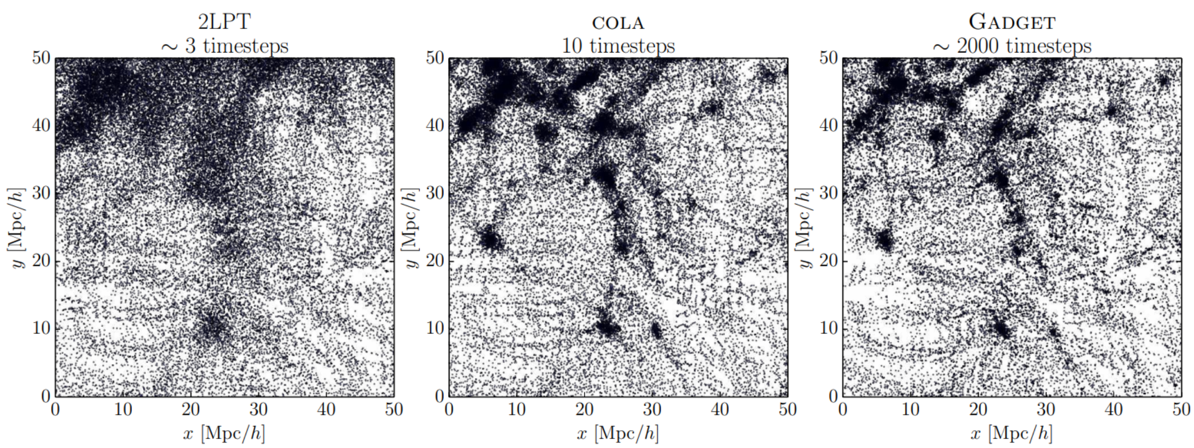


Fig. 5.3 Evolution of the same initial density field with three gravity solvers 2LPT, COLA and GADGET

2.1.4 Likelihood components of BORG

Density field likelihood

- sn are matter tracers - poisson likelihood –> number of sn in voxel

$$\mathcal{P}(N^g_k|\lambda_k) = \prod_k \frac{(\lambda_k)^{N^g_k} e^{-\lambda_k}}{N^g_k!} \tag{5.34}$$

2.1.5 Advantages over summary-statistics analyses

- Retains phase information and non-linearity - analysis through cross-correlations with observables - digital twin of our observable universe may lead to new insights on it

worth citing Richard's paper on H0 to show far we can go with FLI and what does the velocity field adds to density field information

2.1.6 Numerical implmentation

- C++, JAX, HPC

3 Towards a next generation of field level inferences

3.1 Gravity solvers

3.2 Bias models

3.3 Parameter sampling

3.4 Scalability

CHAPTER 6

BORG applications on ZTF simulations

Contents

We presented in the previous chapter the motivations and potential outcomes of conducting a field inference as well as the BORG algorithm. BORG has proven to be very successful when applied to galaxy catalogues as well as show great potential regarding weak lensing. Our goal is advance its applications to type Ia supernovae to eventually run it on the ZTF data. We will be presenting in this chapter the steps taken in order to achieve the latter. We will start by presenting the ZTF-Uchuu simulations. Then the implementation of a an improved supernova-specific likelihood in BORG. Then we will present its results on self-consistent simulations. Finally its application to ZTF-like simulations.

1 ZTF-Uchuu mocks

The ZTF simulations designed to test the impact of ZTF-supernovae peculiar velocities in a cosmological analysis is called DC PV (Data Challenge Peculiar Velocity). These are generated within the ZTF collaboration mainly to test methods in inferring $f\sigma_8$. The DC PV simulations are done similarly to the DC simulations as described in Chapter 4 with the main differences lie in adding structure information to the simulations. For that, the supernovae are simulated with the help of the Uchuu N-body simulation ([Ishiyama et al., 2021]), a high fidelity N-body dark matter simulation in a $2^3\mathrm{Gpc}.h^{-1}$ box with 2.1 trillion particles which can be seen in Figure 6.1 [1]. The full box is then subdivided into 27 subboxes where the supernovae are simulated up to a redshift of $z = 0.11$, giving 27 quasi-independent realizations of the DC PV. Each supernova must be simulated where a particle is and is assigned the particle's velocity. This allows us to add proper motion to supernovae that is driven by structure formation.

We opted to simplify a few aspects of the simulations in order to have an initial assessment of how well can a BORG analysis run on ZTF-like supernova. Firstly, we have chosen to use only the distance moduli and redshifts of supernovae as inputs in BORG, as if we were doing the analysis after having run the full LEMAITRE pipeline. One could assume that in more complicated analyses that the Tripp relation parameters (α_{Tripp}, β_{Tripp}) would be sampled in BORG. This could lead to more acurate results due to the fact that in BORG the 3-dimensional reconstruction of structures in the nearby Universe could accound for non- Homogenous Malmquist bias. Secondly, we do not impose the ZTF DR2 cosmology quality cuts on photometry nor the ZTF footprint, so all of the supernovae that are generated can be used. For a more realistic set of simulations, we would have to account for the geometry of the survey as well as the Milky-Way dust. This means introducing an angular mask $M(\mathrm{RA}, \mathrm{Dec})$ in the likelihood's selection function. This method was applied in BORG analyses such as [McAlpine et al., 2025] (Manticore-Local) and [McAlpine et al., 2026] (Manticore-Deep). Finally, we draw a set of 1000 randomly selected supernovae up to a redshift of $z = 0.06$ to match the expected number of supernovae in the volume-limited sample from ZTF. We show in Figure 6.2 an example of simulated supernovae on top of the Uchuu simulation.

2 BORG-Peculiar Velocity

The BORG-Peculiar Velocity (BORG-PV) algorithm is the same as the traditional BORG algorithm with the main difference being in the tracer modeling and likelihood. For BORG-PV, we will be using the supernovae distance moduli along with their redshifts as input. The goal will be to then infer the underlying

[1] N-body and hydrodynamical simulations are considered the highest fidelity simulations we can get to describe structure formation as they solving the equations of motion for each particle/fluid without any approximations or grid mesh

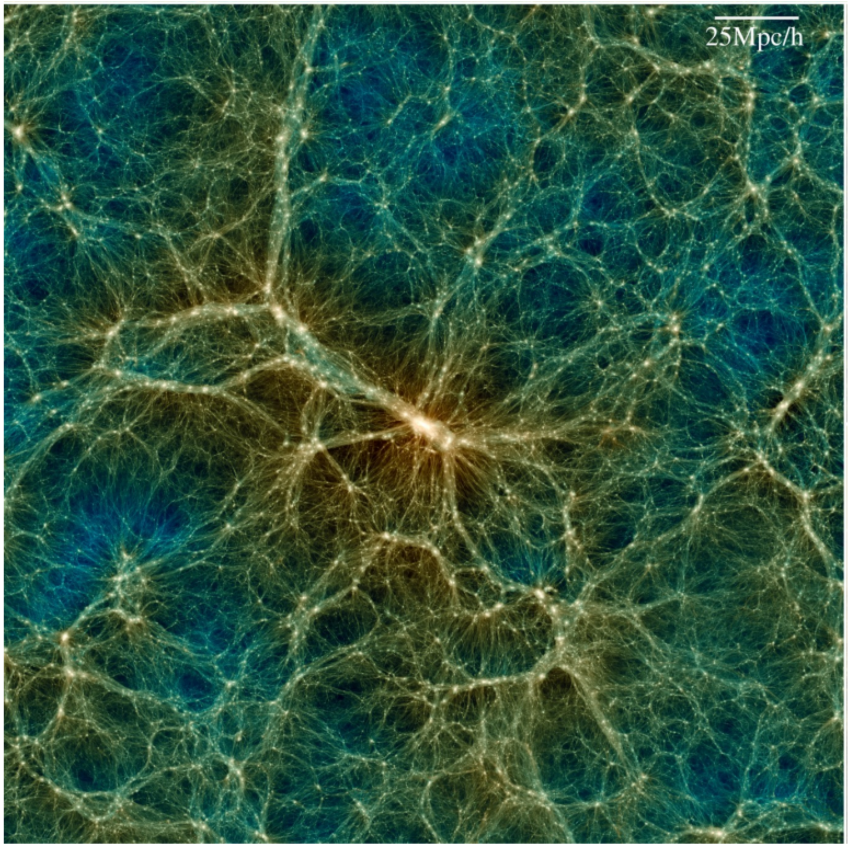


Fig. 6.1 Slice taken from the Uchuu simulation covering a box of 250 Mpc.h^{-1} where each point represents a particle. Taken from [?]

density and velocity field as . One major difference with the study of galaxies is in the implementation of a bias model. The study of galaxy bias models has a long history with many different models having been suggested such as a linear bias ([Kaiser, 1984]), a nonlinear local bias ([Fry and Gaztanaga, 1993]) or an EFTofLSS bias ([Mirbabayi et al., 2015]) and there are still ongoing efforts to improve it. For the case of supernovae we will be relying on a simple phenomonological approach which will be describeed in the following section alongside the supernova modeling and likelihood.

2.1 Initial likelihood

In this section we will be describing the initial version of the BORG-PV likelihood. This likelihood was developped by Deaglan J. Bartelett during the beginning of my PhD and it describes how well we can infer the supernova velocities. We will start by likelihood for one supernova indexed by i. We denote $\mu_{\mathrm{obs},i}$ the observed distance modulus[2], z_{obs} the observed redshift and σ_μ is the intrinsic dispersion of supernovae. The first ingredient we need to determining the likelihood is having a prediction for the observed redshift. This should include the contribution of the supernova peculiar velocities with the help of the inferred velocity field. For objects at low redshift, Equation 2.9 for comoving distances r can be

[2] As we explained in Chapter 2, we do not directly observe the distance modulus but instead it calculated through the observations of supernova lightcurves. In here we assume that that $\mu_{\mathrm{obs},i}$ is the distance modulus obtained after having observed the supernova lightcurves.

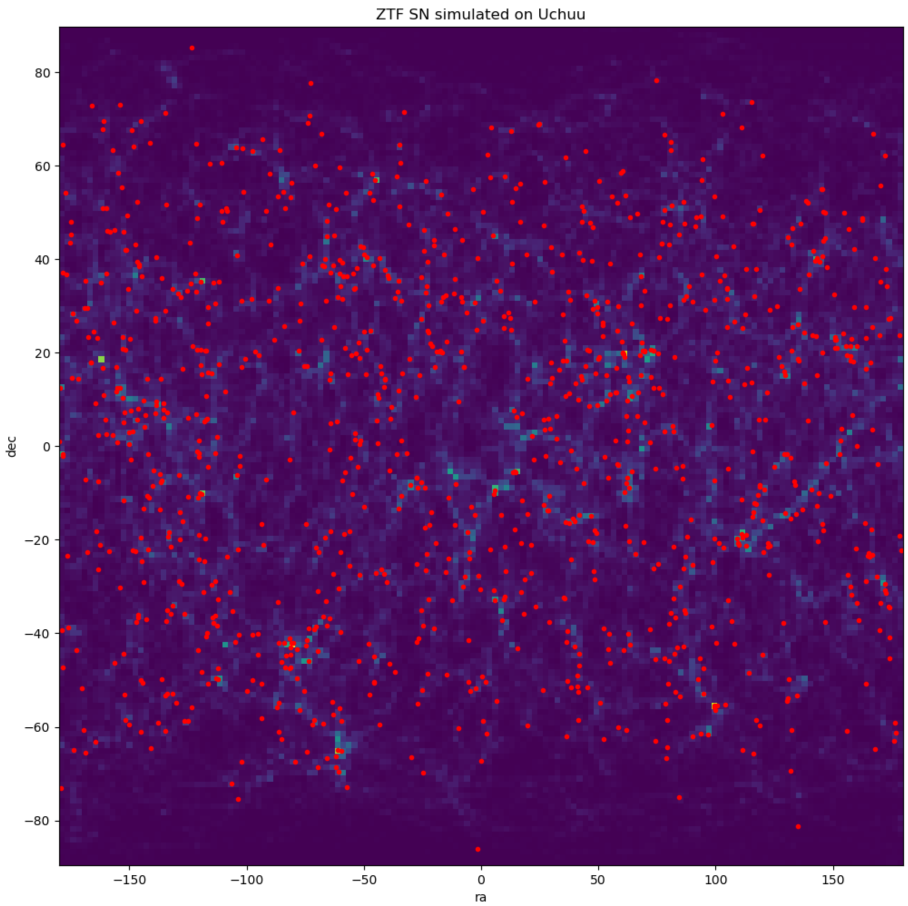


Fig. 6.2 2D full-sky projection of the Uchuu density field at redshift of $z \approx 0.04$ with the ZTF supernovae simulated on top of it represented by red dots.

approximated as:

$$r \approx \frac{c}{H_0}\left[z - \frac{1+q_0}{2}z^2\right] \tag{6.1}$$

with $q_0 = \frac{\Omega_m}{2} - \Omega_\Lambda$. By inverting this relation for z, we get:

$$z = \frac{1 - \sqrt{1 - 2rH_0(1+q_0)/c}}{1+q_0} \tag{6.2}$$

For our analysis, we fix H_0 and Ω_Λ to a fiducial cosmology. We can then infer the cosmological redshift by knowing the distance r, given by inverting the luminosity distance relation in Equation 2.20, and Ω_m. We will be denoting this redshift as z_{cosmo}. We can have a predicted redshift by adding the contribution of the velocity field using the following relation:

$$cz_{\text{pred},i}(r_i) = cz_{\text{cosmo},i}(r_i) + (1 + z_{\text{cosmo},i}(r_i))v_r \tag{6.3}$$

where $z_{\mathrm{pred},i}$ is the predicted redshift and v_r is the radial projection of the velocity field at a distance r_i. We also assume that the peculiar velocities can be affected by sources other than the velocity field which will be modeled as a Gaussian noise $\mathcal{N}(0, \sigma_v)$.

The second ingredient would be to define a prior on the supernova distribution. If we assume a constant density of supernovae, than the expected number of supernova N present at a distance r is purely geometrical giving us:

$$dN \propto 4\pi r^2 dr \tag{6.4}$$

This means that the number of supernovae present between r and $r+dr$ scales as r^2. Moreover, we define a supernova bias model relating the positions of supernovae with the evolved density field as:

$$b_{\mathrm{SN}}(r) = (1+\delta^F(r))^\alpha \tag{6.5}$$

where α is a free parameter. We'll call it the bias parameter. The bias parameter can exhibit four regimes:

- $\alpha = 0$: No bias exists and supernova do not preferential position regarding the dark matter field.
- $\alpha = 1$: Supernova exhibits a linear bias where the number of supernova is proportional to the overdensities.
- $\alpha > 1$: Supernova have a higher preference of being in overdense regions.
- $\alpha < 1$: Supernova have a higher preference of being in underdense regions.

The prior on the supernova positions can then be written as:

$$p_{\mathrm{SN}}(r) = r^2 b_{\mathrm{SN}}(r) = r^2(1+\delta^F(r))^\alpha \tag{6.6}$$

Thirdly, we define a selection function that accounts for the fact that supernovae are not observed infinitely far away. For this, we use the following selection function:

$$p_{\mathrm{selection}}(r) = e^{-\lambda \frac{r}{R_{\max}}} \tag{6.7}$$

where λ is a free parameter controlling the rate of decay and $R_{\max}$ is the maximum comoving distance of the survey. This parametrisation allows for a smooth exponential decay of the expected number of supernovae with distance, avoiding the sharp discontinuity that would arise from a hard cutoff at $r = R_{\max}$.

Finally, we marginalize over the distance r for the supernova and normalize the likelihood, giving us:

$$p_i(cz_{\mathrm{obs},i}|\mu_{\mathrm{obs},i}) = \frac{\left(\int_0^{R_{\mathrm{lim}}} dr \quad e^{-\frac{\left(\mu_{\mathrm{pred},i}(r) - \frac{\mu_{\mathrm{obs},i}}{\mu_A}\right)^2}{2\sigma_\mu^2}} \quad e^{-\frac{(cz_{\mathrm{pred},i}(r) - cz_{\mathrm{obs},i})^2}{2\sigma_v^2}} \quad p_{\mathrm{SN}}(r) \quad e^{-\frac{\lambda}{R_{\max}} r}\right)}{\left(\int_0^{R_{\mathrm{lim}}} dr \quad e^{-\frac{\left(\mu_{\mathrm{pred},i}(r) - \frac{\mu_{\mathrm{obs},i}}{\mu_A}\right)^2}{2\sigma_\mu^2}} \quad p_{\mathrm{SN}}(r) \quad e^{-\frac{\lambda}{R_{\max}} r}\right)} \tag{6.8}$$

Where R_{lim} is the integration limit which is greater than the furthest supernova, μ_A the distance uncertainty and $\mu_{\mathrm{pred}}(r)$ is calculated with:

$$\mu_{\mathrm{pred},i}(r_i) = 5\log_{10}\left((1+z_{\mathrm{cosmo},i}(r_i))r_i\right) + 25 \tag{6.9}$$

If do that for all supernovae, then the full BORG-PV likelihood then becomes

$$\mathcal{L} = \prod_{i=1}^{N_{\mathrm{SN}}} p_i(cz_{\mathrm{obs},i}|\mu_{\mathrm{obs},i}) \tag{6.10}$$

2.2 Generating self-consistent mocks

The self-consistent supernova mocks are simulated to mach the model description above. We start by generating a random Gaussian initial density field δ^{IC} and let it evolve using a gravity solver until we obtain a final density field δ^F and a velocity field $\mathbf{v}$. We then define an amplitude function Φ containing the bias model and selection effects:

$$\Phi(r,\phi,\theta) = (1+\delta(r,\phi,\theta))^{\alpha}.e^{-\frac{\lambda}{R_{\max}}r} \tag{6.11}$$

The supernova are then generated by a Poisson process where a discrete generation of the cartesian coordinates from the amplitude function Φ is done. The number of supernovae generated is chosen beforehand. A mask is applied to ensure that the supernova generated are within the limit R_{lim}. We then calculated the observed redshift using:

$$cz_{\mathrm{obs}} = cz_{\mathrm{cosmo}} + (1+z_{\mathrm{cosmo}})v_r + \mathcal{N}(0,\sigma_v) \tag{6.12}$$

where z_{cosmo} is calculated using Equation 6.2 and v_r is the radial projection of the velocity field at r for each supernova. The distance modulus is obtained the same way as Equation 6.9 and is then scattered so we get an observed distance modulus with:

$$\mu_{\mathrm{obs}} = \mu_{\mathrm{true}} + \mathcal{N}(0,\sigma_\mu) \tag{6.13}$$

Finally, the right ascension and declanation angles are reported by converting the cartesian coordinates into spherical coordinates.

2.3 Issues with intial likelihood

2.3.1 Selection function

While testing different parameter configurations for the previously mentioned selection function, we have noticed a strong dependency in the λ parameter that can bias the sampling of the other model parameters as well as the cosmology. We plot in Figure 6.3 the likelihood profile of a self-consistent simulation for α, the z-axis component of the bulkf flow V_z, Ω_m and σ_8 for 4 values of α : 1, 3, 5 and 7 while $R_{\max}$ is fixed at 370Mpc.h^{-1}.

What we notice is that all parameters are very sensitive to value of λ, more specifically that the smaller the value of λ, the more biased the parameters are. Therefore an incorrect initial estimation of λ can be greatly problematic. Moreover, this choice of modeling will fail if we decide to apply our BORG-PV model on the ZTF DR2 volume limited sample as the selection function does not match the hard cutoff at a redshift of $z = 0.06$. We can even attempt to set λ to 0 since the volume limited sample should be a complete sample but this results in the same issue as seen in Figure (ref).

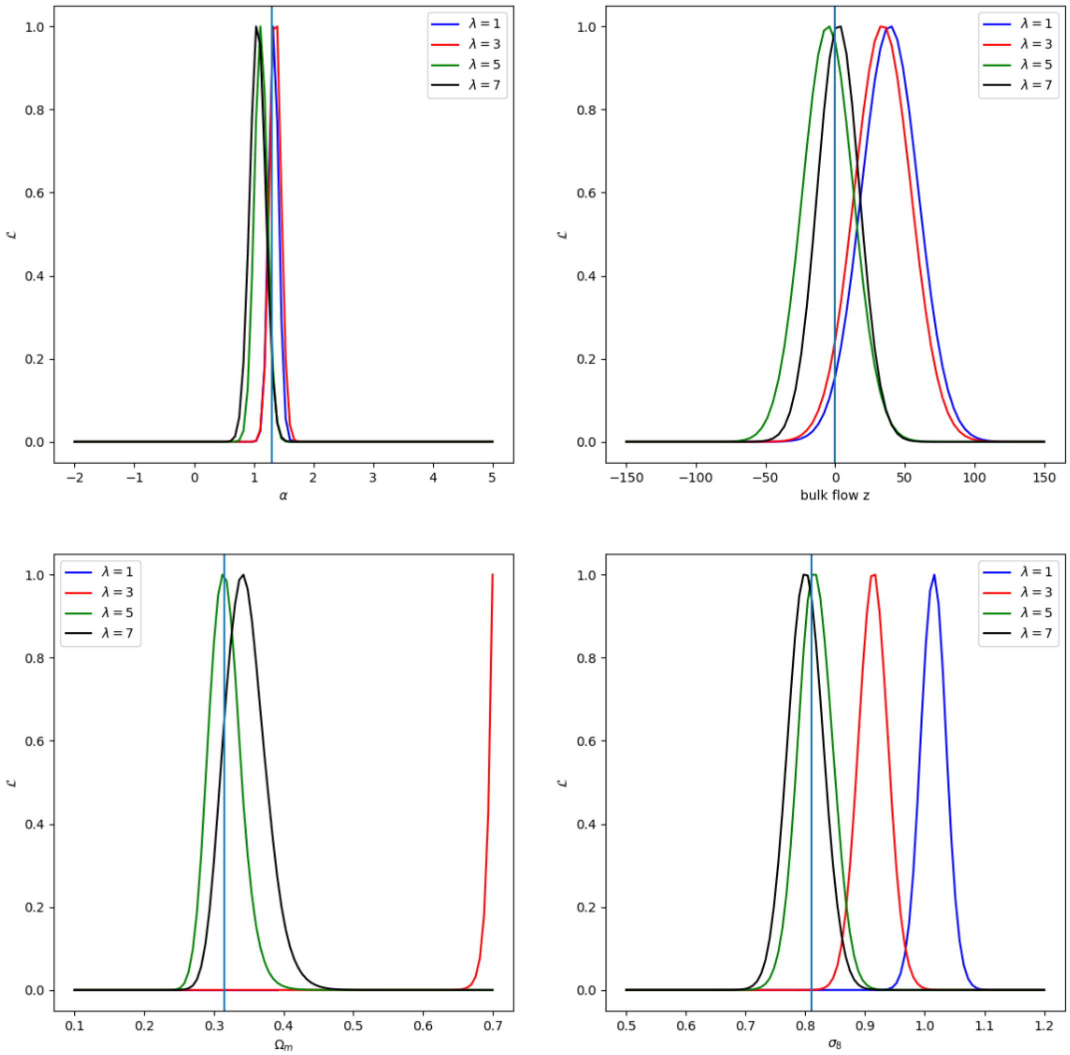


Fig. 6.3 Likelihood profiles using initial likelihood for different values of λ used in the generation of the mocks (1 for blue, 3 for red, 5 for green and 7 for black). The light blue vertical line represents the truth.

2.3.2 Supernova prior

Recent results from [Lordet et al., 2026] have shown that supernovae tend to be found in overly dense regions in the Universe with a large number of supernovae being present at a redshift of $z \approx 0.02 - 0.03$ creating what we see as a "bump" in the supernovae distribution accross redshift. This means that the assumption that the supernovae density is constant at low redshift (example of what was done in [Perley et al., 2020]) is not the most accurate description of the supernova distribution. The number of supernova $N_{\rm SN}$ observed by a survey between a distance r and $r + dr$ can be expressed as:

$$dN = n(r) \times 4\pi r^2 dr \tag{6.14}$$

where $n(r)$ is the number density of supernovae. This can be related to the observed rate as:

$$R(r) = \mathrm{S}(\phi, \theta, r) \frac{n(r)}{T_{\rm obs}} \tag{6.15}$$

where T_{obs} is the time of observation and S is a selection function that would account for the survey's footprint, galactic extinction and any instrumental effects that would decrease the number of observed supernova. For our applications on self-consistent simulations and the ZTF-Uchuu mocks (Sections ref and ref), we will be generating supernovae on the full-sky and will not include any Malmquist bias effects. Therefore S becomes unity. If we assume a constant rate, then $dN \propto r^2 dr$ giving the r^2 geometric dependence in the prior described earlier. If we assume that this is not the case, then we get:

$$dN(r) \propto n(r) r^2 dr \tag{6.16}$$

In practice, we do not have direct access to $n(r)$ but we can infer $n(z)$. In fact, we can relate the two number densities with:

$$n(r) \propto \frac{n(z)}{4\pi r^2 \times \frac{dr}{dz}} \tag{6.17}$$

The $\frac{dr}{dz}$ term can be deduced from Equation 2.9 and is entirely determined by the cosmology. We get $\frac{dr}{dz} = \frac{c}{H(z)}$. Finally, we can present a more realistic prior on supernovae expressed as:

$$p_{\text{SN}}(r) \propto r^2\, n(r)\, (1+\delta^F)^{\alpha}, \quad \text{with } n(r) \propto \frac{n(z)}{4\pi r^2 \times \frac{dr}{dz}}. \tag{6.18}$$

Finally, we fit the inferred $n(r)$ with a smooth differentiable function to be then implemented in the likelihood as seen in Figure 6.4.

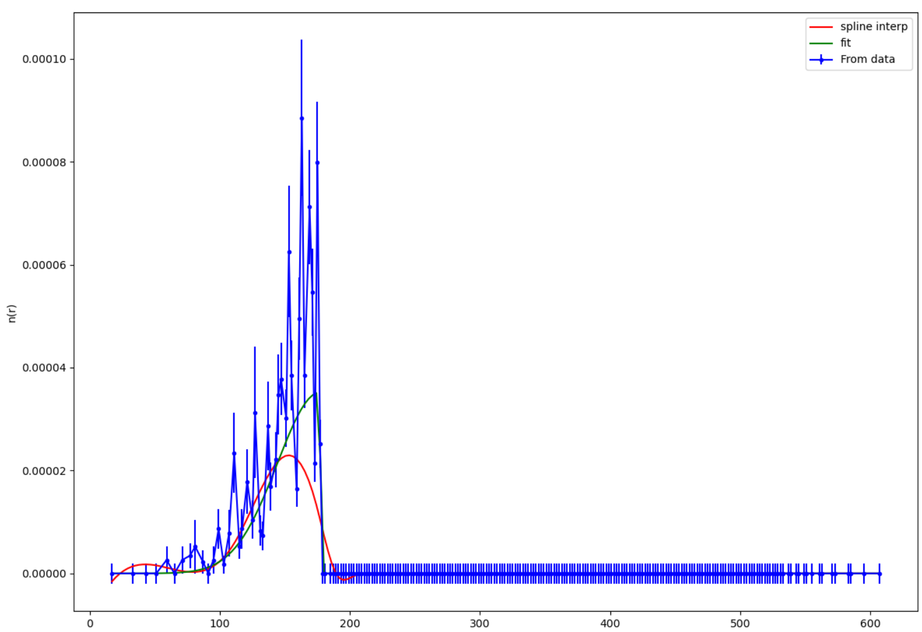


Fig. 6.4 (PLOT NEEDS TO BE REMADE) The number density as a function of the comoving distance from the observer. The red line represents a cubic spline fit. The green line represents a fit using a CDF function convoluted with a Gaussian.

Other motivations for this new prior lies in the ZTF-Uchuu simulations themselves. We could assume that such effects can be captured by the bias model. We compare self-consistent simulations with the ZTF-Uchuu simulations containing the exact same number of supernovae. We generated a set of self-consistent mocks each with a different value for α. We bin the number of supernovae based on the overdensity. If the simulations are generated in the same way, we would expect to see a match between the bins for a given α value. What we noticed that there isn't a singular value of α for which the self consistent simulations match the ZTF-Uchuu simulations. We obtain a best match with $\alpha = 1.4$ presented in Figure 6.5. Moreover, we seem to notice a regime change based on the over density for the ZTF-Uchuu simulations, which could possibly hint that the difference can be capured using two bias model parameters. During this PhD I have not explored this avenue but instead relied on the new supernova prior to capture the differences between the simulations.

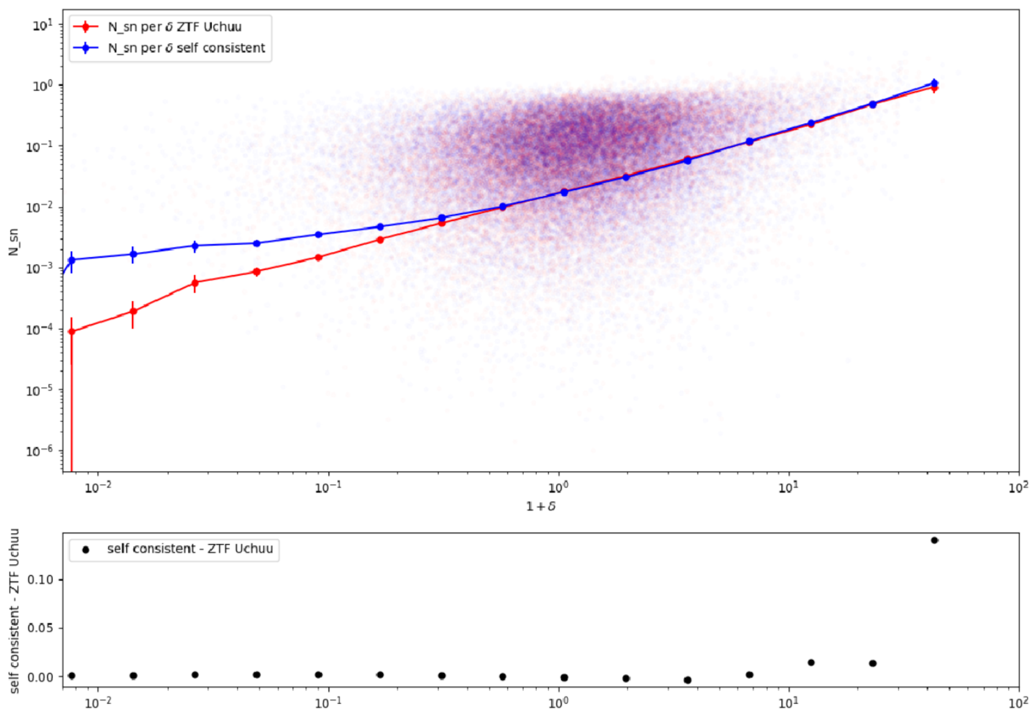


Fig. 6.5 (PLOT NEEDS TO BE REMADE) Comparing a self-consistent simulation generated with $\alpha = 1.4$ with the ZTF-Uchuu simulation by plottinh the number of supernova in a voxel with respect to the overdensity of that voxel. The blue dots are measured on the self consistent simulation while the red dotsare measured on the ZTF-Uchuu simulation. The thicker dots are the averaged value of the number of supernovae obsereved.

2.4 Improved likelihood

Our first improvement to the likelihood is in the implementation of the prior $p_{\rm SN}$ presented in Equation 6.18. The second improvement was done by replacing the selection function $p_{\rm selection}(r)$. We were inspired by the work done in [Stiskalek et al., 2025]. The method derived in the paper largely follows what was done in [Kelly et al., 2008]. Given a an observed dataset $\mathbf{d}_i$ for a supernova i described by a parameters $\boldsymbol{\theta}$ (such as α, σ_v, etc...) and a selection function S such that $S = 1$ is the data is observed and $S = 0$ if not, then the general expression of the likelihood marked by a Poisson process is given by:

$$\mathcal{L} \propto \prod_{i=1}^{N_{\rm SN}} \frac{p(\mathbf{d}_i, S=1|\boldsymbol{\theta})}{p(S=1|\boldsymbol{\theta})} \tag{6.19}$$

where $p(\mathbf{d}_i, S=1|\boldsymbol{\theta})$ denotes the likelihood for an individual supernova and $p(S=1|\boldsymbol{\theta})$ is the probability for any supernova to be selected. We can show using Bayes' theorem that:

$$p(\mathbf{d}_i, S=1|\boldsymbol{\theta}) = p(\mathbf{d}_i|r_i,\boldsymbol{\theta})p(S=1|r_i,\boldsymbol{\theta})p_{\rm SN}(r_i) \tag{6.20}$$

and since our dataset only contains supernova below the redshift threshold then $p(S=1|r_i,\boldsymbol{\theta}) \approx 1$. Moreover, we marginalize over the distance as we did before and given that the observable is the supernova redshift we get:

$$p(cz_{{\rm obs},i}, S=1|\boldsymbol{\theta}) = \int_0^{R_{\rm lim}} p(cz_{{\rm obs},i}|r,\boldsymbol{\theta},\mu_{{\rm obs},i})p_{\rm SN}(r)dr \tag{6.21}$$

The paper also gives the exact formulation when the selection is done on redshift, which is exactly the case for the ZTF DR2 volume limited sample. In that case, the fraction of detected samples is given by Equation (33) in the paper:

$$p(S=1|\boldsymbol{\theta}) = \int p_{\rm SN}(\mathbf{r})\Phi_{\rm CDF}\left(\frac{cz_{\rm thresh} - cz_{{\rm pred}(r)}}{\sigma_v}\right) d^3\mathbf{r} \tag{6.22}$$

where $z_{\rm thresh}$ is the limiting redshift ($z_{\rm thresh} = 0.06$ in our case) and $\Phi_{\rm CDF}$ the normal cumulative distribution function. The full likelihood is therefore given by:

$$\mathcal{L} = \prod_{i=1}^{N_{\rm SN}} \frac{p(cz_{{\rm obs},i}|\mu_{{\rm obs},i})}{p(S=1)} \tag{6.23}$$

Or more simply, we can take the negative logarithm of the likelihood which gives us:

$$\boxed{-\log\mathcal{L} = -\sum_{i=1}^{N_{\rm SN}} p(cz_{{\rm obs},i}|\mu_{{\rm obs},i}) + N_{\rm SN}p(S=1)} \tag{6.24}$$

Where $p(cz_{{\rm obs}_i}|\mu_{{\rm obs}_i})$ is the likelihood of each supernova with no selection effects which is given by

$$\boxed{p(cz_{{\rm obs}_i}|\mu_{{\rm obs}_i}) = \int_0^{R_{\rm lim}} dr \quad e^{-\frac{\left(\mu_{{\rm pred},i}(r) - \frac{\mu_{{\rm obs},i}}{\mu_A}\right)^2}{2\sigma_\mu^2}} \quad e^{-\frac{(cz_{{\rm pred},i}(r) - cz_{{\rm obs},i})^2}{2\sigma_v^2}} \quad r^2\, n(r)\,(1+\delta^F(r))^\alpha} \tag{6.25}$$

3 Applications on self-consistent mocks

Before running BORG-PV on the ZTF-Uchuu mocks, we need to ensure that if we run it on simulations that are generated with the exact same model as the one used in BORG-PV that it would not introduce systematic biases in the reconstructed density and velocity fields as well as the model parameters.

3.1 Forward model analysis

We conduct an initial test on the forward model where we realize likelihood profiles for each model parameter and cosmological parameters. We do that by evaluating the the forward model on the initial conditions δ^{IC} used to generate the supernova after having evolved while varying each parameters and evaluating the likelihood. What expect to see is a peak in the likelihood profiles of each parameter at the value used to generate the mock. We show in Table 6.1 the configuration used to generate the self consistent simulations.

We obtain the following Figure 6.6 that show the likelihood profiles for 1 simulation. We notice that

Tab. 6.1 Set of parameters used to generate the "synthetic" N-body BORG simulation.

Parameters	Values
Gravity model	LPT or COLA
Resolution	10.9 Mpc.h^{-1}
α	1.4
μ_A	1.00
σ_v	150 km.s^{-1}
V_x	0 km/s
V_y	0 km/s
V_z	0 km/s
Ω_m	0.315
σ_8	0.81

the recovered parameters match the input values within 1σ with the exception of the σ_v parameter. It is inconclusive to say that our likelihood is biased given that this was only done on 1 realization. *This part is yet to be done so no results yet but is easy to be done* To assess the statistical robustness of the forward model, we generate 100 self-consistent mock realisations of the supernova catalogue, each drawn from a different set of initial conditions while keeping the model parameters fixed at their fiducial values. For each realization, we evaluate the likelihood profile for every model and cosmological parameter, and compute the mean and standard deviation of the profile peaks across the 100 realizations. We expect that the averaged parameters would coincide with the fiducial input values within the expected standard deviation. Any systematic offset would indicate a bias in the forward model that must be understood and corrected before applying the pipeline to real data. We plot the results in Figures ().

3.2 Results of the field level inference on self-consistent simulations

We present the results of applying BORG-PV on self consistent simulations. In Figure 6.7 we show the generated and reconstructed density field. We notice that the field reconstructed with BORG-PV manages to capture some of the underlying structure visible in generated by comparing the overdense and underdense regions within the red-dashed circle where the supernovae have been generated. Notably, it

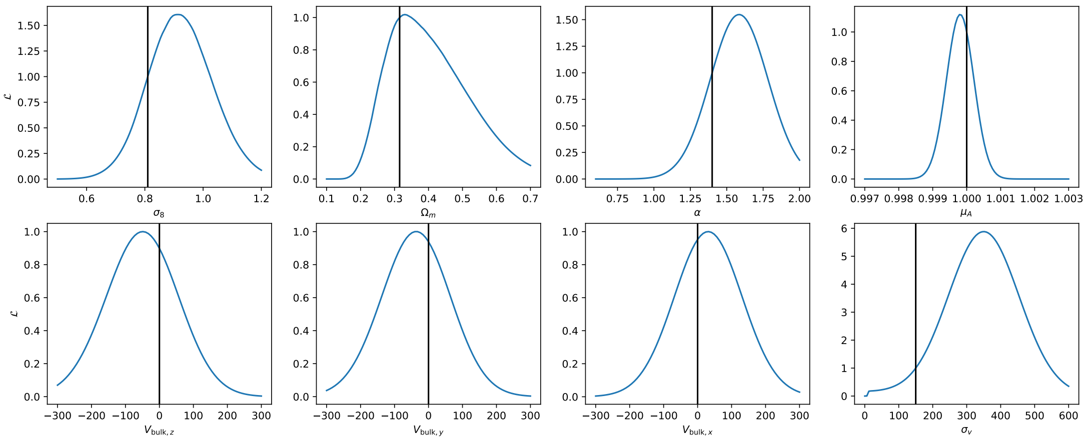


Fig. 6.6 Profile likelihoods from 1 realization of the self-consistent simulation for the model and cosmological parameters.

is the largest structures that are recovered while small scales structures seem to be missed.

A more quantitative result is shown in Figure 6.8 where we plot the power spectra for the initial density field δ^{IC} and final density field δ^{F} calculated in the largest box within the region the supernovae were generated. We notice that our density fields' power spectra are in good agreement with the one from the generated mock at large scales. At very small scales this is not true anymore.

We can also look at the reconstructed model and cosmological parameters in Figure 6.9. We notice the parameter sampling's mean is consistent with the input values within 2σ. This is to be expected since we are only running on one realization of the self consistent simulations.

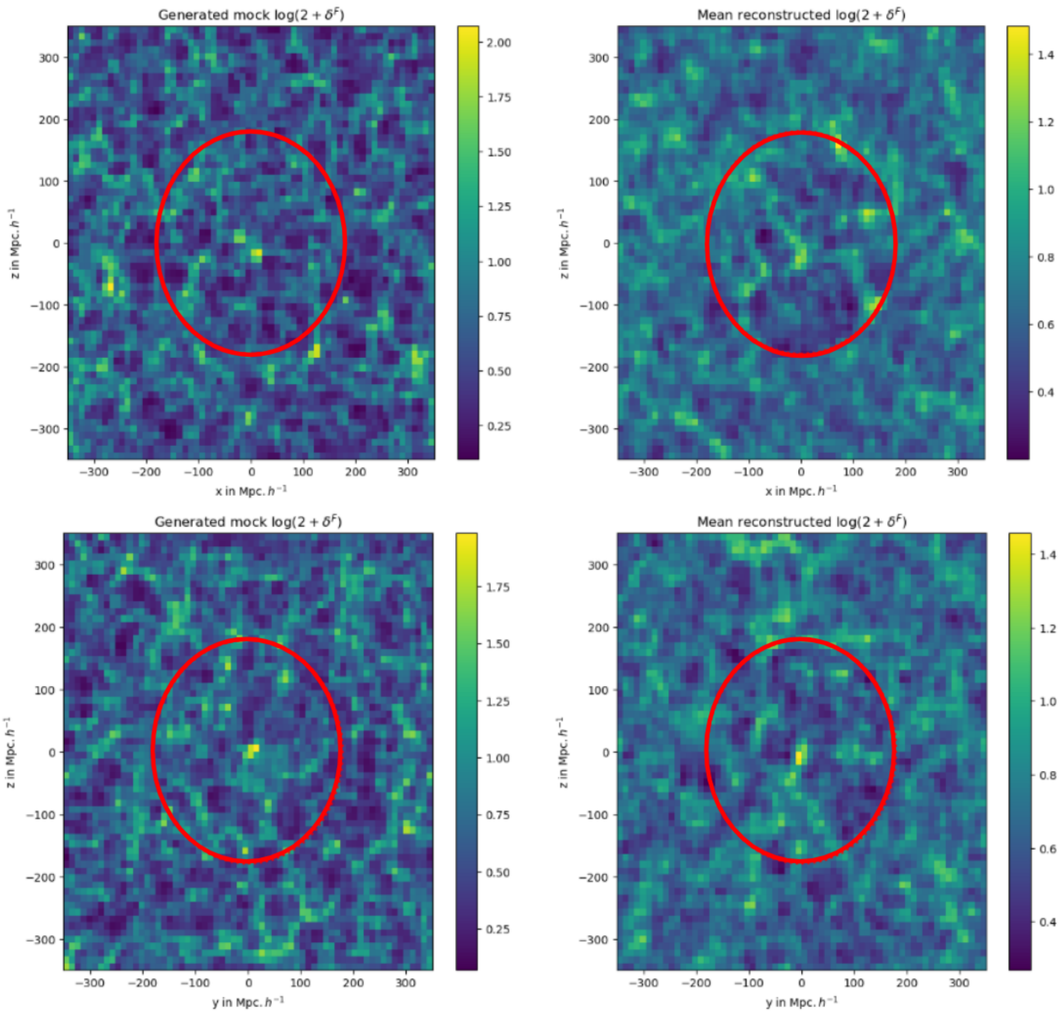


Fig. 6.7 (PLOT NEEDS TO BE REMADE) Two slices of the generated and reconstructed density fields using self-consistent simulations in each row.

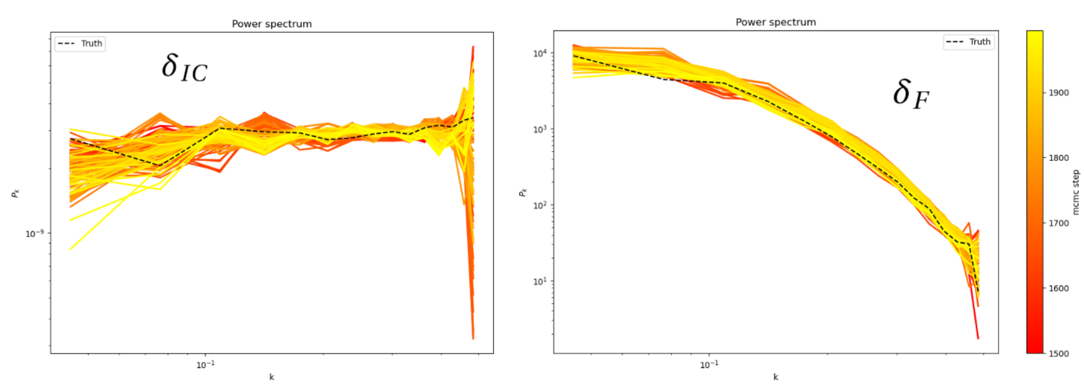


Fig. 6.8 (PLOT NEEDS TO BE REMADE) Reconstructed power spectra at each mcmc step vs the true power spectrum in black for the initial conditions (Left) and final density field (Right).

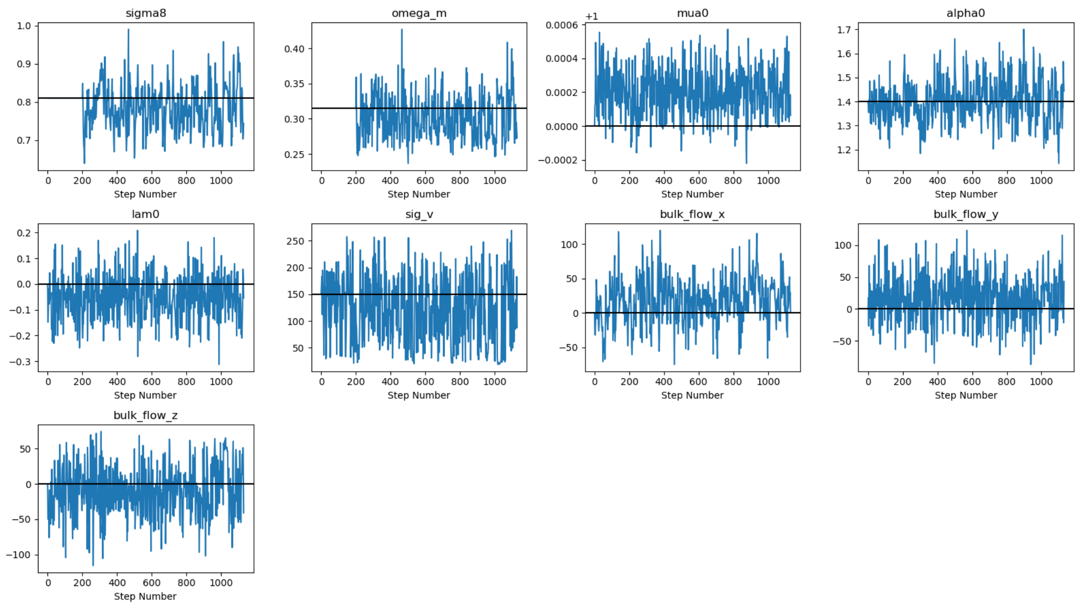


Fig. 6.9 (PLOTS FROM DEAGLAN REDO THEM) Model and cosmological parameter sampling for the self-consistent simulation run.

4 Applications on Uchuu-ZTF mocks

4.1 Supernova forward-model analysis

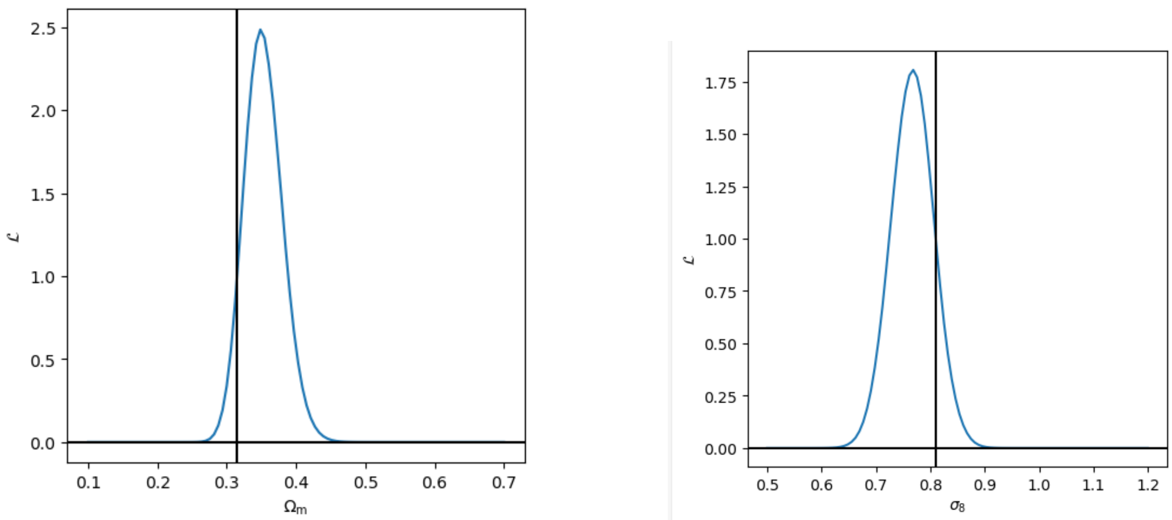
The methodology used to generate supernova in the self consistent simulations and DC PV are different as we have mentioned earlier. We want to ensure that we can quantify any systematic biases introduced by those differences. To do this we start by generating an intial density field using BORG $\delta_{IC}^{\mathrm{BORG}}$ using a Planck18 cosmology ([Aghanim et al., 2020]) and known bulk flow $\mathbf{V}$. The configuration of the generated field is given in Table 6.2. We then let that field evolve using the LPT gravity model giving us the evolved density field δ_F^{BORG} and velocity field $\mathbf{v}_{IC}^{\mathrm{BORG}}$. The ZTF simulations are designed to generate the supernovae on top of an N-body simulation, so from the evolved density field we create a "synthetic" N-body simulation. The simplest way to do so would be taking each voxel-i and generate within it N_i particles proportional to $(1 + \lceil \delta_F^{\mathrm{BORG}} \rceil)_i$ where i denotes the voxel's index. We take the cieling of the evolved density field to ensure we get an integer number of particles and to have at least 1 particle in empty voxels. We also generate less particles than the Uchuu simulation, around 200 million, due to memory and storage limiations.

Tab. 6.2 Set of parameters used to generate the "synthetic" N-body BORG simulation.

Parameters	Values
Gravity model	LPT
V_x	0 km/s
V_y	0 km/s
V_z	0 km/s
Ω_m	0.315
σ_8	0.81

We then use this "synthetic" N-body simulation to generate ZTF-like supernova and then evaluate the BORG-supernova likelihood described in Equation (ref) while varying the cosmological and model parameters in BORG. Essentially creating a likelihood profile for each parameter. This is particularly important not just to determine whether the ZTF-like supernova simulations introduce systematic biases in BORG but also help us understand how we can relate some of the BORG-supernova model package-sarameters, such as α to the ZTF simulations as they are not explicitly specified.

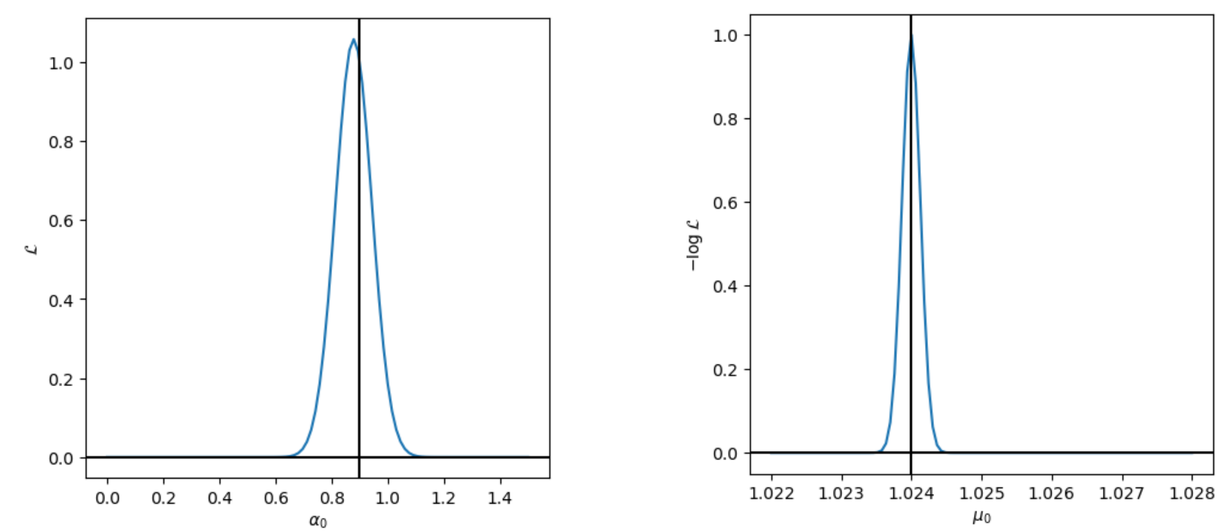
We obtain the following likelihood profiles for 1 synthetic N-body simulation in Figures 6.10, 6.11, 6.12 and 6.13. We notice that the cosmological parameters Ω_m and σ_8 are unbiased and we recover the input cosmology. The bulk flow V parameters are compatible with 0 km/s, indicating that no additional bulk flow is added with the ZTF simulation. The bias model parameter α seems to prefer values around $0.9 - 1.0$. This is not unexpected, if we assume that the ZTF supernova are generated where there are dark matter particles without a preference on whether then the number of supernovae would simply scale as δ_F. The velocity dispersion is prefered around 150 km/s with a large scatter. Measurements from (ref smth regarding sigv like velocity olympics) getting such values is possible and is indeed physical. The μ_A parameter seem to prefer a value of 1.024 indicating that an offset of about 2.4% regarding how distances are described in BORG. (check where does it come from but it is not an issue for our needs).



(a) Likelihood profile for Ω_m

(b) Likelihood profile for σ_8

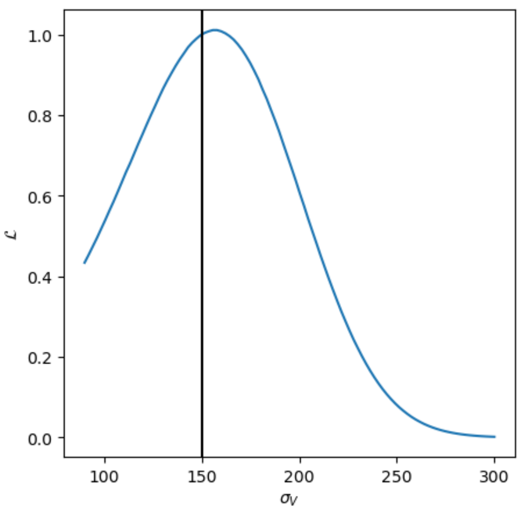
Fig. 6.10 Likelihood profiles of the cosmological parameters for a synthetic N-body simulation generated with LPT



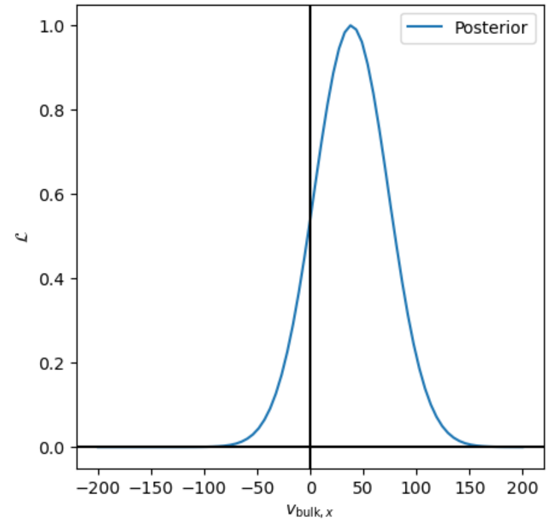
(a) Likelihood profile for α

(b) Likelihood profile for μ_A

Fig. 6.11 Likelihood profiles of α and μ_A for a synthetic N-body simulation generated with LPT

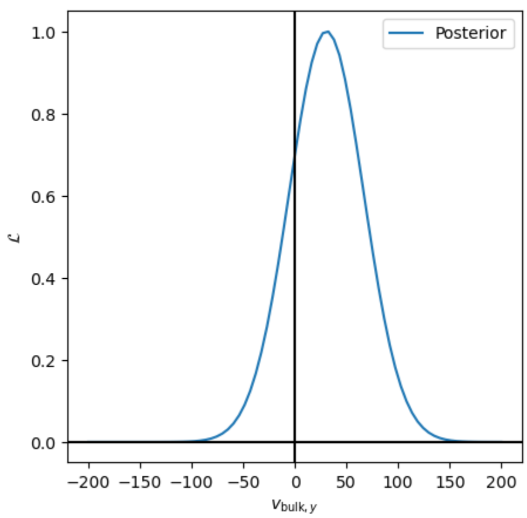


(a) Likelihood profile for σ_v

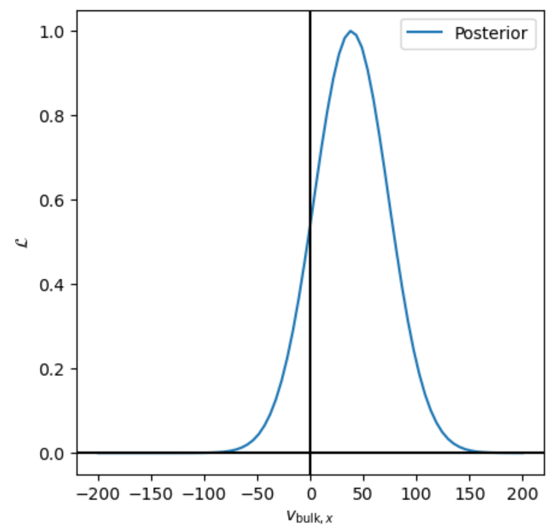


(b) Likelihood profile for V_x

Fig. 6.12 Likelihood profiles σ_v and V_x for a synthetic N-body simulation generated with LPT



(a) Likelihood profile for V_y



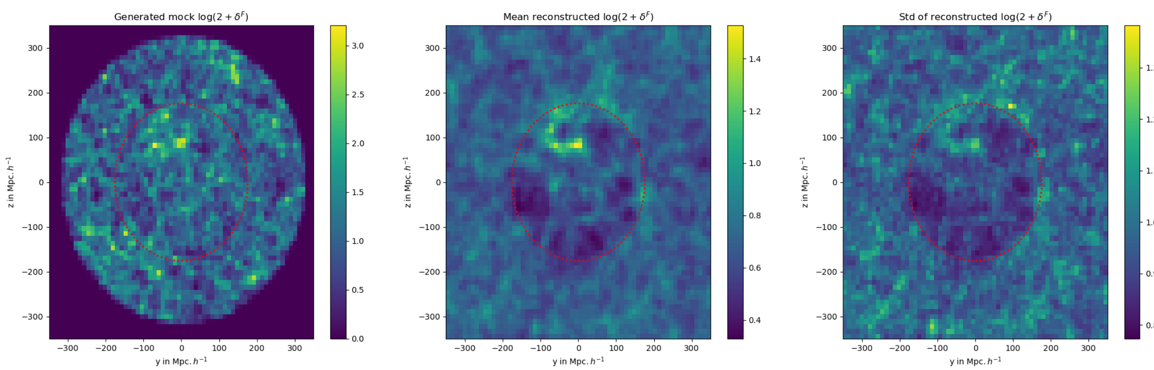
(b) Likelihood profile for V_z

Fig. 6.13 Likelihood profiles V_z and V_z for a synthetic N-body simulation generated with LPT

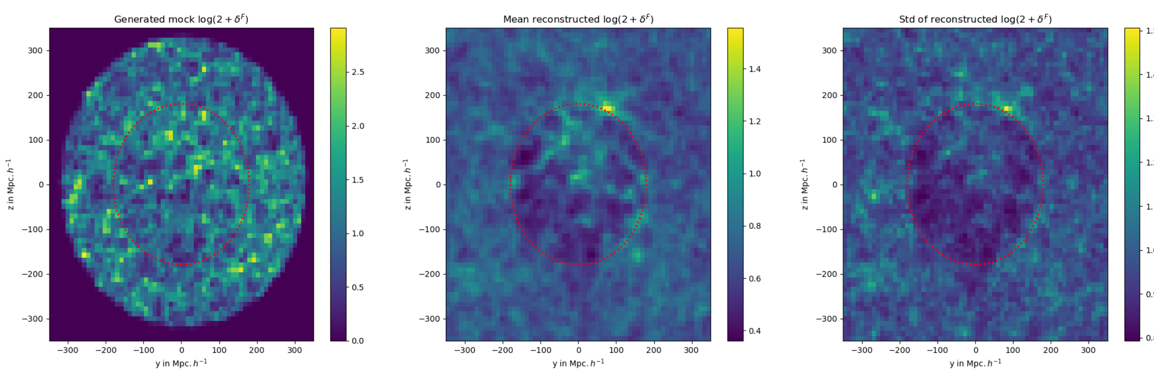
5 Results of the field level inference on Uchuu-ZTF simulations

We present in this section the results obtained on the DC PV. The proceeder follows closely what was done on the self consistent simulations. We start our MCMC sampling of intial conditions for 1000 step while keeping the cosmological and supernova model parameters fixed in order to get an initial estimate of the true underlying field. We then sample the model parameters α and μ_A alongside the bulk flow $\mathbf{V}$ and the velocity dispersion σ_v for another 1000 step all while continuing to sample the initial conditions. Finally, we sample Ω_m and σ_8 alongside the other parameters until we reach a state of convergence.

Firstly, we present the results obtained when using an LPT gravity solver. We show in Figures 6.14 slices of the final reconstructed density field with BORG in comparison with the Uchuu density. The red-dashed circle represents the limit of where the supernova are being generated (up to $z = 0.06$).



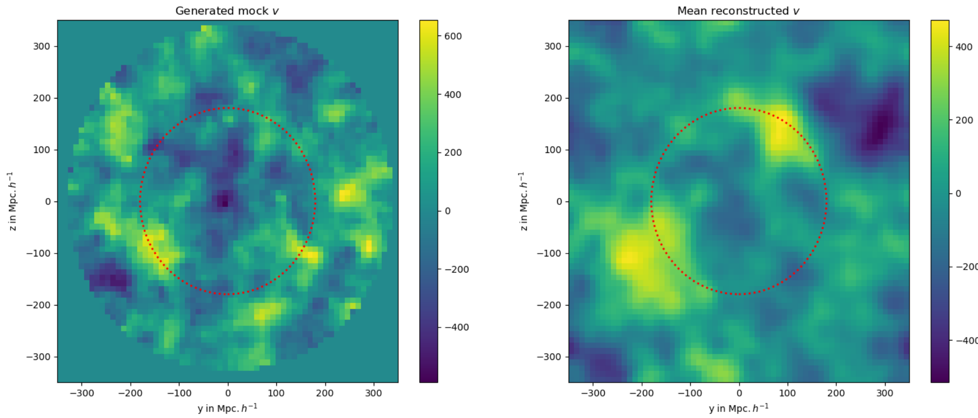
(a)



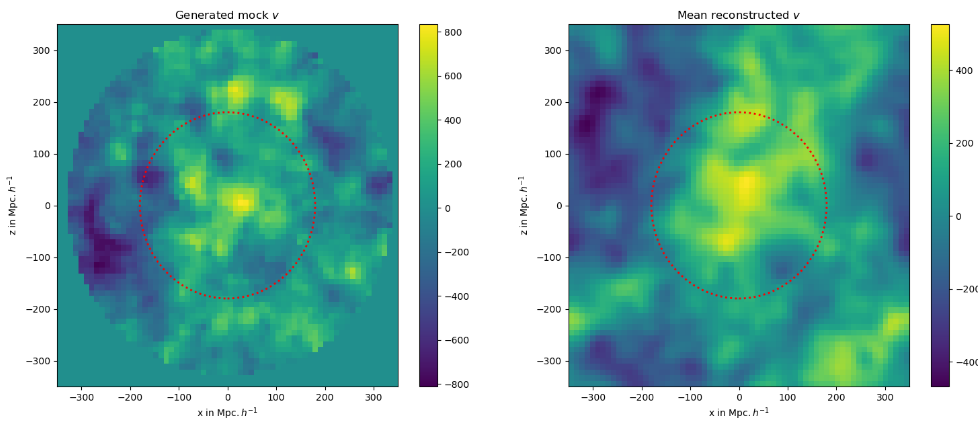
(b)

Fig. 6.14 Slices of the density field from Uchuu and BORG reconstruction after having gone a CIC mass assignment. The left column shows the Uchuu density field. The center column shows the mean posterior from BORG. The right column shows the standard deviation. Each row is a different slice in the box.

We notice that the field reconstructed with BORG-PV manages to capture some of the underlying structure visible in Uchuu by comparing the overdense and underdense regions within the red-dashed circle. Notably, it is the largest structures that are recovered while smal scales structures seem to be missed. We show in Figures 6.15 slices of the final reconstructed velocity field with BORG in comparison with the Uchuu velocity.



(a)



(b)

Fig. 6.15 Slices of the velocity field from Uchuu and BORG reconstruction after having gone a CIC mass assignment. The left column shows the Uchuu velocity field. The reight column shows the mean posterior from BORG. Each row is a different slice in the box. The first row is a slice in the x axis. The second row is a slice in the y axis.

Similarly to the density field reconstruction, the BORG velocity field seem to match the Uchuu velocity at large scales. It's important to note that the velocity field at smal scales does not present the same variation as the density field, which could explain why the velocity field seems to be reconstructed more accurately than the density field. To quantify this, we compare the power spectra of the density and velocity field components v_x, v_y and v_z from Uchuu to the one inferred by BORG in Figures (ref) as well as the cross correlation between them r in Figures (ref). We notice that the power spectrum of the matter density matches the Uchuu power spectrum up to a wavenumber of $k \approx$. For the velocity fields we notice a matching result up to a wavenumber of $k \approx$. This confirms our qualitative results mentioned previously when looking directly at the reconstructed fields.

Secondly we look at the sampled parameters.

NEED SOME COLA RESULTS TO BETTER WRITE OFF THIS PART
- results - covariance additions ? - full dataset with selection?

6 Comparison with other methods for $f\sigma_8$

As with the self consistent simulation, we are restricted to a $\Lambda - CDM$ cosmology. Therefore if assume that $f\sigma_8$ is expressed as:

$$f\sigma_8 = \Omega_m^{0.55}\sigma_8 \tag{6.26}$$

by calculating this quantity, we can compare with other methods within the ZTF collaboration used to calculate $f\sigma_8$. Currently there is one ongoing work in doing so conducted by Rafel Kebadian (Kebadian et al. in prep) where $f\sigma_8$ is determined using a maximum likelihood method on the DC PV supernovae. We present in Figure (ref) the results from the different methods. We (hope) to see that using a field-level inference we obtain tighter constraints on $f\sigma_8$. However as mentioned before, we are limited to ΛCDM, so any deviations from the expected value do not indicate what the strength of the underlying gravitational theory should be as much as it is a test on knowing whether the assumed gravitational theory (GR) is correct or not. The maximum likelihood method however does not present that issue, but the $f\sigma_8$ constraints are looser.

Titre :

Titre ici

Résumé *(en français)* **:**

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Mots-clés :

Supernova, galaxie

Title:

Title here

Abstract *(in english)***:**

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Keywords:

Supernova, galaxy

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